Reaching for Yield in the Dark

Smaller Banks, Local Government Debt, and Household Savings in China

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Abstract

Household savers in China's RMB 33 trillion market for bank wealth-management products respond strongly to the returns that products report and to how much those returns fluctuate, but not to the risk behind them. Using the products' holdings, we show that smaller banks, which lack the reputation to attract savers otherwise, exploited this bias to compete for savings. Their products reached for yield by holding riskier debt than larger banks' products, and the largest source of the extra yield was the debt of local government financing vehicles (LGFVs) in the banks' home provinces. Yet smaller banks rated these products as safe, locked savers in with restricted redemptions, and reported returns that barely moved, so the risk stayed in the dark. The strategy needed both the risky yield and the concealment of its risk, so it was fragile. China's 2023 debt-resolution program absorbed much of the LGFV risk and lowered the yield with it. Smaller banks relied on that yield the most, so they lost returns and fundraising. When banks have little reputation to lose, they compete through a fragile strategy that exploits the behavioral bias of households.

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1 Introduction

Bank wealth-management products (WMPs) are a major alternative to bank deposits for Chinese households. At the end of 2025 they held RMB 33 trillion, nearly a quarter of China's GDP. They had about 143 million investors, and almost all of them were individuals.[1] Since 2018 these products cannot guarantee principal, so their losses fall on savers. Most of these savers are not financially sophisticated. They see a product through what its bank reports and not through the assets it holds. The bank reports the product's returns, how much those returns fluctuate, and a risk rating that it sets itself and that decides who may buy the product. The banks that sell these products differ widely in reputation, from large national banks with nationwide franchises to small city and rural banks with little reputation to lose. We ask how banks with little reputation compete for savers who see only what is reported, and what risk this competition places in household savings.

To answer these questions, we link the holdings of China's WMPs to their reported returns, ratings, and investor flows. Savers respond strongly to the returns that products report and to how much those returns fluctuate, but not to the risk behind them. Smaller banks cannot compete on reputation, so they exploit this bias. Their products reach for yield by holding riskier debt than larger banks' products. The largest source of their extra yield is hidden local government debt, the debt of local government financing vehicles (LGFVs) in their home provinces (Chen et al. 2020, Acharya et al. 2025). Yet smaller banks rate these products as safe, lock savers in with restricted redemptions, and report returns that barely move, so the risk stays in the dark. In the 2022 bond sell-off, their holdings lost about 1.6% of their value at market prices in two months, while their reported returns only dipped to about zero. Smaller banks need this strategy to survive. Without the risky yield they cannot draw savers away from more reputable banks, and if the risk showed, savers would run. The strategy needs both the risky yield and the concealment of its risk, so it is fragile. China's

[1] Industry statistics are from the Banking Wealth Management Registration and Depository Center's annual market reports (2025, 2026), referred to below as the annual WMP market reports. China's 2025 GDP of RMB 140.2 trillion is the National Bureau of Statistics' preliminary estimate.

2023 debt-resolution program absorbed much of the LGFV risk and lowered the yield with it. Smaller banks relied on that yield the most, so they lost returns and fundraising. A model of competition between banks that differ in reputation explains these patterns.

Our data are the quarterly portfolio disclosures of 60,396 WMPs issued by 171 banks. The disclosures run from 2022Q1 to 2025Q3 and list each product's largest holdings. We link them to each product's reported returns, risk rating, and investor flows, to the issuers of the bonds it holds, and to its bank's balance sheet, branches, and WMP balances. Figure 1 shows the main patterns in the raw data for national banks, larger local banks, and smaller local banks. We call a bank smaller when its assets at the end of 2021 are at or below the median of the banks in our main sample, which is RMB 0.44 trillion. All smaller banks are local banks. Before the 2023 program, smaller banks' WMPs held bonds issued at higher spreads than those held by larger local banks and national banks, and they held more home LGFV bonds. Their holdings implied the most return volatility, yet their products reported the least. National banks' products reported nearly all of the volatility their holdings implied, whereas smaller banks' products reported about half. Nearly all of smaller banks' products restricted redemptions, and nine in ten carried a safe rating.

Savers reward the returns and stability they see, not the risk behind them (Section 3). Net inflows into products in the top fifth of past returns exceed those into the bottom fifth by 12% of assets. Savers also punish small dips below the usual quarterly return of 0.5 to 1%. Daily-open products that reported 0.25 to 0.5% had net flows 43 percentage points of beginning assets below those of products of the same bank that reported the usual return. The gap holds within the same product, and flows do not respond to reported returns where savers cannot redeem. The 2022 bond sell-off provides a natural experiment. It cut WMP balances from about RMB 31 trillion to 25 trillion within five months (Liu and Xie 2024), and the same market losses appeared to very different degrees in the returns that products reported. Savers ran from the losses that products reported, not from the losses their holdings took. Among products of the same bank, each 10 basis points a month of reported loss in

November and December 2022 came with withdrawals of 1.2% of assets in the next quarter, whereas the loss of the holdings at market prices made no significant difference.

Smaller banks earn their higher returns by holding riskier debt (Section 4.1). Their products report about 0.5 percentage points a year more than larger banks' products with the same redemption terms and rating. The holdings of smaller banks' products carry a credit spread over Treasury yields of 205 basis points, against 149 at national banks. Home LGFV bonds are the largest source of this gap, as smaller banks earn 61 basis points more than national banks from them. They also earn 14 basis points more from the capital bonds of other local banks and 14 more from corporate bonds of their home provinces, whereas national banks earn 26 basis points more from private credit and asset-management plans. LGFV bonds made up 43.1% of the bonds and private credit held by local banks' WMPs, against 12.6% at national banks, and most of local banks' LGFV bonds came from their home provinces.[2] Home LGFV bonds also earned the largest return premium of any asset class we examine.

Smaller banks' products hold more risk than larger banks' products, yet smaller banks rate them as safe, lock savers in, and report returns that barely move (Section 4.2). A saver cannot tell apart products with the same redemption terms and rating. Among such products, a product of a bank half as large holds bonds issued at spreads 16 basis points higher and its holdings imply 5% more return volatility, yet it reports 10% less volatility. Official credit ratings make this debt look safer than its price indicates. AAA-rated LGFV bonds were issued at an average spread of 170 basis points, against 75 for other AAA-rated bonds. The banks' own product ratings do the same. About four in five smaller banks rate every fixed-income product they sell as safe, against about one in three larger banks. In addition, the average smaller bank restricts withdrawals on 98% of its products, against 89% at larger banks. In the 2022 sell-off, smaller banks' products showed a third of the fall in the value of their holdings, and larger banks' products showed three-fifths. Products that

[2]Shares are by holding value over 2022Q1–2023Q3, relative to identified bonds and private credit with known borrower locations, excluding central-government, central-bank, and policy-bank obligations (Section 4.1).

lock savers in showed the least, and smaller banks' closed-end products still reported positive returns in the sell-off months.

This was reaching for yield, not better local information (Section 4.3). We compare banks headquartered in the same city, where the larger bank has more branches and the two banks hold similar shares of their portfolios in the city's LGFV bonds. Yet a bank half as large holds more bonds of the city's riskier county-level issuers, and in other cities of its home province it holds bonds with coupons 24 basis points higher. In other provinces neither bank is at home, and there the coupon gap is 53 basis points. Those out-of-province issuers also had more credit events later.

Debt resolution removed the main source of the yield on which smaller banks competed (Section 5.1). After the 2023 program, average LGFV issuance spreads over Treasury yields fell from about 200 basis points to 97. Before the program, a product with 20 percentage points more of its holdings in home LGFV bonds earned about 21 basis points more a year, and after the program it earned only 5 more.

Only banks with enough savers could replace the lost yield. Smaller banks relied on it the most, and they lost returns and fundraising (Section 5.2). Larger local banks replaced part of the yield with private credit to LGFVs, whereas the smallest banks did not. Private credit is a negotiated loan, so only a bank with many savers can cover the fixed cost of arranging and monitoring it. A bank whose initial home LGFV bond share was ten percentage points higher lost about 12 basis points more of annualized return and raised 15.3% less through newly launched WMPs. Bank size predicts these losses on its own, but it no longer does once we hold fixed a bank's reliance on home LGFV bonds and its use of private credit.

The program also absorbed the risk that smaller banks' products had hidden (Section 5.3). Nearly all smaller banks run their WMPs without a wealth-management subsidiary. On the home LGFV holdings of local banks without a subsidiary, provincial LGFV spreads fell from 297 basis points in June 2023 to 87 a year later. National banks' products took over the borrowers that the most reliant banks left, and these borrowers now pay lower coupons.

A simple model of competition for savings explains these patterns (Section 6). Savers choose a bank by its reputation and by the returns and volatility its products report. Cautious savers care more about reputation, so they are more likely to choose the larger, more reputable banks. WMPs do not guarantee principal, so a bank bears the credit risk in its products only through the damage that a default would do to its reputation. This damage is larger at banks whose savers are more cautious. A bank with less reputation therefore needs higher reported returns to attract savers, and it has less to lose if the risk behind them materializes. Smaller banks thus take more of the risk that savers cannot see, and they do so without better information. Debt resolution lowers risk and yield together, and only banks with enough savers can cover the fixed cost of replacing the yield with private credit.

The broader lesson is that when banks have little reputation to lose, they compete through a fragile strategy that exploits the behavioral bias of households, who care about returns and observed volatility but not underlying risk. Implicit support for borrowers can sustain such a strategy for years, but it works only while the risk is both present and hidden, so reporting at market prices and removing the risk both end it.

Our paper contributes to three strands of the literature. First, we contribute to work on what retail investors respond to and how intermediaries cater to them. Fund flows chase past performance (Sirri and Tufano 1998) and simple, visible signals such as ratings and raw returns rather than risk-adjusted performance (Del Guercio and Tkac 2008, Barber et al. 2016, Evans and Sun 2021, Ben-David et al. 2022). Investors demand claims that look safe (Gennaioli et al. 2013), rely on intermediaries they trust (Gennaioli et al. 2015), and buy retail products whose costs and risks are hard to see (Gabaix and Laibson 2006, Henderson and Pearson 2011, Célérier and Vallée 2017). The reported returns of illiquid portfolios are smooth (Getmansky et al. 2004, Cassar and Gerakos 2011), and flow sensitivity makes funds fragile when losses appear (Chen et al. 2010, Schmidt et al. 2016, Goldstein et al. 2017, Falato et al. 2021). We show that savers in a market of RMB 33 trillion respond strongly to reported returns and their fluctuations and punish even small dips below the usual return. In the

2022 sell-off they ran from the losses that products reported and not from the losses that the holdings took. Banks with little reputation exploited this bias by rating risky products as safe, locking savers in, and reporting returns that barely moved.

Second, we contribute to work on reaching for yield and on what restrains it. Competition (La Spada 2018, Choi and Kronlund 2018), regulatory constraints (Becker and Ivashina 2015), and low interest rates (Di Maggio and Kacperczyk 2017, Lian et al. 2019) encourage reaching for yield, and money market funds took more risk when investors rewarded higher yields (Kacperczyk and Schnabl 2013). Franchise value restrains banks' risk taking (Keeley 1990, Hellmann et al. 2000). We show that reputation determines which banks reach for yield in products sold to households, and that the strategy is fragile because it needs both the risky yield and the concealment of its risk. Proximity and small organizations can give lenders and investors an information advantage (Petersen and Rajan 2002, Degryse and Ongena 2005, Hauswald and Marquez 2006, Agarwal and Hauswald 2010, Stein 2002, Berger et al. 2005, Coval and Moskowitz 2001, Kacperczyk et al. 2005). In our setting, larger banks from the same city have at least the same access to local borrowers, so smaller banks' riskier holdings reflect reaching for yield and not such an advantage.

Third, we connect China's local government debt and shadow banking to banks' competition for household savings, and study how government support for borrowers affects the intermediaries that hold their debt. Existing work traces off-budget fiscal expansion through LGFVs (Bai et al. 2016), the refinancing of stimulus loans through bonds and shadow banking (Chen et al. 2020), and the crowding out of private investment (Huang et al. 2020). It also shows how deposit competition (Acharya et al. 2025), liquidity rules (Hachem and Song 2021), monetary tightening (Chen et al. 2018), competition with insured deposits (Xiao 2020), and implicit guarantees (Allen et al. 2023, Huang et al. 2023) shaped shadow banking in China. Governments influence credit allocation (Sapienza 2004, Khwaja and Mian 2005) and banks' holdings of government debt (Becker and Ivashina 2018, Ongena et al. 2019). In China, creditor political power and expected government support shape defaults and credit pricing

(Gao et al. 2021, Geng and Pan 2024). Geng et al. (2025) show that debt resolution narrows local banks' subordinated-bond spreads while reducing profitability at banks in riskier cities. We show that local government debt was the largest single source of the extra yield with which smaller banks competed for household savings. We also show that debt resolution absorbed the risk hidden in their products, lowered that yield, and moved both savings and borrowers toward larger banks.

2 Institutional Background and Data

2.1 Wealth-management products

A wealth-management product pools savers' money in a separately accounted portfolio run by a bank or its wealth-management subsidiary. WMPs are a major alternative to bank deposits for Chinese households and a central part of the country's shadow banking system (Acharya et al. 2025, Chen et al. 2020). Since the 2018 asset-management reform, products must report net asset values (NAVs) and cannot guarantee principal or returns.[3]

Savers see a product mainly through what its bank reports. Each product advertises a performance benchmark, which is the return it aims to deliver. It also publishes its NAV, which shows past returns and how much they fluctuate. The issuing bank rates each product's risk on a five-level scale from 1 for the safest to 5 for the riskiest. We call ratings 1 and 2 safe ratings. The bank sets this rating itself under its own method. The rules name qualitative criteria such as the portfolio, the past performance of similar products, and the risk level, and they require neither an external rater nor a quantitative standard (China Banking and Insurance Regulatory Commission 2018b, Article 27). Savers see the rating in the risk disclosure statement of the sales documents when they buy (China Banking and Insurance Regulatory Commission 2021, Article 49). The rating also decides who may buy. Banks rate each saver's risk tolerance on at least five levels and may sell a product only to

[3]See the asset-management reform guidelines (2018).

savers whose tolerance is at or above its rating (China Banking and Insurance Regulatory Commission 2018b, Articles 28–29), so a safe rating opens a product to the most cautious savers. The rules require a second, independent rating only when a product is sold by an institution other than its issuer, and the higher of the two ratings is then disclosed (China Banking and Insurance Regulatory Commission 2021, Article 30). In practice the scale is coarse. In April 2026, 26,709 of the roughly 33,900 rated products in the national registry were rated 2, and products with very different portfolios shared a grade (Li 2026). Savers do not see the risk of the underlying holdings directly. Periodic reports list the largest holdings by name, but a saver would have to identify each issuer and its location to learn how much of a portfolio depends on one local government's credit.

Redemption terms vary. Daily-open products allow withdrawals on any business day, periodic-open products only on set dates, and closed-end products only at maturity. Most of the market consists of fixed-income products, which regulation requires to keep at least 80% of their assets in debt claims.

National banks are the large state-owned banks and the nationwide joint-stock banks. They have extensive branch networks and nationwide name recognition. Local banks are the provincial and city commercial banks and the rural banks. They compete for savings with national banks and with each other. Most national banks and the largest local banks run their WMPs through wealth-management subsidiaries. The other local banks are mostly smaller and run their WMPs directly.

WMPs invest mainly in fixed-income and similar assets such as government and policy-bank bonds, deposits, negotiable certificates of deposit, and corporate debt (Figure A1).[4] Most readily accessible investments offer limited credit spreads, and the public high-yield corporate bond market is narrow (Schipke et al. 2019).

LGFV bonds are one source of higher yield. LGFVs are state-owned companies that finance local infrastructure and public services. They borrow in their own names and outside

[4]Article 35 of the commercial-bank WMP rules lists permitted investments (2018b).

formal government debt. Repayment depends on project assets, local fiscal resources, and expected government support (Hendy et al. 2024, Yang and Jiang 2021). In the issue-level sample of Table A6, average pre-program fixed-rate issuance yields were 4.26% for LGFVs, 3.03% for other nonfinancial issuers, and 2.77% for financial issuers. These differences persist after controlling for issue characteristics and ratings. Because these bonds trade nationally, a bank can hold LGFV bonds from any province. We estimate that WMPs held about RMB 3.1 trillion in LGFV bonds at year-end 2024. This amount is 10.2% of total WMP net assets and 28.1% of outstanding LGFV bonds (Figure A2 and Internet Appendix D.2).

WMPs can also finance the same borrowers through private credit (PC), which includes identified trust loans, leasing receivables, and claims held through trust or asset-management plans. Unlike a marketable bond, private credit must be arranged, documented, and monitored without a continuously observable trading price, and this work is costly even for a small investment.

Regulatory caps on private credit were far from binding for local banks without a subsidiary. Chinese rules treat private credit as non-standard debt and cap it at 35% of a product's net assets. For WMPs that a bank runs itself, a second cap limits it to 4% of the bank's total assets (China Banking and Insurance Regulatory Commission 2018b,a). At the median bank without a subsidiary in June 2023, the second cap allowed private credit of up to 61% of the bank's WMP balance. Private credit made up about 3% of the holdings these banks disclosed, so the caps cannot explain why they made little use of it (Section 5.2). Supervisory guidance addressed the size of these banks' WMP business directly. In September 2021, the financial press reported that banks without a subsidiary would not be allowed to expand this business and would wind down existing balances (Fang 2021). In June 2024, regulators in several provinces asked these banks to shrink it, in some cases to zero by the end of 2026 (Li 2024, Li and Zhang 2024). Our finding that a bank's initial home LGFV bond share predicts its later fundraising holds in a sample ending before this second round of guidance (Section 5.2).

2.2 Debt resolution and data

The policy announcement of July 24, 2023, called for comprehensive local-debt resolution. Provinces began issuing special refinancing bonds in October and raised about RMB 1.4 trillion in 2023Q4 (Xinhua News Agency 2023, Wang 2023, Wong 2024). The proceeds repaid selected obligations within a broader program that also restructured loans and restricted new borrowing. The refinancing bonds are provincial-government obligations, not the LGFV bonds that WMPs hold. An expansion announced in November 2024 raised the debt-swap limit by RMB 6 trillion for 2024–2026 (Ministry of Finance 2024). The program strengthened expected government support without formally guaranteeing every LGFV claim. The resulting fall in spreads raises the prices of bonds already held and lowers the yields available when banks reinvest. We date implementation to 2023Q4 in quarterly tests and to October 2023 in monthly tests.

Our data start from product-level WMP holdings in Wind, a Chinese financial database. The holdings identify each product's largest disclosed positions at the individual-asset level. We link them to NAVs, bond issuance, and borrower-side private credit disclosures from a commercial data vendor. We add product types and redemption terms from product filings, banks' branch locations by city from a bank-branch registry, and issuers' defaults, rating warnings, and guarantor events. Quarterly disclosures over 2022Q1–2025Q3 contain 2,279,252 holding-level records for 60,396 products and 253,484 product-quarters. The 171 banks are headquartered in 30 provinces and include state-owned and joint-stock national banks, provincial and city commercial banks, and rural banks. Tables A1–A3 define the variables and report sample coverage and descriptive statistics.

Figure 2 describes the banks by size, which is highly skewed. The median national bank had RMB 7.8 trillion of interest-bearing assets at the end of 2021, against RMB 0.19 trillion for the median local bank. National banks hold 76% of sample WMP assets. Most banks are thus small local banks, while most savings sit with the few large national banks.

A bank's holdings and product returns are those of its WMPs, whether run directly or through a subsidiary. An investment is home when its borrower is in the bank's headquarters province, so national and local banks' WMPs can both hold home and non-home assets. In the product regressions, i denotes a product, a its issuing bank or subsidiary, b the parent bank, q a quarter, m a month, and p_b the bank's headquarters province. The parent bank is the issuing bank itself when the bank runs its products directly. Log bank size$_b$ always refers to fixed FY2021 bank size, whereas log bank assets in the control rows vary over time. Because bank size is so skewed (Figure 2), we compare banks by size in three ways, by log size across all banks, by a split at the median size, and by size terciles among local banks. We take the median over the banks in the product sample of Section 4.2, where it is RMB 0.44 trillion of FY2021 assets. We call the banks at or below it smaller banks, and all of them are local banks. Local banks without a wealth-management subsidiary include nearly all smaller banks.

Figure 1 previews the patterns that the rest of the paper examines. It splits the banks above the median size into national banks and larger local banks, and it compares each group with the smaller local banks. Before the 2023 program, smaller local banks' WMPs held the bonds with the highest spreads. From 2022Q1 to 2023Q3, the bonds they held had been issued at an average of 224 basis points over Treasury yields, against 183 at larger local banks and 126 at national banks (Panel A). After the program these spreads fell for all three groups, to 131, 115, and 91 basis points by 2025Q3. The holdings of smaller local banks also implied the most return volatility, yet their products reported the least. From April 2022 to June 2023, their holdings implied a standard deviation of monthly returns of 46 basis points, against 44 at larger local banks and 35 at national banks. The products themselves reported 24 basis points, against 32 and 34 (Panel B). National banks' products thus reported nearly all of the volatility their holdings implied, whereas smaller local banks' products reported about half. Smaller local banks also held the most home LGFV bonds. These bonds made up 31% of the bonds and private credit held by their products, against 19% at larger local

banks and almost none at national banks, which held 24% in private credit (Panel C). Finally, nearly all of smaller local banks' products restricted redemptions and carried a safe rating. At the average smaller local bank, 98% of fixed-income products restricted redemptions and 90% carried a safe rating, against 86% and 78% at the average national bank (Panel D).

3 What Savers See

Savers chase the returns that products report, and they penalize reported volatility and any dip below the usual return. They do not respond to the risk behind these returns, which they cannot see (Section 2). This is the bias that smaller banks went on to exploit. The bond sell-off of 2022 provides a natural experiment. In it, savers ran from the losses that products reported, not from the losses their holdings took.

We rank products' prior-quarter returns within the same risk rating and withdrawal category and pool products of local and national banks. Following the performance–flow approach of Sirri and Tufano (1998) and La Spada (2018), we estimate

$$\begin{aligned}\text{Net flow}_{iq} = &\sum_{k=2}^{5} \beta_k \mathbb{1}\{\text{Return quintile}_{i,q-1} = k\} \\ &+ \gamma' \,\text{Controls}_{i,q-1} + \lambda\, \mathbb{1}\{\text{Local bank}\}_b + \alpha_{g(i,q-1),q} + \varepsilon_{iq}.\end{aligned} \tag{1}$$

The outcome Net flow_{iq} is asset growth net of investment gains, as a percentage of beginning assets. $\text{Return quintile}_{i,q-1}$ is the prior return quintile, and the bottom quintile is omitted. The index $g(i, q-1)$ identifies the prior risk and withdrawal category. Category–quarter effects hold common demand conditions fixed.[5]

Flows rise across the performance ranks (Table A5). Net inflows into top-quintile products exceed those into bottom-quintile products by 12.0% of beginning assets, and top-quintile

[5]The regression includes the local-bank indicator and eleven controls. The controls are three product characteristics (size, risk rating, and redemption terms), four bank characteristics (log bank assets, nonperforming-loan ratio, cost–income ratio, and net interest margin), and four home-city characteristics (GDP, GDP growth, LGFV debt/GDP, and urban income). The sample is 2022Q3–2025Q3, and errors cluster by parent bank.

products are 8.0 percentage points more likely to receive positive flows. Among products of banks in the same home province and quarter, the flow difference is a similar 12.6%.

To separate what savers can see from what they cannot, we add three variables to equation (1). One is the volatility of the product's reported monthly returns over the preceding year. The other two measure the risk in its disclosed portfolio at the end of the prior quarter. The first is its home LGFV credit share, which is home-province LGFV bonds plus private credit to home-province LGFVs as a percentage of disclosed holdings. The second is the value-weighted coupon of its LGFV bonds. Each variable enters per standard deviation.

Flows respond to reported volatility but not to the risk held (Table 1). Among products with disclosed holdings, the top-to-bottom quintile flow difference is 11.2% of beginning assets without the two holdings-risk measures and 11.3% with them, and the two measures are jointly insignificant (column 2). One standard deviation of the home LGFV credit share is 16.7 percentage points of disclosed holdings. A share that is higher by this amount is associated with flows 0.26% of beginning assets lower, and the 95% confidence interval excludes reductions larger than 0.8%. By contrast, a one-standard-deviation higher volatility of past reported returns lowers flows by 1.6% of beginning assets, while the holdings-risk measures remain insignificant (column 3).

Savers punish any dip below the usual return, even when the return stays positive. We group products by the return they reported for the quarter. We then compare their flows with those of products of the same bank in the same quarter that reported the usual 0.5 to 1%, using the controls and category effects of Table 1 (Table 2, Panel A). Daily-open products that reported 0.25 to 0.5% had net flows 43 percentage points of beginning assets lower in the same quarter, and those that reported smaller gains or losses had net flows 46 to 52 points lower. The pattern also holds for flows in the quarter after the return was reported, so the outflows did not cause the low returns. In that quarter, products that had reported 0.25 to 0.5% had net flows 19 points lower, and those that had reported losses had net flows 28 to 34 points lower. The pattern holds within the same product as well. Product fixed

effects compare each product with itself in other quarters. With them, daily-open products that reported 0.25 to 0.5% had net flows 30 points lower in the same quarter and 16 points lower in the next.

The response comes from savers who can withdraw. In Panel B of Table 2, the slope of net flows on the reported quarterly return differs by redemption type. For daily-open products, each percentage point of reported return comes with net flows 53 percentage points of beginning assets higher in the same quarter and 58 points higher in the next. With product fixed effects the two slopes are 21 and 27 points. Savers can redeem periodic-open products only on set dates, and for these products the slopes are at most 8 points. Savers cannot redeem closed-end products before maturity, and for these products the slopes are close to zero and insignificant in every column. Net flows subtract the reported return from asset growth, so an error in a reported return would link the two mechanically for every redemption type. The flat slopes of closed-end products show that such errors do not drive our estimates. These results imply that a product whose reported returns showed its holdings' losses would have lost savers.

The bond sell-off of late 2022 is our natural experiment. The losses came from a market-wide fall in bond prices, and the same market losses appeared to very different degrees in the returns that products reported (Section 4.2). Because WMPs report net asset values (Section 2), a fall in bond prices can show up in the returns that savers see. In the sell-off, provincial LGFV spreads rose by about 110 basis points. Between the end of October and mid-December 2022, the one-year government bond yield rose by 60 basis points and the three-year yield by 36. The losses that products reported were small. In early December 2022, 4,932 of 28,038 WMPs had net asset values below par, and more than 90% of these were below par by less than 2% (Wang 2022). The withdrawals were large. China International Capital Corporation (CICC) estimates seasonally adjusted net redemptions of about RMB 720 billion in November 2022 alone and reports that the most risk-averse savers led them (Wang et al. 2022). Redemptions forced funds and WMPs to sell a record RMB 1.3 trillion

of interbank bonds that month (Bloomberg News 2022). According to industry estimates, WMP balances stood at RMB 31.3 trillion in October 2022 before the redemptions began and at 25.0 trillion at the end of March 2023 (Liu and Xie 2024). The official market report puts them at 27.65 trillion at the end of 2022 (Banking Wealth Management Registration and Depository Center 2023). Our data show the same outflow. Products that report in consecutive quarters lost 12.6% of their assets in 2022Q4 and another 14.7% in 2023Q1. Daily-open products took in 6.2% of their assets in 2022Q3 and then lost 10.9% and 15.9% to net withdrawals in the two following quarters.[6]

Savers ran from the losses that products reported, not from the losses their holdings took. In 2022Q4, daily-open products that reported a quarterly loss lost 25 to 37% of their assets, while those that reported the usual 0.5 to 1% took in 9% (Table D3). To measure the losses that holdings took, we mark every disclosed asset each month to the market series that fits it. For an LGFV bond, this series is the Treasury yield of its maturity plus the LGFV spread of the issuer's province. We call the resulting monthly returns implied returns, and Section 4.2 gives the construction. By the next quarter the losses had been reported. In that quarter, products that had reported larger losses in November and December lost more of their assets than other products of the same bank, whereas the losses in their implied returns made no significant difference (Table D4). Each 10 basis points a month of reported loss came with net outflows larger by 1.2% of beginning assets, and by 8% among daily-open products. Savers also penalized the volatility they saw, not the volatility of the holdings at market prices. We measure reported and implied volatility over the same preceding months and relate both to flows in 2022Q4–2023Q3. A one-standard-deviation higher reported volatility lowers flows by 2.4% of beginning assets, whereas the same increase in implied volatility is associated with flows 1.1% higher (Table 1, column 4).

[6]Our sample covers about four-fifths of the market. It holds RMB 21.7 of 27.65 trillion at the end of 2022 and 19.7 of 25.0 trillion at the end of March 2023. The 12.6% decline in 2022Q4 is close to the fall of about 12% in industry balances from October to December 2022, and total sample assets fell 9.0% in 2023Q1, against about 9.4% in the industry figures (Liu and Xie 2024).

Savers' sensitivity to what they see gave banks a strong incentive to offer products whose reported returns are high and fluctuate little. A product that reports a high return attracts savings, and one that shows even a small dip loses them, whatever it holds. Banks with little reputation can compete for savers only through the returns they report. They needed a higher return than more reputable banks, and they needed its risk not to show.

4 Reaching for Yield in the Dark

Savers reward the returns and stability that products report and do not see the risk behind them (Section 3). A bank with little reputation can therefore attract savings only with a higher reported return, and it keeps them only if the risk behind that return does not show. This section shows that smaller banks did both. Section 4.1 shows that they earned their higher returns by holding riskier debt than larger banks, and that the largest source of the extra yield was the debt of LGFVs in their home provinces. Section 4.2 shows that they rated these riskier products as safe, locked savers in with restricted redemptions, and reported returns that barely moved, so the risk stayed in the dark. Section 4.3 shows that this was reaching for yield rather than better local information.

4.1 Higher returns from local credit

Smaller banks' products reported about 0.5 percentage points a year more than larger banks' products with the same label (Section 4.2). They earned these higher returns by holding riskier debt throughout their portfolios, and the largest source of the extra yield was the debt of LGFVs in their home provinces. Their WMPs held more LGFV bonds and less private credit than larger banks' WMPs, concentrated these bonds at home, and held them in products that restrict withdrawals. Home LGFV bonds carried the largest return premium of any asset class we examine.

Where the yield comes from. Smaller banks hold riskier debt throughout their portfolios, but most of their extra yield comes from credit in their home provinces. To locate the extra yield, we price every disclosed holding by its credit spread over Treasury yields. A bond carries its issue spread over the Treasury yield of the same maturity. Private credit carries its contract rate over the Treasury yield of its remaining maturity. Asset-management plans and funds carry the mean issue spread of AA+ or AAA bonds with one to three years to maturity, and cash, deposits, and repurchase agreements carry zero. A product's all-in spread is the value-weighted mean over its holdings. It is therefore the sum over asset classes of each class's holding share times its spread. Its size gradient thus splits exactly into the contributions of the classes (Table 3). Section 4.2 uses the same all-in spread to measure the risk a product holds.

Per log point of bank size, the all-in spread falls by 12.8 basis points (Table 3, column 1). Home LGFV bonds drive this gradient. Their contribution to the all-in spread falls by 17.0 basis points per log point of size. Two other kinds of credit work in the same direction. The contribution of other local banks' capital bonds falls by 4.3 basis points, and that of bonds of non-LGFV corporate issuers from the home province falls by 4.6. Larger banks earn more of their yield elsewhere. The contributions of private credit and of asset-management plans rise by 8.2 and 5.2 basis points per log point of size, so the all-in gradient is smaller than the home LGFV component alone. The same pattern appears in levels. The holdings of smaller local banks' products carry an all-in spread of 205 basis points, against 149 at national banks. Smaller local banks earn 61 basis points more than national banks from home LGFV bonds. They also earn 14 basis points more from other local banks' capital bonds and 14 more from home corporate bonds. National banks earn 26 basis points more from private credit and asset-management plans and 6 more from the remaining classes (columns 2–6). Smaller banks also hold higher-spread claims within most asset classes. Their home LGFV bonds carry 317 basis points against 236 at national banks, their home corporate bonds 270 against 121, and

their private credit 406 against 351. Home LGFV bonds are thus the largest single source of smaller banks' extra yield, and the rest of this subsection examines them.

Asset allocation. Two higher-yielding investments are central to the analysis, marketable LGFV bonds and private credit. Before the program, local banks' WMPs relied more on LGFV bonds, whereas national banks' WMPs already made substantial use of private credit. We express each holding as a share of included debt holdings. This is the combined value of identified bonds and private credit with known borrower locations, and it excludes central-government, central-bank, and policy-bank obligations. This denominator isolates the credit whose borrowers and their locations we observe.[7] LGFV bonds accounted for 43.1% of included debt holdings at local banks and 12.6% at national banks, while private credit accounted for 9.3% and 24.1% (Table D1). We then estimate the initial size gradient across national and local banks

$$\text{Asset share}_{iq} = \beta_S \,\text{Log bank size}_b + \gamma' \,\text{Controls}_{iq} + \alpha_q + u_{iq}. \tag{2}$$

The outcome is the total LGFV bond share or the total private credit share over 2022Q1–2023Q3, in percentage points of included debt holdings. Log bank size$_b$ is log FY2021 parent-bank interest-bearing assets in RMB trillion, which the products of a bank and its subsidiary share. The ten variables in Controls$_{iq}$ are three product characteristics (size, risk rating from 1 to 5, and redemption terms), three bank characteristics (nonperforming-loan ratio, cost–income ratio, and net interest margin, in percent), and four home-city characteristics (GDP, GDP growth, LGFV debt/GDP, and urban income).[8]

Smaller banks' WMPs allocate more to LGFV bonds and less to private credit (Table 4). A bank half as large has an estimated LGFV bond share about 4.9 percentage points higher and a private credit share 4.7 points lower. Private credit is also scarce at the smallest third

[7] Internet Appendix D.2 reconciles the samples and denominators, including bond-only shares and shares of all disclosed assets.

[8] Regressions weight by included debt value and cluster by parent bank.

of local banks, where it is only 1.1% of included debt holdings, against 11.4% at the largest third (Table D1, Panel B). Smaller banks thus entered the program more dependent on marketable LGFV bonds and with little use of the privately arranged alternative.

Home bias. Local banks also concentrate their investment in their home provinces. Home LGFV bonds account for more than one-fifth of included debt holdings in every local-bank size group (Table D1, Panel B). For the home LGFV bond share, equation (2) gives a size coefficient of −5.46 percentage points per log point of bank size (Table 4), so a bank half as large has a home LGFV bond share about 3.8 points higher. Most of smaller banks' additional LGFV allocation is at home.

Large home holdings could reflect a large local bond market rather than a preference for nearby borrowers, so we also measure geographic concentration within each asset class. For LGFV bonds, within-category home bias is the product's home LGFV holdings divided by all its LGFV holdings, minus the fraction of nationwide disclosed LGFV holdings issued by borrowers in that province. We define it in the same way for other bonds and for private credit. Over 2022Q1–2025Q3, we estimate

$$\begin{aligned}\text{Within-category home bias}_{iq}^{n} = \beta_n \mathbb{1}\{\text{Local bank}\}_b \\ + \gamma_n' \,\text{Controls}_{iq} + \alpha_q + \varepsilon_{iq}.\end{aligned} \tag{3}$$

Here n identifies the asset class, and β_n measures local banks' additional within-category home bias relative to national-bank products.[9]

Local banks' WMPs invest disproportionately at home on this measure as well (Table 5). Their within-category home bias exceeds that of national-bank products by 32.0 percentage points for LGFV bonds, 16.1 points for other bonds, and 48.9 points for private credit. Their specialization is geographic as well as a choice of asset class.

[9] Internet Appendix D.4 lists the controls and gives the details of this specification and of equation (4).

In addition, local banks place more of their WMP assets in products that restrict withdrawals. Periodic-open and closed-end products hold 80.2% of local banks' WMP assets, compared with 47.3% at national banks (Table A4). Such products give banks more room to hold investments that are difficult to sell quickly. To connect withdrawal terms to investment, we divide a product's home holdings in an asset class by all its included debt holdings and subtract the corresponding nationwide share. This portfolio home overweight reflects both the amount invested in the asset class and its geographic concentration. We estimate

$$\begin{aligned}\text{Portfolio home overweight}^{n}_{iq} = \beta_1 \, \mathbb{1}\{\text{Local bank}\}_b \\ + \beta_2 \, \mathbb{1}\{\text{Restricted redemption}\}_i \\ + \beta_3 \, \mathbb{1}\{\text{Local bank}\}_b \times \mathbb{1}\{\text{Restricted redemption}\}_i \\ + \gamma' \, \text{Controls}_{iq} + \alpha_q + \varepsilon_{iq}.\end{aligned} \tag{4}$$

$\mathbb{1}\{\text{Restricted redemption}\}_i$ identifies products without daily withdrawals. The coefficient β_1 measures the local–national allocation gap among daily-open products, and β_3 measures how much wider that gap is among restricted-redemption products.

Local banks invest more at home than national banks only in products that restrict withdrawals (Table A4). Among daily-open products, the local-bank coefficient is -6.3 percentage points for home investment pooled across asset classes and 0.9 points (standard error 3.5) for LGFV bonds. The interaction with restricted redemption is 20.0 percentage points for pooled home investment, of which 7.7 points are in LGFV bonds and 7.8 points in private credit.

Returns. Before the program, home LGFV bonds earned the largest return premium of the asset classes we examine. To separate home LGFV bonds from bonds elsewhere and from private credit, we relate monthly product returns to holdings at the preceding quarter end

$h(m)$ in

$$\begin{aligned}\text{Monthly return}_{im} = {} & \beta' \,\text{Asset shares}_{i,h(m)} \\ & + \delta' \,\text{Asset shares}_{i,h(m)} \times \mathbb{1}\{\text{Post}\}_m \\ & + \gamma' \,\text{Controls}_{i,h(m)} + \lambda \,\mathbb{1}\{\text{Local bank}\}_b + \alpha_m + \varepsilon_{im}.\end{aligned} \tag{5}$$

The outcome Monthly return$_{im}$ is the change in log distribution-adjusted NAV, in basis points. The five fractions in Asset shares$_{i,h(m)}$ are the shares of included debt holdings in home LGFV bonds, home private credit, home other bonds, non-home LGFV bonds, and non-home private credit. Non-home other bonds are the omitted class. Controls are measured at the same date, and Post starts in October 2023.[10] Before implementation, β measures each asset's return premium relative to non-home other bonds. The interactions δ allow the premiums to change afterward, which Section 5.1 examines.

Reallocating 20 percentage points from non-home other bonds to home LGFV bonds is associated with 20.5 annual basis points more product return, compared with 7.7 basis points for LGFV bonds from other provinces (Table 6, column 2). The home premium is more than two and a half times the non-home premium, and their difference is statistically significant ($p = 0.003$). Private credit also earns more at home. The same 20-point allocation is associated with 17.6 annual basis points for home private credit, against 8.2 for private credit elsewhere. Home other bonds contribute little. This pattern ties the higher returns to LGFV bonds and private credit rather than to home investment generally.

Home LGFV bonds make up a large part of local banks' included debt holdings, with an initial mean allocation of 26% and a 90th percentile of 72% (Table A3). Home LGFV bonds thus supplied most of the extra yield with which smaller banks competed, and they tied their products to the borrowers of a single province. Section 4.2 asks whether savers could see that risk.

[10]The eleven controls add time-varying log bank assets to the ten controls in equation (2). The regression includes a local-bank indicator and month effects, and errors cluster by parent bank. It uses returns from April 2022 to December 2025, winsorized at the 1st and 99th percentiles of all monthly returns.

4.2 Hidden risk

Smaller banks earned their higher returns by holding riskier debt, and the largest source of their extra yield was the debt of LGFVs in their home provinces (Section 4.1). The risk behind that yield stayed in the dark. Their products held more risk than larger banks' products with the same label, yet smaller banks rated them as safe more often, locked savers in with restricted redemptions, and reported returns that barely moved. In the bond sell-off of November and December 2022, their reported returns only dipped to about zero while their holdings lost about 1.6% at market prices.

A saver sees a product's label and the returns it reports. The label is the product's redemption type and its risk rating. We therefore compare products with the same label and ask how what they hold and what they report vary with bank size. The sample comprises products that the filings classify as fixed-income and that have disclosed holdings and at least nine monthly returns between April 2022 and June 2023. For product i of bank b, we estimate

$$\text{Outcome}_i = \beta \,\text{Log bank size}_b + \alpha_{t(i),r(i)} + \varepsilon_i. \tag{6}$$

Redemption-type-by-risk-rating effects $\alpha_{t(i),r(i)}$ compare products with the same redemption type (closed-end, periodic-open, or daily-open) and risk rating, which a saver cannot tell apart by label. Log bank size$_b$ covers national and local banks, so a negative β means the outcome is larger at smaller banks. We also report the difference between smaller and larger banks. Smaller banks are those at or below the median size of this sample, and every national bank is among the larger banks. Each bank receives equal total weight, because the bank designs and rates its products.[11]

We first show that smaller banks' products hold more risk. We then show how that risk stays out of what savers see, in reported volatility, in the 2022 sell-off, in ratings, and in redemption terms. We measure the risk a product holds in four ways, and each uses its

[11] Errors cluster by parent bank. Internet Appendix D.3 gives the sample construction, and Table D2 reports alternative specifications.

disclosed holdings rather than what the bank reports. The first is the home LGFV credit share used in Section 3, which is LGFV bonds and private credit to LGFVs of the bank's home province as a share of disclosed holdings. The second is the issue spread of the bonds held. A bond's issue spread is its coupon minus the same-maturity Treasury yield at issue, and it is the compensation the market required for the bond's risk. The third is the all-in spread of Section 4.1, which extends this compensation to the whole portfolio. The fourth is the volatility that the holdings imply. To compute it, we mark every disclosed asset each month to the market series that fits it. An LGFV bond moves with the Treasury yield of its maturity plus the LGFV spread of the issuer's province. Other bonds move with the Treasury yield of their maturity plus a credit-bond index of the same rating and maturity. Private credit moves with its borrower's provincial LGFV spread or with a credit-bond index at its remaining maturity. Asset-management plans and funds move with a credit-bond index, and cash, deposits, and repurchase agreements do not move. Price changes omit the interest income that reported returns contain. We therefore add to each product's price changes the average difference between its reported returns and its price changes outside November and December 2022. We call the resulting monthly returns implied returns and take their standard deviation.

All four measures are higher at smaller banks (Table 7, Panel A, and Figure 3, Panel A). A bank half as large holds 3.7 percentage points more home LGFV credit, and the bonds it holds were issued at spreads 16 basis points higher. Across its whole portfolio it earns a spread 9 basis points higher, and its holdings imply 5% more return volatility. Compared with larger banks' products, smaller banks' products hold 15.5 percentage points more home LGFV credit. Their bonds were issued at spreads 50 basis points higher, their all-in spread is 33 basis points higher, and their holdings imply 17% more volatility. These gradients keep their sign and significance when we add a local-bank indicator and compare banks only with others of the same type (third row of each panel of Table 7).

Issuers' official credit ratings do not show this risk. Of the rated LGFV bonds issued between 2021 and mid-2023, 62% carry issuer ratings of AA+ or AAA, and the rest carry AA. Yet AAA-rated LGFV bonds were issued at an average spread of 170 basis points, against 75 for other AAA-rated issuers (Figure 4). Smaller banks hold bonds priced as riskier than their ratings indicate, as Becker and Ivashina (2015) find for the bonds that insurers choose within a rating. Among bonds with the same issue quarter, issuer rating, administrative level, maturity, and placement type, those held by a bank half as large were issued at spreads 12 basis points higher (Table 7, Panel A, column 3).

Yet smaller banks' products report less risk (Table 7, Panel B). The reported return volatility of a bank half as large is 10% lower, and smaller banks' products report 43% less volatility than larger banks' products with the same label (Table 7, Panel B, column 1, and Figure 3, Panel B). Relative to the volatility their own holdings imply, the volatility they report is 15% lower at a bank half as large and 51% lower at smaller banks than at larger banks (Table 7, Panel B, column 2).

In the bond sell-off of November and December 2022, smaller banks' products showed a third of the fall in the value of their holdings, and larger banks' products showed three-fifths. Smaller banks' holdings implied returns of -80 and -76 basis points in the two months, while their products reported -3 and -4 basis points (Figure 5). By construction, both series average about 35 basis points in the other months. Their reported returns thus fell 39 basis points a month below their usual level, while the returns implied by their holdings fell 113 below it. For 2022Q4 as a whole, these products still reported a small gain. Larger banks' holdings implied -72 and -57 basis points, and their products reported -27 and -21. Among products with the same redemption type and risk rating, the part of the fall that reported returns did not show was 33 basis points a month larger at smaller banks than at larger banks. It was 8.1 basis points larger at a bank half as large (Table 7, Panel B, column 4). Panel C of Figure 3 shows the same gap bank by bank. In the market-wide withdrawals of 2022Q4 and 2023Q1, the average smaller bank's products had net outflows of 9% and 8%

of beginning assets, against 14% and 19% at the average larger bank (Figure 5, Panels C and D). Within each bank, the products that reported larger losses lost more in the following quarter (Section 3).

Smaller banks also rate more of their products as safe, although they hold riskier portfolios (Figure 3, Panels A and D). The issuing bank assigns the rating, and a safe rating is what lets a bank sell a product to its most cautious savers (Section 2). Among all fixed-income products that reported returns in this period and have the same redemption terms, a bank half as large gives safe ratings to 3.0 percentage points more of its products (Table 7, Panel B, column 6). At the average smaller bank, 90% of fixed-income products carry a safe rating, against 78% at the average national bank (Figure 1, Panel D). The difference is starkest across banks. About four in five smaller banks give every fixed-income product a safe rating, compared with about one in three larger banks. The same rating also covers very different risk. A rating of 2 is the most common safe rating. Among products with this rating, smaller banks' products imply 13% more volatility than larger banks' products with the same redemption type, yet they report 44% less.[12] What savers do see is the higher return. Smaller banks' products reported 4.0 basis points a month more than larger banks' products with the same label, which is about 0.5 percentage points a year (Table 7, Panel B, column 5).

Bank by bank, the return that savers see rises with the risk held, while the volatility they see does not (Figure 6). Each 100 basis points of all-in spread in a bank's holdings comes with reported returns 6.4 basis points a month higher (standard error 1.0). The volatility implied by the holdings rises with the spread, while the volatility that products report does not (Panel B). In the sell-off, a bank with 100 basis points more spread had holdings that lost 19 basis points a month more at market prices, yet its products reported returns 17 basis points a month higher.

[12]With redemption-type effects, bank-equal weights, and parent-bank clustered errors, the coefficient on the smaller-bank indicator is 0.125 (standard error 0.054) for log implied volatility and −0.572 (0.144) for log reported volatility.

Smaller banks also lock savers in, and their risk sits in the products that do so. Such products need not sell their holdings when markets fall. In the sample of Table 7, the average smaller bank restricts withdrawals on 98% of its products, against 89% at the average larger bank. The local–national gap in home LGFV bonds is 7.7 percentage points wider in restricted products than in daily-open ones (Section 4.1). Products that lock savers in also showed less of the losses in the returns they reported (Table 8). The part of the fall that reported returns did not show was 34 basis points a month larger in closed-end products and 12 larger in periodic-open products than in daily-open products of the same bank (Panel B). Smaller banks' closed-end products showed a quarter of the fall in their holdings' value and still reported positive returns in the sell-off months. The closed-end products of larger local banks and of national banks showed about half of the fall (Panel A). Within each redemption type, smaller banks' products showed less of the fall than the products of larger local banks, although the holdings of the two groups fell about as much. The difference therefore follows bank size rather than the line between local and national banks. The risk that smaller banks took thus does not appear in the returns and ratings their savers see, while the extra return does.

The risk that did not show was large and concentrated. Home LGFV bonds made up 20% of all the holdings that smaller banks' products disclosed, against 1% at national banks (Table 3). They made up 31% of smaller banks' bonds and private credit (Figure 1, Panel C). This exposure depended on the LGFVs of a single province, whose repayment rests on the same local fiscal resources and expected government support. Smaller banks' higher returns rested above all on this risk, which their savers could not see.

4.3 Reaching for yield, not information

Smaller banks hold riskier claims with higher yields than larger banks from the same city, and the larger banks have at least the same access to local borrowers. Smaller banks do so even where neither bank has any local relationship. Because their extra home LGFV

credit is local (Section 4.2), it might seem to reflect local knowledge. However, a larger bank headquartered in the same city operates in the same local market and has more branches there, so a smaller bank should have no information advantage over it. We therefore compare banks headquartered in the same city, first in home LGFV bonds and then in claims where neither bank has a local relationship.

Banks from the same city. For local bank b in quarter q, we group its LGFV bond holdings by issuer city c. A comparison cell g contains the banks headquartered in the same city that hold bonds of issuer city c in quarter q and have the same branch presence in c, which is either a branch there or none. Each cell has one location $\ell(g)$, which is the banks' own city, another city of their home province, or another province. Table 9 reports the comparisons in other cities of the home province and in other provinces, and Table 10 compares holdings in the banks' own city at the bank level. For bank b in cell g, we estimate

$$\text{Bond characteristic}_{bg} = \sum_{\ell} \beta_{\ell} \, \text{Log bank size}_{b} \times \mathbb{1}\{\ell(g) = \ell\} + \alpha_{g} + \varepsilon_{bg}. \tag{7}$$

The outcome is the value-weighted mean over the bonds bank b holds in cell g. Cell effects α_g absorb everything common to the headquarters city, issuer city, quarter, and branch presence, so β_ℓ compares the bonds that banks of different size hold in the same place with the same presence.[13] The yield outcome is the coupon in basis points relative to the mean coupon of held LGFV bonds with the same issue year and maturity group. The risk outcomes indicate county-level issuers, issuers rated below AAA, and issuers with a default, rating warning, or guarantor event between 2022Q4 and 2025Q4.

At home, smaller banks hold riskier LGFV bonds than larger banks from the same city (Table 9, Panel A). In other cities of its home province, a bank half as large holds bonds

[13] Each bank receives equal total weight within each location, and errors cluster by parent bank. Holdings are 2022Q1–2023Q2 unless noted. Internet Appendix D.3 gives the construction.

with coupons 24 basis points higher. Its bonds come from county-level issuers 9.2 percentage points more often and from issuers rated below AAA 2.8 points more often.

These differences do not come from better local access. Among banks headquartered in the same city, a bank twice as large has about 16% more branches there (Table 10, column 1). The two banks hold similar shares of their portfolios in own-city LGFV bonds (column 2), but the smaller bank holds more of the riskiest ones. A bank half as large holds about 1.3 percentage points more of its portfolio in own-city LGFV bonds of county-level issuers, which is a fifth more than the mean of 6.0%. With its own-city branches held fixed, the difference is 1.25 points (column 3).

Assets without local relationships. The same preference appears where no local relationship exists. In other provinces, a bank half as large holds LGFV bonds with coupons 53 basis points higher than a same-city bank with the same presence in the issuer city. Its bonds come from county-level issuers 4.8 percentage points more often and from issuers rated below AAA 2.6 points more often (Table 9, Panel A). This risk materialized. The out-of-province issuers held by a bank half as large were 0.85 percentage points more likely to have a credit event by 2025Q4, against a mean of 2.2% in these cells (column 4). The preference persisted after debt resolution and the 2024 guidance. In 2024Q3–2025Q2, a bank half as large still held LGFV bonds paying 27 basis points more in other cities of its home province and 20 more in other provinces (column 5). By then LGFV bonds offered about half their earlier extra yield over government bonds (Section 5.1).

Smaller banks also prefer weaker issuers in assets unrelated to local government finance. We first look at their holdings of other banks' tier-2 and perpetual capital bonds. Among same-city banks in the same quarter, a bank half as large invests 13 percentage points more of these holdings in the bonds of city and rural banks, and it earns coupons 17 basis points higher (Table 9, Panel B). Interbank negotiable certificates of deposit give a second test, because a local bank has no relationship with issuers in other provinces. Among same-city

banks in the same quarter and maturity group, a bank half as large holds 13.6 percentage points more of its out-of-province certificates in paper of city and rural banks rather than national banks, against a mean of 37% (Table 9, Panel C).

Although smaller banks prefer riskier claims everywhere, most of their risk and most of their extra yield are in their home provinces (Section 4.1). About three-fifths of local banks' LGFV bonds are issued at home (Table D1, Panel A), and home LGFV bonds carried the largest return premium (Section 4.1). Smaller banks lacked the reputations of larger banks, so they competed for savings by reaching for yield, and these home bonds were its largest source. Savers rewarded the yield (Section 3) without seeing its risk (Section 4.2). Both the yield and its risk depended on expected government support for LGFVs, so a change in that support would move them together.

5 Debt Resolution and the Loss of Yield

China's 2023 debt-resolution program removed the main source of the yield on which smaller banks competed, because the government support that lowered the credit risk of LGFV debt also lowered its yield. Sections 3 and 4 show that savers reward what they see and that smaller banks reached for yield. The largest source of their extra yield was home LGFV credit, and their savers could not see its risk. The strategy needed both the risky yield and the concealment of its risk. It would therefore fail if the products had to report at market prices, and it would also fail if the risk and its yield were removed. The debt-resolution program described in Section 2 did the second. The program reduced refinancing pressure and thereby strengthened expected government support for LGFVs. This lowered both the credit risk of local banks' home holdings and the yield those holdings could earn. Section 5.1 documents the loss of the yield, Section 5.2 shows which banks lost the most from it, and Section 5.3 shows where the borrowers and their risk went.

5.1 The loss of yield

The program removed most of the extra return that home LGFV bonds had supplied. We document the compression of LGFV spreads, the loss of the WMP-return premium on home LGFV bonds, and local banks' retreat from these bonds.

Spreads and yields. LGFV bonds lost about half of the extra yield they offered over government bonds. Among newly issued fixed-rate LGFV bonds, the average spread over the Treasury yield fell from 200 basis points in 2022Q1–2023Q3 to 97 basis points in 2023Q4–2025Q4.[14] Spreads fell more where special refinancing was larger relative to pre-program GDP. In a quarterly event study with province and quarter effects, this difference relative to 2023Q2 emerges with implementation and deepens through 2024 (Figure 7). For WMPs competing for savers, home LGFV bonds therefore offered much less extra yield.

Private credit retained more of its yield and offered a substitute. From 2023Q2 to 2024Q4, average yields on new private credit contracts fell from 6.6% to 5.9%, while yields on new LGFV bonds fell from 4.6% to 2.7%. The gap between the two thus widened from about 2.0 to 3.2 percentage points (Figure 8). The widening holds for the same borrower. We compare the private credit and the bonds of the same legal borrower in the same financing month, with borrower–month effects and a separate maturity slope in each quarter.[15] The gap widens from 1.90 percentage points in 2023Q2 to 2.86 points in 2024Q4 and remains 1.06 points above its 2023Q2 level in 2025Q3 (Figure D1).

Return premiums. In basis points, home LGFV bonds lost a larger WMP-return premium than any other asset class, while private credit kept more of its premium. Section 4.1 estimated the initial premiums as β in equation (5). The Post interactions δ in the same regression measure the change in each asset's premium, and $\beta + \delta$ gives its post-program premium (Table 6, column 2).

[14] The averages weight issues equally and do not condition on WMP ownership.
[15] Internet Appendix D.4 gives this specification.

For the common 20-point allocation, the annual home LGFV bond premium falls from 20.5 to 5.0 basis points. This is a loss of 15.5 basis points, or 76%. The non-home LGFV premium falls from 7.7 to 1.5 basis points. The asset with the largest initial premium thus lost the most. The home private credit premium falls from 17.6 to 9.4 annual basis points, which is a loss of 8.2. The non-home private credit premium rises from 8.2 to 9.6. The decline in the home LGFV premium is statistically larger than the change in each of the other asset classes.[16] This difference matters for substitution, because private credit from local borrowers still offered a positive return premium after much of the bond premium had disappeared.

Lower income from the bonds that WMPs hold explains more than half of the loss. When we add each portfolio's average bond issue yield and its Post interaction to the return regression, the estimated annual premium loss falls from about 16 to 7 basis points (Table D5, Panel A). Home LGFV bonds lose their premium and home private credit retains its premium in four further specifications. These use mature products only, add a control for return volatility, average returns within product and month, and drop the winsorization (Table D5, Panel B).

Portfolio retreat. As home LGFV bonds lost their return advantage, local banks investing at home pulled back more than other buyers of bonds from the same city. Lower holdings alone could reflect fewer local bonds rather than a change in banks' preference for them. We therefore compare changes in the allocations of home buyers and other WMP buyers within the same issuer city and quarter, and we estimate

$$\begin{aligned}\text{LGFV bond share}_{acq} = \beta\, \mathbb{1}\{\text{Home buyer}\}_{ac} \times \mathbb{1}\{\text{Post}\}_q \\ + \mu_{ac} + \mu_{aq} + \mu_{cq} + \varepsilon_{acq}.\end{aligned} \tag{8}$$

[16] Bank-clustered tests using the regression's full covariance matrix give $p = 0.016$ for the difference in changes between home and non-home LGFV bonds, $p = 0.041$ between home LGFV bonds and home private credit, $p = 0.016$ against home other bonds, and $p < 0.001$ against non-home private credit.

Each observation is a buying institution a, an LGFV issuer city c, and a quarter q. The outcome is that city's LGFV bonds as a percentage of the institution's total WMP bond holdings. Home equals one for a local bank investing in a city of its headquarters province, and Post begins in 2023Q4. City–quarter effects μ_{cq} absorb changes in local bond supply, buyer–quarter effects μ_{aq} absorb changes across a buyer's portfolio, and buyer–city effects μ_{ac} absorb persistent investment ties.[17]

Home buyers cut their allocation by an additional 0.45 percentage points per issuer city relative to other buyers. This cut is about 12% of their pre-program mean allocation of 3.91% (Table 11). The decline is concentrated in bonds from their own city, and net buying and new purchases also fall. It is larger for cities whose LGFV bonds carried higher spreads before the program, and home buyers show no differential trend before it (Table 11, columns 5 and 6). In an event study for the destination province as a whole, home buyers' allocation relative to other buyers' falls by 2.3 percentage points from 2023Q2 to 2024Q1 and by 6.1 to 2025Q3, while the pre-announcement confidence intervals include zero (Figure 9). Where a province received more special refinancing, the local institutions that had been more concentrated at home cut their home LGFV bond shares by more.[18] Within two years of the program, home LGFV bonds had lost most of the extra return on which smaller banks had competed for savings, and the banks that had relied on them were leaving them.

5.2 Replacing the lost yield

Larger local banks replaced part of the lost home LGFV bond yield with private credit. The smallest banks did not, even in new products. Private credit kept a higher yield, but arranging and monitoring it requires a fixed investment in sourcing, evaluation, and contracting (Section 2). This investment does not shrink with the size of the position, so it

[17] Errors cluster by bank city.

[18] We measure refinancing intensity as cumulative special-refinancing bond issuance divided by 2020 LGFV debt in the home province. When it is ten points higher, an institution whose initial home overweight (Section 4.1) is ten percentage points higher cuts its home LGFV bond share by 0.46 percentage points more (wild-bootstrap $p = 0.010$). Internet Appendix D.4 gives this specification, which compares local-bank products in the same home province and quarter.

favors banks with more savers. Smaller banks fell behind in returns and WMP balances. We measure a bank's reliance on home LGFV bonds by its initial home LGFV bond share. Once this reliance and the bank's substitution into private credit are held fixed, size alone is not statistically significant for either the return losses or the lower balances of newly launched products.

Substitution into private credit. Local banks moved into private credit, which had been concentrated at national banks. Before implementation, national banks held 82.7% of the identified value of private credit in our sample, and most of it was outside their headquarters provinces (Table 12). Local banks' share then rose from 17.3% to 31.3%. Within their portfolios, home private credit increased from 6.1% to 13.9% of included debt holdings, which is more than the increase outside their provinces.

This shift was concentrated among larger local banks. Among banks observed in both periods, the largest third raised home private credit from 7.45% to 18.26% of included debt holdings. This increase more than offset their decline in home LGFV bonds, so the two together rose from 28.1% to 33.2%. At the smallest third, home private credit remained negligible and fell from 0.72% to 0.31%, and the combined share fell from 25.8% to 21.1%. The size groups use fixed FY2021 assets, so no bank is reclassified after the program. The change was largely in the form of financing, not a departure from the LGFV sector. We match borrowers to LGFV bond issuers and screen their names for terms typical of LGFVs. These two steps identify 85% of the largest local banks' post-program home private credit value as LGFV-linked (Internet Appendix D.1).

The size difference remains after accounting for product characteristics, bank conditions, and local economic conditions. In product-quarter regressions for local banks with bank and quarter effects, the largest FY2021 size tercile raises its home private credit share after implementation by 8.21 percentage points more than the smallest tercile and its total private credit share by 10.46 points more. Both differences are significant at the 1% level

(Table 13).[19] The size difference also holds conditional on initial home overweight, which measures banks' initial orientation toward home investments. Among banks facing the same provincial conditions, a bank twice as large increases its home private credit share by about 4.1 percentage points more (Table 14, Panel B). Panel A shows the same pattern when size interacts with refinancing intensity instead of the post-program indicator. Refinancing intensity is cumulative special-refinancing bond issuance divided by 2020 LGFV debt in the home province.

New products. New products let banks choose a fresh portfolio, yet even there the smallest banks made almost no use of the substitute. Because existing portfolios adjust slowly, we compare the first disclosed debt portfolios of WMPs launched in 2023Q4–2025Q3 across the same fixed size groups. The smallest banks launched 253 new products, and only two of them held home private credit. Of the 5,248 new products that the largest banks launched, 1,118 held it, which is about 21% (Table 15). Home private credit made up just 0.78% of the smallest banks' new-product debt holdings, against 27.5% at the largest banks. Home LGFV bonds still accounted for 29.4% at the smallest banks. Legacy bond positions therefore do not explain the limited substitution. It follows the pattern that the bank-level fixed cost of private credit predicts for a bank with fewer savers (Section 6).

Returns and balances. After the program, smaller local banks fell behind larger ones in product returns and in total outstanding WMP balances, which include older products. A size-by-quarter event study relates monthly product returns to initial bank size and compares each quarter with 2023Q3 (Figure 10). It includes issuing-institution and home-province-by-month effects.[20] The size–return slope shows no trend before the program and does not move in the first quarters after implementation. It then widens from 2024Q3, and the gap is statistically significant through 2025Q2. This timing matches the mechanism of Section 2.

[19] Internet Appendix D.4 gives the specifications of this subsection that are described in words.
[20] Internet Appendix D.4 gives the specification.

A fall in spreads first raises the prices of bonds already held, and it lowers returns only as products reinvest at lower yields.

Because savers' money follows reported returns (Section 3), the return gap should also slow the growth of the whole WMP business. To test this, for local bank b at half-year end s we estimate

$$\begin{aligned}\text{Log WMP balances}_{bs} = \beta_S \,\text{Log bank size}_b \times \mathbb{1}\{\text{Post}\}_s \\ + \alpha_b + \alpha_s + \varepsilon_{bs}.\end{aligned} \tag{9}$$

The outcome is log total WMP balances, and Post begins with the December 2023 report. Bank effects α_b remove persistent differences in WMP balances, and half-year effects α_s remove common growth.[21]

Smaller banks' WMP businesses also grew more slowly. After implementation, the WMP balance of a bank twice as large grows about 11% more (Table 16), and about 10.5% more when we compare banks within the same province and half-year. Across the fixed size groups, the largest third grows about 40% more than the smallest. Among the smallest banks reporting in both June 2023 and December 2024, aggregate WMP balances fall by 22.0%. In an event study, the gap for a bank twice as large is small in December 2023 and grows to roughly 11% by December 2024 and 26% by December 2025 (Figure 11). The difference remains when we keep only banks observed at every date, exclude the final period, or give banks with and without wealth-management subsidiaries separate time paths.

Reliance, not size. The size gaps in returns and fundraising follow what banks held and added, not size itself. Among local banks, reliance on home LGFV bonds does not rise steadily as size falls. The middle size tercile held the largest initial home LGFV bond share (Table 12, Panel C), so reliance and size can be separated. We do so by including the initial home LGFV bond share, subsequent private-credit expansion, and initial bank size together

[21]The panel covers 2022H1–2025H2, and errors cluster by bank.

in one regression. For parent bank b in month m, the outcome is its average WMP log return in basis points, and we estimate

$$\begin{aligned}\text{Monthly return}_{bm} = {} & \beta_L \,\text{Initial home LGFV bond share}_b \times \mathbb{1}\{\text{Post}\}_m \\ & + \beta_P \,\Delta\text{Home PC share}_b \times \mathbb{1}\{\text{Post}\}_m \\ & + \beta_S \,\text{Log bank size}_b \times \mathbb{1}\{\text{Post}\}_m \\ & + \beta_H \,\text{Initial home PC share}_b \times \mathbb{1}\{\text{Post}\}_m \\ & + \gamma' \,\text{Controls}_{b,h(m)} + \alpha_b + \alpha_m + u_{bm}.\end{aligned} \tag{10}$$

The initial home LGFV bond and private-credit shares are the bank's home holdings of each as a share of its included debt holdings, with holding values aggregated over 2022Q1–2023Q2. The expansion measure ΔHome PC share$_b$ is the bank's average quarterly home private-credit share in 2023Q4–2025Q3 minus its average in 2022Q1–2023Q2. Bank effects absorb the levels of these four bank-level measures, month effects absorb Post, and the eight prior-quarter bank and home-city controls enter through Controls$_{b,h(m)}$.[22]

On its own, size predicts the return losses. A bank twice as large loses about 15 fewer annualized basis points after the program ($p = 0.017$, Table 17, column 1). Both investment measures matter once they enter. A ten-point higher initial home LGFV bond share predicts about 12 basis points more annualized return decline ($p = 0.017$), while a ten-point larger expansion in home private credit is associated with about 14 basis points less decline ($p = 0.023$, column 2). With these two measures held fixed, the size–Post coefficient halves and is no longer statistically distinguishable from zero, so the return losses follow what banks held and added rather than their size. The initial home LGFV bond share also predicts losses without conditioning on banks' subsequent substitution (Internet Appendix D.4).

Banks with a higher initial home LGFV bond share also raised less money through newly launched WMPs. Our fundraising proxy is the log of the summed balances of the

[22]The sample is July 2022–September 2025, excluding July–September 2023. Each bank-month receives equal weight, and errors cluster by bank.

observed products that local bank b launched in quarter q, measured at the end of quarter $q+1$. We estimate the same two specifications, with launch-quarter effects in place of month effects (Table 17, Panel B). On its own, size again predicts the losses. A bank twice as large raises about 22% more through new products after the program ($p = 0.002$, column 3). With the investment measures added (column 4), a ten-percentage-point higher initial home LGFV bond share predicts 15.3% lower new-product balances ($p = 0.002$), and a ten-point larger increase in home private credit is associated with 14.6% higher new-product balances ($p = 0.034$). The size coefficient falls by more than half and is no longer statistically significant. Because the June 2024 guidance asked banks without a subsidiary to shrink their WMP business (Section 2), we also end the sample before it. The reliance estimate is nearly unchanged, at 14.9% lower balances ($p = 0.001$).

The banks that had competed most on home LGFV yield thus lost the most when debt resolution compressed it. Smaller banks relied on bond yield to compete. They had fewer savers to cover the fixed cost of private credit, so they made little use of it when that yield fell.

5.3 Where the risk went

Debt resolution removed the main source of the yield on which smaller banks competed, and it also absorbed the credit risk that their products had hidden from savers. This subsection shows who took over the borrowers that the most reliant banks left and how far the market's price of the credit risk in their holdings fell. We focus on local banks without a wealth-management subsidiary. This group includes nearly all smaller local banks, and the 2024 guidance asked it to reduce its WMP business (Section 2).

The borrowers. As the most reliant banks withdrew, national banks' products took over their borrowers at lower coupons. We split local banks without a subsidiary at the median initial home LGFV bond share into high-reliance and low-reliance banks. For each LGFV

borrower, we measure the fractions of its WMP holdings that each of the two groups held in 2022Q4–2023Q1. The outcome is the inverse hyperbolic sine of the borrower's bonds held by the WMPs of a group of banks, which approximates the log and keeps zero holdings. We relate it to the two fractions, each interacted with indicators for the periods after implementation and after the June 2024 guidance. Borrower effects remove persistent differences in WMP funding, and borrower-province-by-quarter effects compare borrowers in the same province and quarter.[23]

Consider a borrower funded entirely by high-reliance banks before the program. After implementation, holdings of its bonds by local banks without a subsidiary fell by 1.89 log points, while holdings by national banks' WMPs rose by 1.83 and holdings by local banks with a subsidiary rose by 0.74 (Table 18, Panel A). Both the withdrawal and the national-bank replacement are significantly larger than for borrowers of low-reliance banks ($p = 0.005$ and $p = 0.010$), and private credit follows the same pattern (Panel B). Relative to borrowers that local banks without a subsidiary did not fund, the coupons of new bonds fell by 1.17 percentage points for a borrower funded entirely by high-reliance banks and by 0.62 for one funded by low-reliance banks ($p = 0.001$ for the difference, column 5). The same support that compressed the yield on these borrowers' bonds lowered their funding costs.

The credit risk. A year after the program, the market priced far less credit risk in these banks' home LGFV holdings. We weight each province's LGFV spread over Treasury yields by the home LGFV credit that local banks without a subsidiary held in 2022Q4–2023Q2. This weighted spread fell from 297 basis points in June 2023 to 87 in June 2024.

Debt resolution moved the credit risk that smaller banks' products had accumulated to provincial governments and to national banks' products, which now hold these borrowers at lower yields. Smaller banks' WMP business was fragile either way. Without resolution, credit losses would have fallen on products that savers had bought as safe. With it, these

[23] Internet Appendix D.4 gives the specification and further results, and Internet Appendix D.3 the construction.

banks lost the return advantage on which they competed. Regulators in several provinces have asked for their gradual exit since June 2024. This exit has so far coincided with lower funding costs for the borrowers they financed.

6 Model

The model explains three features of the evidence. First, smaller banks reach for yield by taking credit risk that savers do not see. Second, scale determines which banks can replace bond yield with private credit. Third, smaller banks lose more returns and savings when debt resolution compresses that yield. Three assumptions deliver these results. Savers choose banks by reputation and by the returns and volatility that products report. Banks bear the credit risk in their products only through their reputation, because the losses themselves fall on savers. Arranging private credit requires a fixed cost.[24] Internet Appendix B states the model in full, derives the results, and computes an illustrative equilibrium, and Internet Appendix C gives proofs.

6.1 Savers and banks

Savers see what products report and not what they hold, and banks bear credit risk only through their reputation. A bank in the model is its WMP business, and the investments below belong to its products and not to its balance sheet. A unit mass of savers each invests one unit with a bank or an outside saving option. Savers differ in risk aversion γ_h, which is drawn from a distribution G on a bounded positive interval. Saver h obtains utility

$$U_{hb} = \gamma_h a_b + \beta\left(R_b - \frac{\gamma_h}{2} V_b\right) + \varepsilon_{hb} \tag{11}$$

[24]Savers choose institutions and institutions choose assets, and this two-layer demand follows Darmouni et al. (2024). Heterogeneous saver demand draws on Xiao (2020) and random-coefficient models (Berry et al. 1995), fee-based portfolio incentives follow La Spada (2018), and reputation as a restraint on risk taking follows Keeley (1990).

from bank b. Here R_b and V_b are the excess return and return variance that the bank's products report, and $\beta > 0$ measures how strongly choices respond to the risk-adjusted reported return. In the data, flows respond to both reported returns and their volatility (Section 3). The term a_b captures the bank's reputation and sales network. We assume that cautious savers weight reputation more, because a bank's standing matters more to savers who fear losses. With logit taste shocks ε_{hb}, the share of savers with risk aversion γ choosing bank b is

$$s_b(\gamma) = \frac{\exp\Big(\gamma a_b + \beta(R_b - \gamma V_b/2)\Big)}{1 + \sum_j \exp\Big(\gamma a_j + \beta(R_j - \gamma V_j/2)\Big)}, \tag{12}$$

and the bank's market share is $s_b = \int s_b(\gamma)\, dG(\gamma)$. There are three banks, a national bank N, a large local bank L, and a small local bank S, with reputations $a_N > a_L > a_S$.

Each bank places a share x_b of its products' funds in LGFV bonds, a share n_b in private credit linked to the same borrower sector, and the rest in riskless assets. LGFV bonds stand for the risky credit that supplied most of smaller banks' extra yield (Section 4.1). Bonds and private credit pay excess returns μ_B and μ_{PC}. Their reported returns have variances σ_B^2 and σ_{PC}^2 and covariance $\sigma_{B,PC}$, so the portfolio reports $R_b = \mu_B x_b + \mu_{PC} n_b$ and $V_b = \sigma_B^2 x_b^2 + 2\sigma_{B,PC} x_b n_b + \sigma_{PC}^2 n_b^2$. All banks face the same asset returns and risks, so no bank has better information about the borrowers.

LGFV-linked credit also carries a credit risk that reported returns do not show (Section 4.2). With probability θ, a credit event occurs among the sector's borrowers. Savers then see the losses and lower the bank's reputation by $\kappa \ell_b^2/2$, where $\ell_b = x_b + n_b$ is the bank's exposure to the LGFV sector. Because WMPs do not guarantee principal, the losses fall on savers. A bank earns a fee f per yuan of savings, and initiating private credit requires a one-time fixed cost $F > 0$ for sourcing, evaluation, and contracting. Let $\mathcal{S}_b(a)$ be the bank's market share when its reputation is a. The bank chooses its portfolio to maximize

$$f\,\mathcal{S}_b(a_b) - \theta f\Big[\mathcal{S}_b(a_b) - \mathcal{S}_b\Big(a_b - \frac{\kappa}{2}\ell_b^2\Big)\Big] - F\,\mathbb{1}\{n_b > 0\}\, e_b, \tag{13}$$

where $e_b = 1$ if the bank has not arranged private credit before and $e_b = 0$ otherwise. The second term is the expected loss of fee income from the damage to its reputation.

6.2 Hidden credit risk

Credit risk that savers do not see restrains a bank's bond holdings in the same way as the variance they do see. Both restraints are scaled by the caution of the bank's marginal savers, who are the savers most likely to switch when its offer improves. Weighting each saver type by the response of its choice to the bank's offer gives their average risk aversion,

$$\bar{\gamma}_b = \frac{\int \gamma \, s_b(\gamma)\big(1 - s_b(\gamma)\big) \, dG(\gamma)}{\int s_b(\gamma)\big(1 - s_b(\gamma)\big) \, dG(\gamma)}. \tag{14}$$

Because savers weight reputation by their risk aversion, a loss of reputation costs a bank more savers when its marginal savers are more cautious. To first order, the expected fee loss in equation (13) equals the fee value of a reduction in reported return of $\bar{\gamma}_b \varphi \ell_b^2/2$, where

$$\varphi = \frac{\theta\kappa}{\beta} \tag{15}$$

measures the credit risk the bank bears through its reputation, in units of reported variance. Holding the composition of marginal savers fixed, a bank that holds only bonds chooses

$$x_b^B = \frac{\mu_B}{\bar{\gamma}_b \left(\sigma_B^2 + \varphi\right)}. \tag{16}$$

A lower reported variance σ_B^2 lets a bank hold more bonds without losing savers. Products whose reported returns do not follow market prices have such a lower variance. Smaller banks' products reported less volatility than their holdings implied (Section 4.2).

Lower-reputation banks hold more bonds and therefore more credit risk that their savers do not see. The reason is that reputation determines not only how many savers a bank attracts but also which ones. More risk-averse savers are relatively more likely to choose

the more reputable bank, so higher-reputation banks have more cautious marginal savers (Proposition B.1). These banks both accept less reported variance and have more to lose from a credit event. The higher bond share in equation (16) helps a lower-reputation bank compensate for its weaker reputation in attracting savings.

6.3 Private credit and scale

Scale determines which banks add private credit, because its cost is fixed while its gain grows with the bank's investor base. Private credit earns more than the bonds it displaces, so it raises a bank's risk-adjusted value per unit of savings by an amount g_b, which Internet Appendix B derives. Let Φ_b^B and Φ_b^{PC} be the maximized values of the first two terms of equation (13) without and with access to private credit, holding rivals' portfolios fixed. We break ties in favor of entry. A bank that has not yet arranged private credit then enters if and only if the additional value covers the fixed cost,

$$\Phi_b^{PC} - \Phi_b^B \geq F. \tag{17}$$

To first order, the gain is $f\beta s_b g_b$ (Proposition B.2). It is the fee f times the bank's investor base s_b times βg_b, which is the proportional increase in that base from adding private credit. Less cautious savers accept more risk in exchange for yield, so private credit adds more value per unit of savings at the small bank. However, the larger bank can offer it to a broader investor base, and under the scale condition stated in Internet Appendix B that broader base dominates. For an intermediate fixed cost, only the large local bank enters. The small bank also wants higher returns, but private credit would cost it more than the additional fees it earns, and launching a new product does not remove this bank-level cost. Figure 12 illustrates the resulting entry threshold.

6.4 Debt resolution

Debt resolution takes over part of the credit risk and the bond yield together. It strengthens expected support for LGFVs. We represent this as a fall in the probability of a credit event from θ to $(1-\tau)\theta$ with $0 < \tau < 1$, so the credit risk that banks bear through reputation falls from φ to $(1-\tau)\varphi$. The excess yield on LGFV bonds compensates investors for this risk, so it falls in the same proportion, from μ_B to $(1-\tau)\mu_B$. The return on private credit and all reported variances are unchanged. This simplification matches Section 5.1, where private-credit yields fell far less than bond yields.

Prediction 1. Smaller banks take more credit risk that savers do not see. The bond share in equation (16) falls with reputation, because higher reputation raises the risk aversion of a bank's marginal savers. All banks face the same returns and risks, so the gradient arises without any information advantage, as in Table 9. We take bank size as the empirical counterpart of reputation. Table 4 tests the gradient across all banks, and Table 7 shows the same gradient among products with the same redemption terms and risk rating. In the model, the bonds of every bank have the same reported variance σ_B^2. The lower volatility that smaller banks' products report (Table 7, Panel B) corresponds to a lower σ_B^2 at these banks, which in equation (16) raises their bond share further.

Prediction 2. Scale determines which banks substitute into private credit. Resolution lowers the bond yield and the credit risk that private credit shares with bonds, so the gain from private credit rises and the scale needed to cover the fixed cost falls. The large local bank can cross this threshold while the small bank remains below it, and earlier users expand their positions as bonds become less attractive. The allocation tests measure expansion across size groups (Tables 13 and 14), and the difference also appears in freshly chosen portfolios (Table 15).

Prediction 3. Smaller local banks lose more product return and savings business. Comparing the two initially bond-only local banks when only the large bank enters,

$$\begin{aligned} \Delta R_L - \Delta R_S &= \lambda \frac{\mu_B^2}{\sigma_B^2 + \varphi}\Big(\frac{1}{\bar{\gamma}_S} - \frac{1}{\bar{\gamma}_L}\Big) + \Big(\mu_{PC} - \chi_\tau(1-\tau)\mu_B\Big) n_L^{PC} > 0, \\ \lambda &= 1 - \frac{(1-\tau)^2(\sigma_B^2 + \varphi)}{\sigma_B^2 + (1-\tau)\varphi} \in \Big(0, \tau(2-\tau)\Big]. \end{aligned} \tag{18}$$

Here λ is the proportional fall in a bond-only bank's reported return. Internet Appendix B defines χ_τ and n_L^{PC}, which are the bonds that a unit of private credit displaces and the large bank's private-credit position after resolution. The first term reflects the smaller bank's larger bond holdings before resolution, and the second term reflects the private credit with which only the larger bank replaces part of that income. The small bank therefore loses the return advantage it used to compete against the larger bank's reputation, and it loses savers as a result (Proposition B.5). Because smaller banks both cut their LGFV holdings and lose savers, more of the LGFV credit held by WMPs moves to larger banks' products, as in Section 5.3. In equation (18), reputation affects the return gap only through the banks' initial bond holdings and the larger bank's private credit. The data show the same pattern. Smaller local banks lost more return and grew more slowly (Table 16, Figures 10 and 11). Once the initial home LGFV bond share and private-credit expansion enter, size is no longer statistically significant for the return and fundraising losses (Table 17).

7 Conclusion

Chinese households choose wealth-management products by the returns they report and by how much those returns fluctuate, not by the risk of the assets behind them. In the 2022 bond sell-off they ran from the losses that products reported, not from the losses their holdings took. Smaller banks lack the reputations of national banks, so they could compete for savings only by exploiting this bias. They reached for yield in the dark. Their products held riskier debt than larger banks' products, and the largest source of their extra yield was concentrated

LGFV credit from their home provinces. Yet these banks rated the products as safe, locked savers in with restricted redemptions, and reported returns that barely moved. In the 2022 bond sell-off, their reported returns only dipped to about zero while their holdings lost about 1.6% at market prices. Larger banks from the same city have more branches there, yet they held safer bonds with lower yields, so this was reaching for yield rather than better local information.

Smaller banks needed this strategy to survive. It was fragile because it needed both the risky yield and the concealment of its risk. Savers punish even small dips in reported returns. They would have run if these products had shown the losses that their holdings took in the 2022 sell-off. Instead, China's 2023 debt-resolution program absorbed much of the risk and removed much of the yield with it. Smaller banks relied on that yield the most, so they lost returns and fundraising. Banks with a higher initial home LGFV bond share lost more return and raised less through newly launched WMPs. Once this reliance and the use of private credit are held fixed, bank size itself does not predict the losses. Larger local banks replaced part of the lost yield with private credit, whose fixed costs favor scale. National banks' products took over the borrowers that the most reliant banks left, and these borrowers now pay lower coupons.

Our model traces this pattern to reputation. Banks bear the credit risk in their products only through their reputations, so the banks whose cautious savers value reputation most take the least of it. In China, support for the borrowers absorbed this risk before savers bore it, and it also removed the main source of the yield on which smaller banks competed. Their WMP business is winding down, and the borrowers they financed now borrow at lower cost. When banks have little reputation to lose, they compete through a fragile strategy that exploits the behavioral bias of households, who care about returns and observed volatility but not underlying risk.

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Table 1
Past returns, reported volatility, holdings risk, and investor flows

Dependent variable	Net flow (% initial assets)			
	(1)	(2)	(3)	(4)
$\mathbb{1}\{\text{Return quintile} = 2\}$	2.267***	2.289***	1.643*	−0.746
	(0.833)	(0.832)	(0.899)	(0.940)
$\mathbb{1}\{\text{Return quintile} = 3\}$	3.457***	3.489***	2.688***	0.221
	(0.986)	(0.994)	(0.890)	(0.826)
$\mathbb{1}\{\text{Return quintile} = 4\}$	6.096***	6.145***	5.637***	2.373**
	(1.335)	(1.354)	(1.361)	(1.141)
$\mathbb{1}\{\text{Return quintile} = 5\}$	11.208***	11.299***	8.524***	4.325***
	(3.136)	(3.155)	(2.430)	(1.348)
Reported return volatility (per SD)	—	—	−1.609***	−2.371***
			(0.482)	(0.625)
Home LGFV credit share (per SD)	—	−0.262	−0.177	—
		(0.271)	(0.300)	
Coupon of LGFV bonds held (per SD)	—	0.016	−0.229	—
		(0.642)	(0.477)	
Volatility implied by holdings (per SD)	—	—	—	1.121**
				(0.550)
Joint test of holdings risk (p)	—	0.626	0.757	—
Product, bank, and city controls	Yes	Yes	Yes	Yes
Observations	61,848	61,848	49,574	19,629
Category × quarter fixed effects	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Notes: Each column estimates equation (1) and adds the regressors shown, and each regressor enters per standard deviation. Return quintiles rank prior-quarter returns within risk rating, redemption terms, and quarter, and the bottom quintile is omitted. Reported return volatility is the log standard deviation of the product's monthly reported returns over the prior 12 months (SD 1.00). Home LGFV credit share is home-province LGFV bonds and private credit to home-province LGFVs as a percentage of disclosed holdings (SD 16.7 percentage points). Coupon is the value-weighted coupon of the LGFV bonds held (SD 1.14 percentage points). The joint test covers these two holdings-risk measures. Columns 1–3 use product-quarters with disclosed holdings, and column 3 also requires a 12-month return history. Column 4 uses products of every type with disclosed holdings and at least nine monthly returns between April 2022 and June 2023. In this column, reported and implied volatility (SD 0.55) are measured over the same months. Controls are those of equation (1) and a local-bank indicator. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2022Q3–2025Q3 (columns 1–3), 2022Q4–2023Q3 (column 4).

Table 2
Reported returns and investor flows

	Flows in the same quarter		Flows in the next quarter	
	(1)	(2)	(3)	(4)
Panel A. Daily-open products, by reported quarterly return				
Loss of more than 0.5%	−47.5***	−11.8*	−33.9***	−3.9
	(9.0)	(6.8)	(10.3)	(7.8)
Loss of 0.1% to 0.5%	−46.1***	−23.2***	−32.2***	−12.8**
	(8.6)	(8.2)	(7.3)	(6.1)
Loss of less than 0.1%	−51.1***	−31.5***	−28.1***	−12.2**
	(7.4)	(7.9)	(5.2)	(4.6)
Gain of less than 0.25%	−51.7***	−35.7***	−27.0***	−14.7**
	(9.7)	(10.6)	(5.7)	(5.6)
Gain of 0.25% to 0.5%	−43.4***	−30.4***	−18.5***	−15.7***
	(4.8)	(5.3)	(2.3)	(2.7)
Gain of 1% or more	17.6***	0.7	52.5***	31.2***
	(5.1)	(7.4)	(11.8)	(8.2)
Observations	22,106	21,449	22,106	21,449
Panel B. All products, by redemption type				
Return × daily-open	52.5***	20.7**	57.7***	26.6**
	(16.4)	(8.6)	(18.7)	(11.7)
Return × periodic-open	3.8***	−2.0**	8.2***	3.2***
	(0.8)	(0.9)	(1.4)	(1.1)
Return × closed-end	−0.3	1.1	−0.7	−0.5
	(0.9)	(0.8)	(0.7)	(0.6)
Observations	129,554	122,194	129,554	122,194
Controls	Yes	Yes	Yes	Yes
Risk rating × redemption type × quarter fixed effects	Yes	Yes	Yes	Yes
Bank × quarter fixed effects	Yes	Yes	Yes	Yes
Product fixed effects	No	Yes	No	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Notes: The dependent variable is net flow, which is asset growth net of investment gains as a percentage of beginning assets. The reported return is that of the same quarter in columns 1 and 2 and of the preceding quarter in columns 3 and 4. In Panel A, each row is an indicator for the product's reported quarterly return, and the omitted group reported 0.5% to 1%. In Panel B, the reported quarterly return is interacted with redemption-type indicators. It is in percentage points and capped at its 1st and 99th percentiles. All regressions include the lagged controls of Table 1, risk rating by redemption type by quarter effects, and bank-by-quarter effects, and columns 2 and 4 add product effects. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2022Q3–2025Q3.

Table 3
Where smaller banks' extra yield comes from

	Size gradient (bp per log point)	Share of holdings (%)		Spread (bp)		Smaller local minus national (bp)
		Smaller local	National	Smaller local	National	
	(1)	(2)	(3)	(4)	(5)	(6)
Home LGFV bonds	−17.0*** (3.3)	19.8	0.9	317	236	60.8
Other LGFV bonds	−2.7* (1.4)	10.9	8.0	202	210	5.2
Private credit	8.2*** (2.6)	5.4	10.2	406	351	−14.0
Local banks' capital bonds	−4.3*** (0.9)	8.6	1.7	191	153	13.8
National banks' capital bonds	1.1* (0.6)	3.6	8.0	125	113	−4.5
Other financial bonds	0.6 (0.8)	5.7	8.1	126	126	−3.0
Home corporate bonds	−4.6*** (1.4)	5.8	1.5	270	121	13.9
Other corporate bonds	1.4*** (0.5)	3.5	8.1	136	142	−6.7
Government and policy-bank bonds	−0.4** (0.2)	5.9	3.5	24	11	1.0
Certificates of deposit	−0.4 (0.4)	3.7	2.7	112	88	1.8
Plans, funds, and other wrappers	5.2 (3.4)	22.9	30.6	193	186	−12.4
Cash, deposits, and repos	0.0	4.1	16.7	0	0	0.0
All holdings	−12.8*** (2.9)	100.0	100.0	205	149	55.8

Notes: The sample is the products of Table 7, with holdings averaged over 2022Q2–2023Q2. Each holding is priced at its credit spread over Treasury yields. Bonds carry their issue spread, and private credit carries its contract rate over the Treasury yield of its remaining maturity. Plans, funds, and other wrappers carry the mean issue spread of AA+ bonds with one to three years to maturity (AAA for bond funds and rate products). Certificates of deposit carry the spread of short-term bonds of the issuer's rating, and cash, deposits, and repurchase agreements carry zero. The all-in spread is the value-weighted mean, which equals the sum over classes of holding share times spread. Column 1 regresses each class's contribution on log bank size as in Table 7, so its entries sum to that table's all-in spread coefficient. Columns 2–5 average products within a bank and then banks. Column 6 is the contribution at smaller local banks minus that at national banks, where smaller local banks are those at or below the median size. Capital bonds are perpetual and tier-2 bonds, and home corporate bonds are bonds of non-LGFV, non-financial issuers from the home province. Stars indicate 10%, 5%, and 1% significance.

Table 4
Initial credit specialization by bank size

Dependent variable	Total LGFV bond share (pp) (1)	Total private credit share (pp) (2)	Home LGFV bond share (pp) (3)
Log bank size	−7.130***	6.831***	−5.458***
	(1.778)	(1.803)	(1.123)
Log product assets	−3.028***	−3.480***	−0.677
	(0.828)	(0.914)	(0.431)
𝟙{Restricted redemption}	2.405	16.943***	−0.264
	(2.138)	(4.113)	(1.323)
Product risk rating	−3.065**	4.162	0.392
	(1.316)	(2.956)	(0.867)
Bank nonperforming-loan ratio	−7.968	10.855**	−3.180
	(6.171)	(5.471)	(4.021)
Bank cost–income ratio	−0.245*	−0.582***	−0.085
	(0.131)	(0.184)	(0.092)
Bank net interest margin	−5.013	9.229	2.795
	(7.203)	(8.397)	(4.126)
Home-city GDP (RMB trillion)	0.064	−1.051	0.810
	(3.585)	(1.945)	(2.118)
Home-city GDP growth (%)	−0.381	−0.182	0.527
	(1.211)	(1.046)	(0.768)
Home-city LGFV debt/GDP	27.483**	3.752	22.502***
	(11.977)	(4.867)	(6.930)
Home-city income (RMB 10,000)	1.325	3.696**	−1.132
	(3.213)	(1.595)	(2.049)
Wild p: log bank size	0.002	0.009	< 0.001
Observations	68,331	68,331	68,331
Quarter fixed effects	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank

Notes: Each column regresses the stated allocation share (pp) of national- and local-bank products on log FY2021 parent-bank interest-bearing assets in RMB trillion, without standardization. Zero holdings are retained, and all ten other controls are displayed. Fixed size replaces current bank assets. The regressions include quarter effects and no national-bank indicator or bank effects, and they weight by included debt value. Parentheses contain parent-bank clustered errors, which group each bank with its wealth-management subsidiary. Stars indicate 10%, 5%, and 1% significance. Wild tests use 4,999 Rademacher draws. Estimation sample: 2022Q1–2023Q3.

Table 5
Within-category home bias by bank type and asset class

Asset class	All included debt	LGFV	PC	Other bonds
Dependent variable	Within-category home bias (fraction) (1)	Within-category home bias (fraction) (2)	Within-category home bias (fraction) (3)	Within-category home bias (fraction) (4)
$\mathbb{1}\{\text{Local bank}\}$	0.2713***	0.3205***	0.4885***	0.1611***
	(0.0316)	(0.0449)	(0.0594)	(0.0292)
Log product assets	−0.0004	−0.0019	0.0016	−0.0083*
	(0.0051)	(0.0092)	(0.0123)	(0.0048)
$\mathbb{1}\{\text{Restricted redemption}\}$	−0.0182	−0.0189	−0.0285	−0.0107
	(0.0147)	(0.0218)	(0.0321)	(0.0093)
Product risk rating	0.0059	0.0035	0.0067	−0.0021
	(0.0089)	(0.0155)	(0.0182)	(0.0087)
Log(1 + disclosed positions)	−0.0965***	0.0077	0.0001	−0.0629
	(0.0337)	(0.0543)	(0.0298)	(0.0386)
Total disclosed weight (%)	0.0007*	0.0016**	0.0008	0.0003
	(0.0004)	(0.0007)	(0.0006)	(0.0004)
Log average product maturity	0.0203**	0.0337**	0.0178	0.0217***
	(0.0086)	(0.0145)	(0.0187)	(0.0080)
Average product risk rating	−0.0513	−0.1744	0.1068	−0.1189***
	(0.0598)	(0.1515)	(0.0931)	(0.0443)
Bank nonperforming-loan ratio (%)	−0.0371	0.0296	0.0490	−0.0142
	(0.0374)	(0.0651)	(0.0745)	(0.0367)
Bank cost–income ratio (%)	−0.0008	−0.0003	0.0019	−0.0001
	(0.0015)	(0.0023)	(0.0032)	(0.0016)
Bank net interest margin (%)	0.0939	0.2127***	0.2460***	0.0412
	(0.0577)	(0.0785)	(0.0926)	(0.0458)
Home-city GDP growth (%)	0.0188*	0.0310*	0.0158	0.0088
	(0.0105)	(0.0160)	(0.0188)	(0.0103)
Mean of dependent variable	0.1851	0.2562	0.1944	0.1305
Observations	186,731	108,313	43,803	158,812
Quarter fixed effects	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Notes: PC denotes private credit. Estimates use equation (3). The unit of observation is a product-quarter with positive holdings in the class. The dependent variable is the product's home-province share within the asset class minus the home province's share of nationwide disclosed holdings in the class, as a fraction. The first column pools all included debt, and the others restrict it to LGFV bonds, private credit, and other bonds. Local bank equals one for banks other than state-owned and joint-stock commercial banks. Product size and the number of disclosed positions enter as $\log(1 + x)$. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2022Q1–2025Q3.

Table 6
Home credit exposure and product returns before and after the program

Dependent variable	Monthly NAV return (bp)	
	(1)	(2)
Home LGFV bond share	10.734***	8.543***
	(1.967)	(1.916)
Home LGFV bond share × 𝟙{Post}	−6.728***	−6.473***
	(1.720)	(1.667)
Home private credit share	8.262***	7.328***
	(1.590)	(1.450)
Home private credit share × 𝟙{Post}	−3.719**	−3.408**
	(1.715)	(1.542)
Home other-bond share	1.944	1.310
	(1.760)	(1.626)
Home other-bond share × 𝟙{Post}	−1.222	−0.978
	(1.863)	(1.723)
Non-home LGFV bond share	3.603***	3.188***
	(1.377)	(1.208)
Non-home LGFV bond share × 𝟙{Post}	−2.925**	−2.559*
	(1.416)	(1.362)
Non-home private credit share	3.243***	3.434***
	(1.109)	(0.969)
Non-home private credit share × 𝟙{Post}	0.616	0.580
	(0.936)	(0.873)
𝟙{Local bank}	1.700*	0.355
	(0.934)	(0.771)
Log product assets	−0.665***	−0.603***
	(0.165)	(0.152)
𝟙{Restricted redemption}	3.930***	3.721***
	(0.774)	(0.711)
Product risk rating	−2.686***	−2.391***
	(0.809)	(0.694)
Log bank assets	—	−0.160
		(0.456)
Bank nonperforming-loan ratio	—	1.329
		(1.277)
Bank cost–income ratio	—	0.061***
		(0.023)
Bank net interest margin	—	1.000
		(1.573)
Home-city GDP (RMB trillion)	—	−0.006
		(0.601)
Home-city GDP growth (%)	—	−0.065
		(0.296)
Home-city LGFV debt/GDP	—	3.750**
		(1.658)
Home-city income (RMB 10,000)	—	−0.338
		(0.612)
Observations	567,855	567,855
Month fixed effects	Yes	Yes
Standard errors	Parent bank	Parent bank

Notes: Monthly log adjusted-NAV returns (bp) are regressed on all five lagged share levels, their Post interactions, and controls. Returns are winsorized at the 1st and 99th percentiles of all monthly returns. Month effects absorb Post, which starts in October 2023. Column 1 uses product controls, and column 2 adds bank and home-city controls. Both columns use the column 2 sample. Shares are fractions, and non-home other bonds are the reference. Multiply coefficients by 0.2 for a 20-point allocation difference and by 12 to annualize. Column 2's post-program premiums are 2.070 (SE 1.809) for home LGFV bonds and 3.920 (1.070) for home private credit. The difference in their premium changes is −3.065 (1.489). Centrally administered issuers are never home. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance. Estimation sample: April 2022–December 2025.

Table 7
Bank size, risk held, and risk reported

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. What the product holds						
Dependent variable	Home LGFV credit share (pp)	Issue spread of bonds held (bp)	Issue spread relative to same rating (bp)	All-in spread of holdings (bp)	Log volatility implied by holdings	
Log bank size	−5.394***	−22.374***	−17.079***	−12.792***	−0.074***	
	(0.874)	(3.511)	(4.084)	(2.866)	(0.016)	
𝟙{Below-median size}	15.509***	50.431***	48.555***	32.983***	0.157***	
	(3.512)	(14.552)	(14.121)	(11.214)	(0.049)	
Log bank size, with 𝟙{Local bank}	−4.280***	−19.807***	−22.247***	−9.914**	−0.054**	
	(1.345)	(6.705)	(6.889)	(4.753)	(0.025)	
Observations	4,165	3,991	3,263	4,165	4,165	
Fixed effects	Type × rating	Type × rating	Type × rating	Type × rating	Type × rating	
Panel B. What the product reports						
Dependent variable	Log SD of monthly return	Log reported minus log implied SD	Return, Nov.–Dec. 2022 (bp)	Reported minus implied, Nov.–Dec. 2022 (bp)	Mean monthly return (bp)	Safe rating, 1 or 2 (pp)
Log bank size	0.154***	0.228***	−6.228***	−11.656***	−1.413***	−4.266***
	(0.036)	(0.040)	(1.669)	(2.320)	(0.297)	(1.595)
𝟙{Below-median size}	−0.565***	−0.722***	20.999***	33.023***	3.952***	6.982*
	(0.134)	(0.144)	(5.995)	(7.713)	(1.028)	(4.135)
Log bank size, with 𝟙{Local bank}	0.137**	0.191***	−6.144**	−9.881**	−1.357**	−4.589**
	(0.055)	(0.062)	(2.594)	(3.898)	(0.535)	(1.994)
Observations	4,165	4,165	4,165	4,165	4,165	11,580
Fixed effects	Type × rating	Type × rating	Type × rating	Type × rating	Type × rating	Type
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank

Notes: Except in column 6 of Panel B, the sample is fixed-income products with disclosed holdings and at least nine monthly returns. Holdings are averaged over 2022Q2–2023Q2, and fixed effects compare products with the same redemption type and risk rating. Column 6 of Panel B uses every fixed-income product with a monthly return in the period and compares products with the same redemption type. Its outcome equals 100 for a safe rating of 1 or 2. Log bank size is log FY2021 parent-bank interest-bearing assets in RMB trillion. Below-median size equals one for smaller banks. These are the banks at or below the median size of the banks in the sample of columns 1–5, which is RMB 0.44 trillion. The third row adds a local-bank indicator. Home LGFV credit share is LGFV bonds and private credit to LGFVs of the bank's home province as a percentage of disclosed holdings. Issue spread is the coupon minus the same-maturity Treasury yield at issue, and column 3 of Panel A nets the mean of bonds with the same issue quarter, issuer rating, administrative level, maturity group, and placement type. The all-in spread is defined in Table 3. Implied returns mark every disclosed asset to its market series (Section 4.2), and column 2 of Panel B is log reported minus log implied volatility. Reported returns are monthly changes in log distribution-adjusted NAV, and each series is winsorized at the 1st and 99th percentiles of all its monthly values. Each bank receives equal total weight. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance. Estimation sample: April 2022–June 2023.

Table 8
Sell-off losses reported, by redemption type

Panel A. Fall below the usual monthly return, November–December 2022 (basis points a month)

	Products	Reported	Implied by holdings	Share reported
Closed-end products				
Smaller local banks	601	−28.5	−117.9	0.24
Larger local banks	422	−54.7	−115.3	0.47
National banks	727	−51.8	−100.4	0.52
Periodic-open products				
Smaller local banks	533	−47.1	−106.8	0.44
Larger local banks	420	−63.6	−107.9	0.59
National banks	1,035	−60.8	−87.0	0.70
Daily-open products				
Smaller local banks	6	−22.7	−89.3	0.25
Larger local banks	24	−40.7	−71.2	0.57
National banks	397	−44.0	−61.8	0.71
All products				
Smaller local banks	1,140	−39.4	−113.6	0.35
Larger local banks	866	−56.3	−109.0	0.52
National banks	2,159	−57.5	−87.9	0.65

Panel B. Reported minus implied return in the sell-off, relative to other months, within bank

	(1)	(2)
Closed-end	33.9***	29.0***
	(6.5)	(6.4)
Periodic-open	12.3**	10.4*
	(6.0)	(5.8)
Mean, daily-open products	28.0	27.1
Observations	4,163	4,153
Bank fixed effects	Yes	—
Bank × risk rating fixed effects	—	Yes
Standard errors	Parent bank	Parent bank

Notes: The sample is the products of Table 7. Smaller local banks are at or below the median FY2021 size of the banks in that sample, larger local banks are the local banks above it, and every national bank is above it. In Panel A, the fall below the usual return is the mean monthly return in November and December 2022 minus the mean in the other months from April 2022 to June 2023. It averages products within a bank and then banks. Share reported is the reported fall divided by the implied fall. Returns are constructed as in Figure 5. In Panel B, the dependent variable is the reported minus implied return in November and December 2022 minus the same difference in the other months, in basis points a month. Daily-open products are the omitted group. Each bank receives equal total weight. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance.

Table 9
Bank size and the yield and risk of bonds held

Dependent variable	Coupon (bp)	County-level issuer (pp)	Issuer below AAA (pp)	Issuer credit event (pp)	Coupon after June 2024 (bp)
	(1)	(2)	(3)	(4)	(5)
Panel A. LGFV bonds held					
Log bank size × 𝟙{Rest of home province}	−34.293***	−13.241*	−4.093**	−1.686	−39.405***
	(11.030)	(6.986)	(1.954)	(1.601)	(13.656)
Log bank size × 𝟙{Other provinces}	−76.785***	−6.993***	−3.704***	−1.219**	−29.276***
	(22.493)	(2.566)	(1.367)	(0.549)	(6.992)
Observations	1,674	1,606	1,562	1,652	1,144
Panel B. Other banks' capital bonds held					
Dependent variable	Issued by city or rural banks (pp)			Coupon (bp)	
Log bank size	−19.218***			−24.668***	
	(3.125)			(3.637)	
Observations	196			196	
Panel C. Other banks' certificates of deposit held					
Dependent variable	Out-of-province certificates of deposit issued by city or rural banks (pp)				
Log bank size	−19.693***				
	(7.095)				
Observations	199				
Comparison cell fixed effects	Yes	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank

Notes: Each entry is a coefficient on log bank size among local banks headquartered in the same city. Log bank size is log FY2021 interest-bearing assets in RMB trillion. In Panel A, a comparison cell holds the banks that hold LGFV bonds of the same issuer city in the same quarter with the same branch presence there. The outcome is the value-weighted mean over the bank's bonds in the cell, and banks receive equal total weight within each location. Coupon is net of the mean coupon of held LGFV bonds of the same issue year and maturity group. Credit event equals 100 if the issuer has a default, rating warning, or guarantor event between 2022Q4 and 2025Q4. Panel B uses other banks' tier-2 and perpetual bonds within the same home city and quarter, with coupons net of issue-year and instrument-type means. Panel C uses other provinces' negotiable certificates of deposit within the same home city, quarter, and maturity group, and the outcome is the percentage issued by city or rural banks (mean 37%). Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2022Q1–2023Q2, and 2024Q3–2025Q2 in column 5 of Panel A.

Table 10
Own-city LGFV bonds and local presence

Dependent variable	Own-city branches (log)	Own-city LGFV bonds (pp of portfolio)	Own-city county-level LGFV bonds (pp of portfolio)
	(1)	(2)	(3)
Log bank size	0.217**	−0.006	−1.934***
	(0.094)	(1.544)	(0.627)
Log bank size, own-city branches held fixed	—	0.063	−1.804***
		(1.533)	(0.540)
Mean of dependent variable	5.462	9.829	6.050
Observations	204	202	202
Home city × quarter fixed effects	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank

Notes: Each entry is the coefficient on log bank size from a separate regression that compares local banks headquartered in the same city in the same quarter. The unit of observation is a bank-quarter. Column 1 is log one plus the number of the bank's branches in its headquarters city, from a bank-branch registry. Column 2 is the bank's LGFV bonds issued in its headquarters city as a percentage of its disclosed bonds and private credit, and column 3 is the same percentage for bonds of county-level issuers. The second row adds log one plus the number of own-city branches as a control. The sample comprises local banks headquartered in cities with at least two sample banks, and each bank receives equal weight. Log bank size is log FY2021 interest-bearing assets in RMB trillion. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2022Q1–2023Q2.

Table 11
Home-province LGFV holdings after the program

Specification	Baseline	Own-city split	Net buying	New purchases	City-risk interaction	Pre-period trend
Dependent variable	LGFV holdings / bond book (pp)	LGFV holdings / bond book (pp)	Net holdings change / current book (pp)	Log(1 + new purchases)	Log(1 + new purchases)	LGFV holdings / bond book (pp)
	(1)	(2)	(3)	(4)	(5)	(6)
$\mathbb{1}\{\text{Home buyer}\} \times \mathbb{1}\{\text{Post}\}$	-0.453^{**}	—	-0.127^{**}	-0.158^{**}	0.203^{*}	—
	(0.193)		(0.053)	(0.079)	(0.117)	
$\mathbb{1}\{\text{Own city}\} \times \mathbb{1}\{\text{Post}\}$	—	-1.936^{***}	—	—	—	—
		(0.626)				
$\mathbb{1}\{\text{Rest of home province}\} \times \mathbb{1}\{\text{Post}\}$	—	-0.242	—	—	—	—
		(0.179)				
$\mathbb{1}\{\text{Home buyer}\} \times \mathbb{1}\{\text{Post}\} \times$ city risk	—	—	—	—	-0.149^{***}	—
					(0.051)	
$\mathbb{1}\{\text{Home buyer}\} \times$ trend	—	—	—	—	—	0.003
						(0.051)
Mean of dependent variable	0.784	0.784	-0.052	0.937	0.945	0.816
Observations	73,089	73,089	73,089	73,127	72,196	32,540
Bank and city controls	*Absorbed*	*Absorbed*	*Absorbed*	*Absorbed*	*Absorbed*	*Absorbed*
Buyer × issuer city fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Buyer × quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Issuer city × quarter fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors	Bank city	Bank city	Bank city	Bank city	Bank city	Bank city

Notes: Estimates use equation (8). The unit of observation is a bank, LGFV issuer city, and quarter. A bank that holds a city at some point has zero holdings in its quarters without positions there (73,635 cells in 248 cities). Home equals one when a local bank invests in a city of its own province, and Post equals one from 2023Q4. Holdings are the bank's LGFV holdings in the city as a percentage of its total WMP bond holdings in the quarter. Net holdings change includes entries and exits and is winsorized at the 1st and 99th percentiles, and new purchases enter as $\log(1 + \text{RMB million})$. Column 2 separates the bank's own city from the rest of its home province. Column 5 interacts Home × Post with the city's pre-program spread of one-to-five-year LGFV bonds in percentage points, so its Home × Post coefficient applies at a zero spread. Column 6 replaces Post with a linear quarterly trend over 2022Q1–2023Q3. The mean holdings share in home cells is 3.91 before 2023Q4 and 3.43 afterward. Parentheses contain bank-city clustered errors. Stars indicate 10%, 5%, and 1% significance.

Table 12
Home LGFV bonds and private credit by bank type and size

	National banks		Local banks	
Asset category	Pre	Post	Pre	Post
Panel A. Composition of included debt holdings, percent				
Home-province LGFV bonds	0.4	0.3	24.7	17.4
Home-province private credit	2.1	1.9	6.1	13.9
Private credit from other provinces	22.0	26.6	3.2	5.7
Product-quarters	34,595	59,691	36,194	57,516
Panel B. Share of the sector's holdings in local banks' products, percent				
All private credit			17.3	31.3
Panel C. Local banks present in both periods, by pre-program size tercile				
Home-province private credit, smallest tercile			0.72	0.31
Home-province private credit, middle tercile			3.45	4.74
Home-province private credit, largest tercile			7.45	18.26
Home-province LGFV bonds, smallest tercile			25.1	20.8
Home-province LGFV bonds, middle tercile			37.7	25.5
Home-province LGFV bonds, largest tercile			20.7	15.0

Notes: PC denotes private credit, which includes trust loans, leasing, and other identified private credit. The sample is product-quarters with positive included debt holdings over 2022Q1–2025Q3. Pre ends in 2023Q3, and holdings are weighted by value. Panel A reports class shares of each group's included debt holdings, and Panel B reports the fraction of each class's holdings across all sample WMPs that local banks' products hold. Guangfa Bank is a nationwide joint-stock bank, so it and its subsidiary count as national. Panel C keeps local banks observed in both periods, in fixed terciles of FY2021 interest-bearing assets.

Table 13
Bank size groups and private-credit allocation

Dependent variable	Home PC share (pp) (1)	Total PC share (pp) (2)
𝟙{Middle tercile} × 𝟙{Post}	1.536	1.355
	(1.404)	(1.752)
𝟙{Largest tercile} × 𝟙{Post}	8.212***	10.464***
	(2.906)	(3.376)
Log product assets	−1.952**	−2.986***
	(0.862)	(1.091)
𝟙{Restricted redemption}	5.778*	10.572***
	(3.019)	(3.749)
Product risk rating	3.842	4.604
	(2.914)	(3.846)
Bank nonperforming-loan ratio (%)	−1.448	−3.888
	(5.666)	(6.473)
Bank cost–income ratio (%)	−0.891	−1.222*
	(0.581)	(0.725)
Bank net interest margin (%)	9.202	10.547
	(5.931)	(6.540)
Home-city GDP (RMB trillion)	3.357	20.673
	(16.132)	(19.721)
Home-city GDP growth (%)	−1.041*	−1.234*
	(0.611)	(0.669)
Home-city LGFV debt/GDP	24.427	28.158
	(22.700)	(22.963)
Home-city income (RMB 10,000)	17.075*	22.989*
	(10.240)	(11.976)
Log bank assets	−10.152	−25.650
	(15.857)	(19.351)
Observations	85,181	85,181
Bank fixed effects	Yes	Yes
Quarter fixed effects	Yes	Yes
Standard errors	Parent bank	Parent bank

Notes: PC denotes private credit. Each column is a product-quarter regression for local banks. Private-credit shares are percentages of included debt holdings, not total WMP assets. The omitted group is the smallest FY2021 bank-asset tercile, and the groups follow Table 12. Post begins in 2023Q4. Bank effects absorb fixed group indicators, and quarter effects absorb Post. Regressions use included debt value as weights, so the estimates describe the allocation of investment value. The displayed product, bank, and city controls include contemporaneous log bank assets. Parentheses contain bank-clustered standard errors. Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2022Q1–2025Q3.

Table 14
Bank size and private credit substitution

Dependent variable	Panel A. Refinancing intensity		Panel B. Post-program	
	Home PC share (pp)	Non-home PC share (pp)	Home PC share (pp)	Non-home PC share (pp)
	(1)	(2)	(3)	(4)
Log bank size × refinancing intensity	24.205***	14.531***	—	—
	(8.531)	(4.161)		
Initial home overweight × refinancing intensity	5.021	13.152*	—	—
	(17.269)	(6.792)		
Log bank size × 𝟙{Post}	—	—	5.853***	2.329***
			(1.338)	(0.752)
Initial home overweight × 𝟙{Post}	—	—	7.157	4.363**
			(5.827)	(1.984)
Log product assets	−2.172**	−1.122***	−2.151**	−1.115***
	(0.872)	(0.418)	(0.870)	(0.417)
𝟙{Restricted redemption}	5.538*	4.252**	5.456*	4.218**
	(2.860)	(1.647)	(2.877)	(1.638)
Product risk rating	3.442	0.907	3.563	0.961
	(2.636)	(1.372)	(2.634)	(1.375)
Log bank assets	−30.179	−23.281**	−25.758	−22.166**
	(21.762)	(9.606)	(20.394)	(9.513)
Bank nonperforming-loan ratio	−2.621	−4.356*	−0.336	−3.246
	(4.782)	(2.225)	(4.101)	(2.211)
Bank cost–income ratio	−0.417	−0.407	−0.370	−0.376
	(0.582)	(0.306)	(0.510)	(0.309)
Bank net interest margin	16.147***	1.141	13.616***	0.567
	(5.003)	(2.505)	(4.406)	(2.474)
Home-city GDP (RMB trillion)	51.899***	15.888**	37.848**	11.336
	(17.370)	(7.969)	(16.216)	(7.128)
Home-city GDP growth (%)	−3.070***	−0.469	−1.711*	0.181
	(1.150)	(0.388)	(0.960)	(0.436)
Home-city LGFV debt/GDP	3.490	13.626*	−12.207	4.744
	(23.197)	(7.933)	(20.837)	(6.382)
Home-city income (RMB 10,000)	−43.525*	−17.132**	−21.873	−11.227
	(26.121)	(7.902)	(22.859)	(7.048)
Observations	83,027	83,027	83,027	83,027
Institution fixed effects	Yes	Yes	Yes	Yes
Home province × quarter fixed effects	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Notes: PC denotes private credit. Each column is one regression for local-bank WMPs. Size is log FY2021 interest-bearing assets. Panel A interacts size and initial home overweight with cumulative refinancing divided by the home province's 2020 LGFV debt, and Panel B interacts them with Post. Institution effects absorb their levels, and province-by-period effects absorb intensity or Post. All eleven controls are displayed, and regressions use investment-value weights. Errors cluster by parent bank. Stars indicate 10%, 5%, and 1% significance. The sample covers 2022Q1–2025Q3.

Table 15
Private credit in new products, by bank size

First eligible portfolio	Smallest tercile	Middle tercile	Largest tercile
New products	253	1,877	5,248
Products holding home private credit	2	108	1,118
Products holding any private credit	2	134	1,586
Home private credit share (%)	0.78	6.31	27.46
Non-home private credit share (%)	0.00	2.73	14.74
Total private credit share (%)	0.78	9.04	42.20
Home LGFV bond share (%)	29.36	27.94	11.11
Home private credit share, first disclosure within one quarter (%)	0.59	5.75	27.29
Products in that restriction	234	1,737	4,611

Notes: PC denotes private credit. This table compares the first observed portfolios of products launched after the program by local banks in each size tercile. Launches in 2023Q4–2025Q3 are identified from income-start dates, when products begin accruing returns, and not from first disclosure. Each product contributes its first eligible portfolio, which is the first disclosed portfolio with positive included debt holdings. Shares are holding-value weighted percentages of included debt holdings, and counts are unweighted. Size terciles use fixed FY2021 parent-bank interest-bearing assets and the common local-bank cutoffs. Products of a bank's wealth-management subsidiary are assigned to the parent bank. The final two rows restrict to products whose first portfolio is disclosed in the launch quarter or the next quarter.

Table 16
Bank size and WMP business growth after the program

Dependent variable	Log WMP balance		
Specification	Baseline	Province–time	Size terciles
	(1)	(2)	(3)
Log bank size × 𝟙{Post}	0.152***	0.144***	—
	(0.039)	(0.052)	
𝟙{Middle size tercile} × 𝟙{Post}	—	—	0.184*
			(0.093)
𝟙{Largest size tercile} × 𝟙{Post}	—	—	0.335***
			(0.088)
Wild-bootstrap p	< 0.001	0.008	< 0.001
Observations	571	494	571
Bank fixed effects	Yes	Yes	Yes
Half-year fixed effects	Yes	No	Yes
Home province × half-year fixed effects	No	Yes	No
Standard errors	Parent bank	Parent bank	Parent bank

Notes: Each column is one bank-half-year regression of log observed WMP balances after source corrections. Balances combine the bank and its subsidiary. Size is log FY2021 interest-bearing assets in RMB trillion, and Post begins at year-end 2023. Column 2 replaces half-year effects with home-province-by-half-year effects, and column 3 uses fixed size terciles, with the smallest omitted. Parentheses contain bank-clustered errors. Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2022H1–2025H2.

Table 17

Initial home LGFV bond share, private-credit expansion, and WMP performance

Dependent variable	Panel A. Returns		Panel B. New-product balances	
	Monthly WMP return (bp)		Log new-product balances	
	Size only	Add investments	Size only	Add investments
	(1)	(2)	(3)	(4)
Initial home LGFV bond share × 𝟙{Post}		−10.182**		−1.662***
		(4.186)		(0.528)
Home private-credit expansion × 𝟙{Post}		11.432**		1.363**
		(4.936)		(0.630)
Log FY2021 bank size × 𝟙{Post}	1.808**	0.900	0.292***	0.138
	(0.744)	(0.735)	(0.091)	(0.109)
Initial home private credit share × 𝟙{Post}		3.980		1.547**
		(5.231)		(0.711)
Lagged log bank assets	−1.274	−0.217	−0.390	−0.213
	(6.173)	(5.972)	(0.717)	(0.735)
Lagged nonperforming-loan ratio	−3.147	−3.269	−0.340	−0.504*
	(3.112)	(2.985)	(0.305)	(0.285)
Lagged cost–income ratio	−0.350	−0.383	0.007	−0.010
	(0.323)	(0.325)	(0.030)	(0.032)
Lagged net interest margin	2.816	2.177	0.309	0.252
	(3.359)	(3.293)	(0.290)	(0.289)
Lagged city GDP (RMB trillion)	13.398**	14.211**	1.175	1.697
	(6.205)	(6.319)	(1.071)	(1.025)
Lagged city GDP growth	−0.757**	−0.695*	−0.070	−0.080
	(0.364)	(0.391)	(0.051)	(0.052)
Lagged city LGFV debt/GDP	14.810	15.145	2.979***	3.033***
	(12.724)	(12.202)	(1.121)	(1.035)
Lagged urban income (RMB 10,000)	12.129***	9.822***	1.112**	0.894*
	(4.437)	(3.681)	(0.536)	(0.488)
Observations	1,851	1,851	641	641
Bank fixed effects	Yes	Yes	Yes	Yes
Month fixed effects	Yes	Yes	No	No
Launch-quarter fixed effects	No	No	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

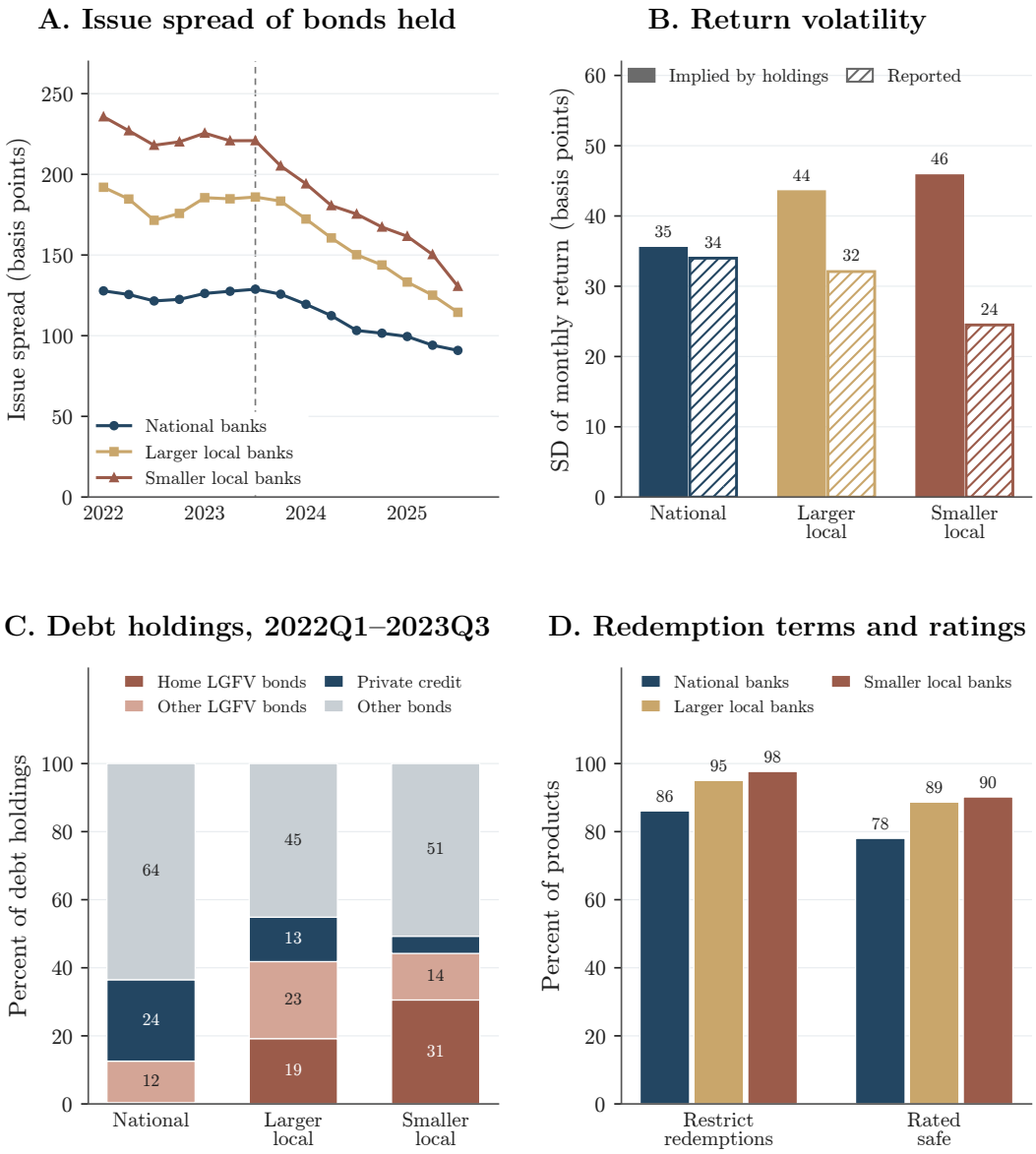
Notes: Each column is a separate regression. Panel A estimates equation (10) for bank-average monthly log returns over July 2022–September 2025, winsorized at the 1st and 99th percentiles. Panel B estimates equation (D.5) over 2022Q2–2025Q2. Its outcome is the log total balance of the products a local bank launched in a quarter, measured at the end of the following quarter. Both omit 2023Q3, and Post begins in 2023Q4. Columns 1 and 3 include bank size and the lagged controls. Columns 2 and 4 add the initial home LGFV bond and private-credit shares from 2022Q1–2023Q2 holdings and the change in the home private-credit share to 2023Q4–2025Q3. Size is log FY2021 interest-bearing assets. Bank effects absorb the levels, and time effects absorb Post. Parentheses contain bank-clustered errors. Stars indicate 10%, 5%, and 1% significance. Internet Appendix D.4 gives the construction.

Table 18
Who funded the borrowers that reliant local banks left

Dependent variable	Holdings by the WMPs of (inverse hyperbolic sine)				Coupon of new bonds (pp)
	Local banks without subsidiary	Local banks with subsidiary	National banks	All banks	
	(1)	(2)	(3)	(4)	(5)
Panel A. WMP holdings of the borrower's LGFV bonds, by holder					
High-reliance banks' share × 𝟙{Post}	−1.894***	0.735***	1.833***	−0.163	−1.166***
	(0.231)	(0.184)	(0.198)	(0.202)	(0.091)
Low-reliance banks' share × 𝟙{Post}	−0.904**	1.150***	1.190***	0.822***	−0.624***
	(0.352)	(0.258)	(0.276)	(0.318)	(0.160)
High-reliance banks' share × 𝟙{Late}	−0.899***	0.837***	1.065***	−0.475**	−0.141***
	(0.208)	(0.176)	(0.184)	(0.186)	(0.046)
Low-reliance banks' share × 𝟙{Late}	−0.764**	0.360	0.718***	−0.218	−0.019
	(0.304)	(0.228)	(0.261)	(0.287)	(0.059)
High = low reliance, Post (p)	0.005	0.054	0.010	0.002	0.001
Observations	17,271	17,271	17,271	17,271	12,541
Panel B. WMP holdings of the borrower's private credit, by holder					
High-reliance banks' share × 𝟙{Post}	−3.962***	0.809**	2.281***	−0.572	—
	(0.571)	(0.368)	(0.397)	(0.672)	
Low-reliance banks' share × 𝟙{Post}	−1.610***	0.286	1.140***	0.091	—
	(0.582)	(0.419)	(0.407)	(0.720)	
High-reliance banks' share × 𝟙{Late}	−1.739***	−0.229	0.360	−0.551	—
	(0.439)	(0.318)	(0.351)	(0.466)	
Low-reliance banks' share × 𝟙{Late}	−1.334*	0.580*	1.992***	1.457**	—
	(0.720)	(0.328)	(0.514)	(0.663)	
High = low reliance, Post (p)	0.004	0.336	0.038	0.495	—
Observations	8,244	8,244	8,244	8,244	—
Borrower fixed effects	Yes	Yes	Yes	Yes	Yes
Borrower province × quarter fixed effects	Yes	Yes	Yes	Yes	No
Borrower province × month fixed effects	No	No	No	No	Yes
Maturity group fixed effects	No	No	No	No	Yes
Standard errors	Borrower	Borrower	Borrower	Borrower	Borrower

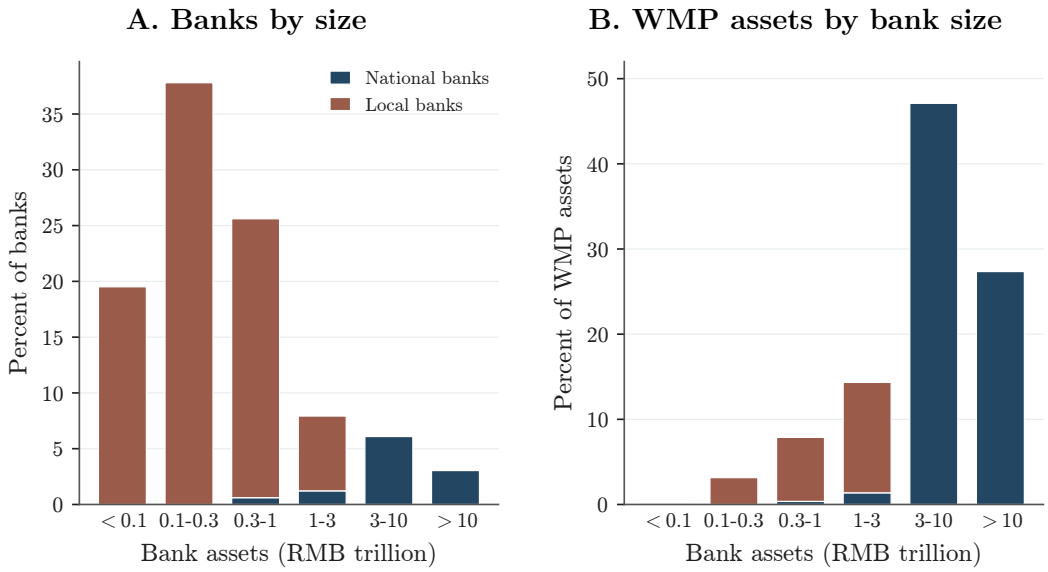
Notes: The unit of observation in columns 1–4 is a borrower-quarter. The outcome is the inverse hyperbolic sine of the borrower's bonds (Panel A) or private credit (Panel B) held by the stated group's WMPs, in RMB 10,000. High- and low-reliance banks' shares are the fractions of the borrower's WMP holdings held in 2022Q4–2023Q1 by local banks without a wealth-management subsidiary whose initial home LGFV bond share is above or below the median. Post begins in 2023Q4 and Late in 2024Q3, after the June 2024 guidance. Column 5 uses new LGFV bonds issued by the same borrowers and controls for log maturity. Parentheses contain borrower-clustered errors. Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2023Q2–2025Q2.

Figure 1
Holdings, return volatility, and product terms by bank group



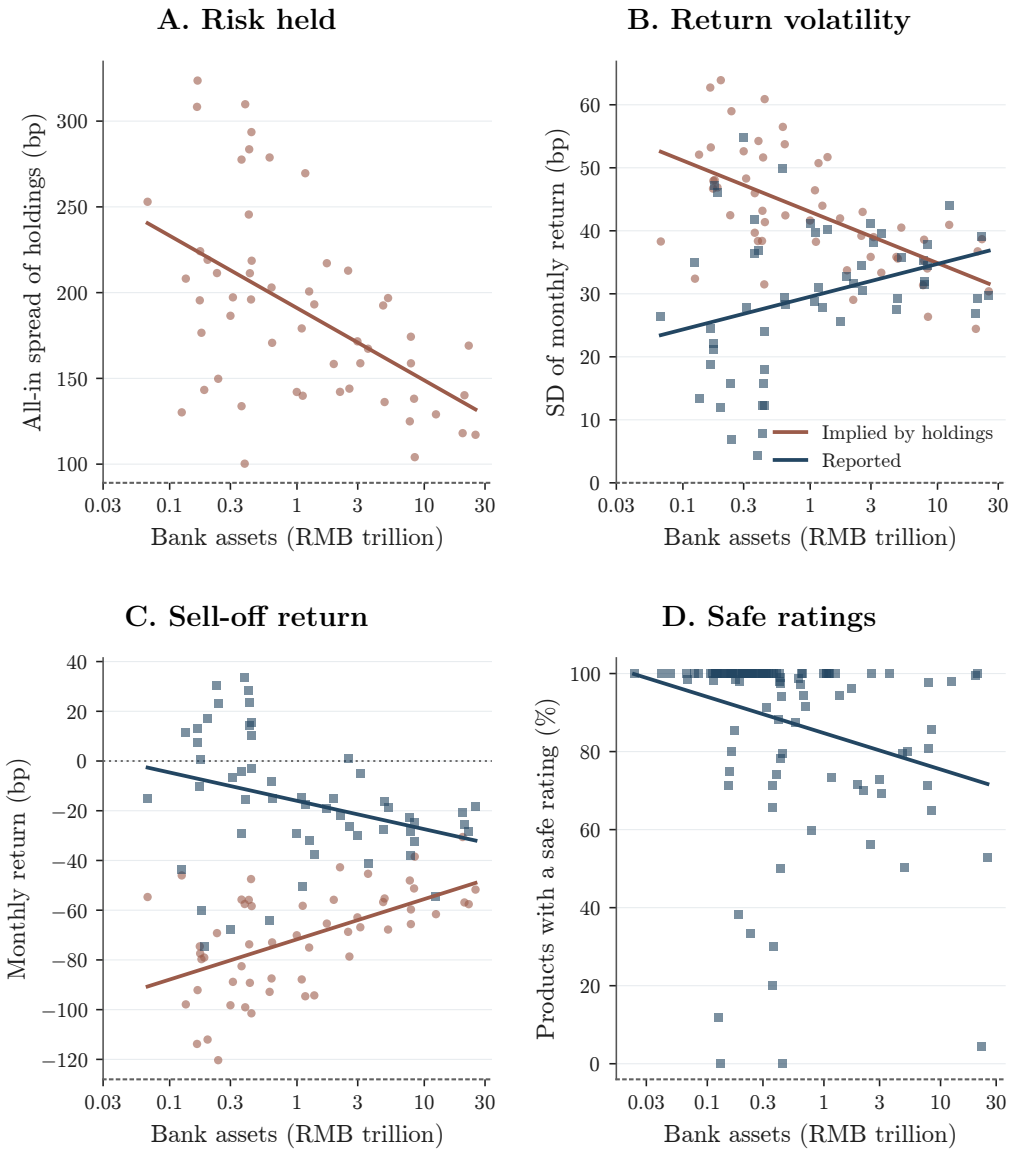
Notes: Smaller local banks are the banks at or below the median FY2021 parent-bank assets of the sample of Table 7, which is RMB 0.44 trillion. Larger local banks are the local banks above it. Panel A shows the value-weighted issue spread of the bonds held by each group's WMPs by quarter. The issue spread is the coupon minus the same-maturity Treasury yield at issue. Unlike the all-in spread of Table 3, it covers bonds only and leaves out central-government, central-bank, and policy-bank bonds. The dashed line marks the 2023Q3 announcement of the debt-resolution program. Panel B shows the implied and the reported standard deviation of monthly returns between April 2022 and June 2023. It uses the products of Table 7 and averages products within a bank and then banks. Implied returns mark every disclosed asset each month to a market series, and reported returns are monthly changes in log distribution-adjusted NAV. Each series is winsorized at the 1st and 99th percentiles of all its monthly values. Panel C shows the composition by value of the bonds and private credit with known borrower locations that each group's WMPs held over 2022Q1–2023Q3, as in Section 4.1. Panel D uses the fixed-income products that reported a monthly return between April 2022 and June 2023 and averages over banks. It shows the percentage of a bank's products that are closed-end or periodic-open, which restrict redemptions, and the percentage with a safe rating of 1 or 2 on the five-level product risk scale. Internet Appendix D.3 gives the construction of implied returns.

Figure 2
Bank size and WMP assets in the sample



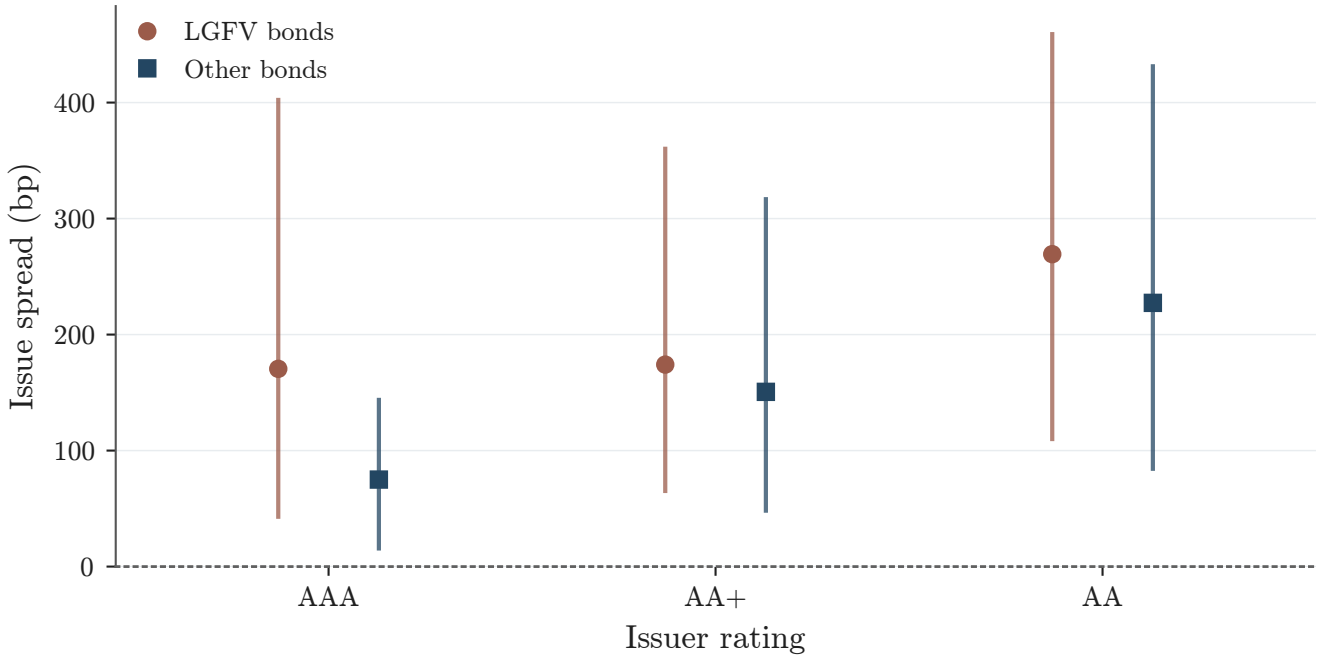
Notes: Both panels sort banks by FY2021 parent-bank assets and distinguish national from local banks. Panel A shows the percentage of banks in each asset range, and Panel B shows the percentage of sample WMP assets. A bank's WMP assets are the mean quarterly net assets of its products over 2022Q1–2023Q3, as implied by their disclosed holdings. Banks without FY2021 balance sheets are left out.

Figure 3
Risk held and risk reported, by bank size



Notes: Each panel plots one point per national or local bank against FY2021 bank assets on a log scale, with linear fits. Panels A to C use the banks with at least five fixed-income products in the sample of Table 7. Panel D uses every bank with fixed-income products that reported a monthly return between April 2022 and June 2023, which is the sample of column 6 of Panel B of Table 7. Panel A shows the all-in spread of the bank's holdings, defined in Table 3. Panel B shows the standard deviation of monthly returns between April 2022 and June 2023 averaged over the bank's products, and Panel C shows the mean monthly return in November and December 2022. Both panels show the value implied by holdings and the value reported. Panel D shows the percentage of the bank's fixed-income products with a safe rating of 1 or 2 on the five-level product risk scale. In bank-level fits to the plotted points, the all-in spread falls by 18.3 basis points per log point of assets (standard error 3.7), and the safe-rating share falls by 4.0 percentage points (standard error 1.6). Smaller banks are those at or below the median size. About four in five of them give every fixed-income product a safe rating, against about one in three larger banks. Implied returns mark every disclosed asset each month to a market series and add the product's mean difference between reported returns and price changes in the other months, which reflects interest income net of fees. Reported returns are monthly changes in log distribution-adjusted NAV, and each series is winsorized at the 1st and 99th percentiles of all its monthly values. Internet Appendix D.3 gives the construction.

Figure 4
Issuer credit ratings and issue spreads



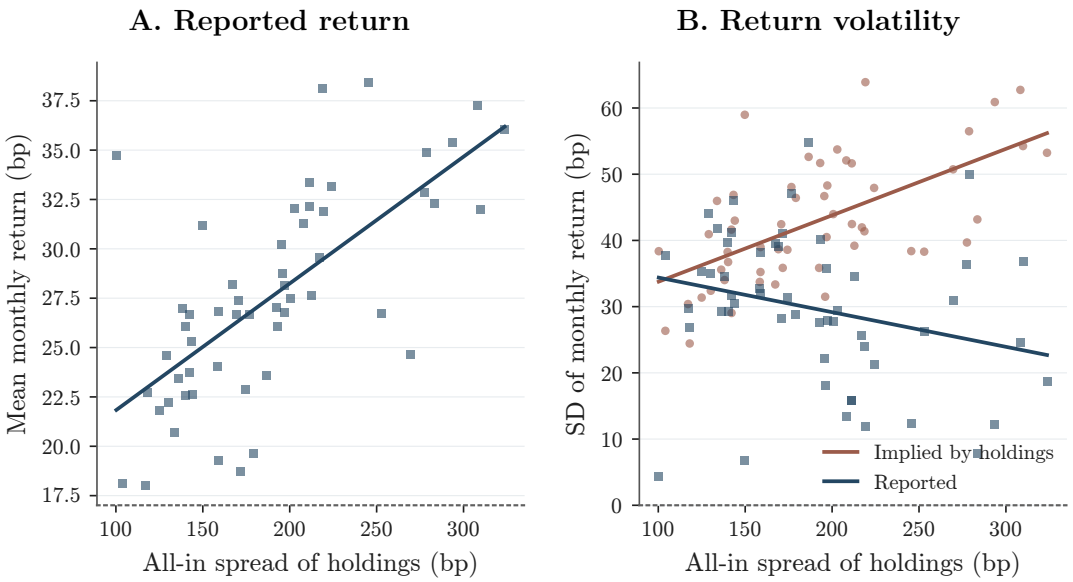
Notes: The figure shows the mean issue spread of LGFV bonds and of other bonds issued between 2021Q1 and 2023Q2 by issuer rating, together with the range from the 10th to the 90th percentile. The issue spread is the coupon minus the Treasury yield of the same maturity at issue. The sample comprises medium-term notes, short-term financing bills, and corporate bonds with an issuer rating of AAA, AA+, or AA. Internet Appendix D.3 gives the construction.

Figure 5
Returns implied by holdings, returns reported, and investor flows



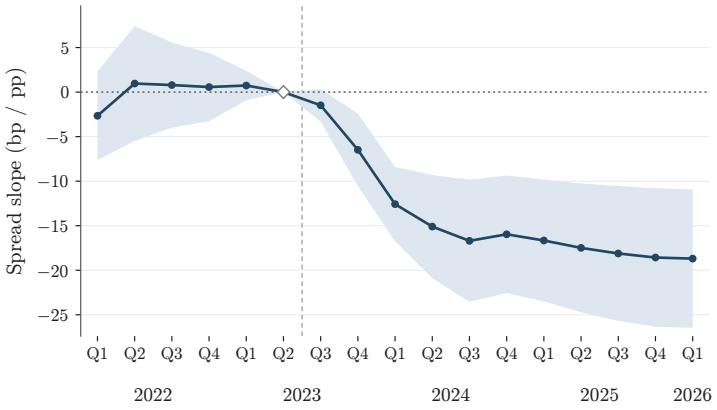
Notes: The panels plot the products in the sample of Table 7 between April 2022 and June 2023, separately for smaller and larger banks. Smaller banks are those at or below the median size. They are all local banks, and larger banks include all national banks. The panels average products within a bank and then banks. Panels A and B show monthly returns. Panels C and D show quarterly net flows for 2022Q2 to 2023Q2. Net flow is asset growth net of investment gains as a percentage of beginning assets, as in Table 1. Each quarterly flow is plotted at the middle month of its quarter and averages a bank's products weighted by beginning assets. Implied returns mark every disclosed asset each month to a market series. Each product's implied return adds to the month's price change the product's mean difference between reported returns and price changes in the other months. This difference reflects interest income net of fees, so the two series agree on average outside November and December 2022. Reported returns are monthly changes in log distribution-adjusted NAV. Reported and implied returns are each winsorized at the 1st and 99th percentiles of all their monthly values. Internet Appendix D.3 gives the construction.

Figure 6
What savers see and the risk held



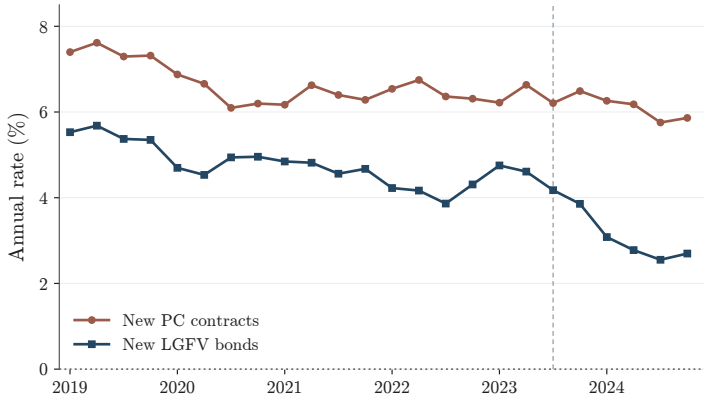
Notes: Each panel plots one point per national or local bank with at least five fixed-income products in the sample of Table 7 against the mean all-in spread of its products' holdings, with linear fits. The all-in spread is defined in Table 3. Panel A shows the mean reported monthly return between April 2022 and June 2023, and Panel B shows the standard deviation of monthly returns as implied by holdings and as reported. Per 100 basis points of all-in spread, the bank-level slope of the mean return is 6.4 basis points a month (standard error 1.0). In November and December 2022 the slope is −19 for the implied return and +17 for the reported return. Internet Appendix D.3 gives the construction.

Figure 7
Provincial LGFV spreads by special refinancing relative to GDP



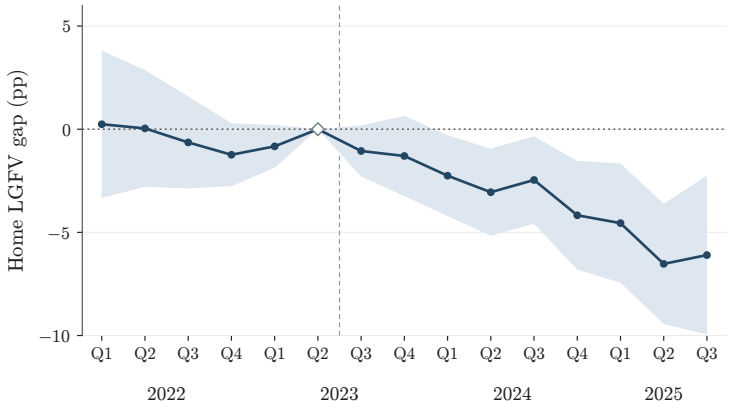
Notes: This event study plots quarter-specific coefficients β_k. Provincial LGFV secondary-market spreads in basis points are regressed on special refinancing relative to GDP interacted with quarter indicators. The regression includes province and quarter fixed effects, and 2023Q2 is the omitted quarter. Special refinancing relative to GDP is cumulative special-refinancing issuance at the end of the series as a percentage of 2020 provincial GDP. Coefficients are basis points per percentage point of special refinancing relative to GDP, without standardization. Spreads and special refinancing are winsorized at the 1st and 99th percentiles, and the sample covers 31 provinces over 2022Q1–2026Q1. The shaded bands are pointwise 95% confidence intervals based on standard errors clustered by province, and the dashed line separates 2023Q2 and 2023Q3.

Figure 8
Contractual rates on new LGFV financing



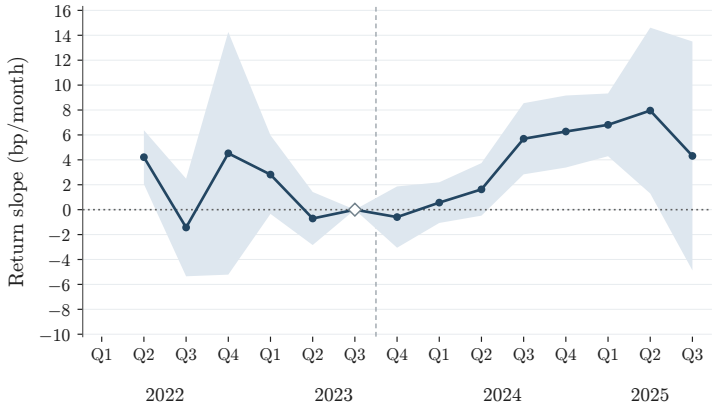
Notes: PC denotes private credit. The series are annual contractual yields in percent on new PC contracts and new LGFV bond issues, by quarter over 2019Q1–2024Q4. Each series uses financing reported by LGFV issuers in the deduplicated financing records and weights rate-visible observations by positive known original financing amounts. Unknown amounts are not imputed. Figure D1 compares the same borrowers. The dashed vertical line marks the 2023Q3 announcement. Incomplete 2025 observations are excluded.

Figure 9
Home-province gap in local banks' LGFV holdings by quarter



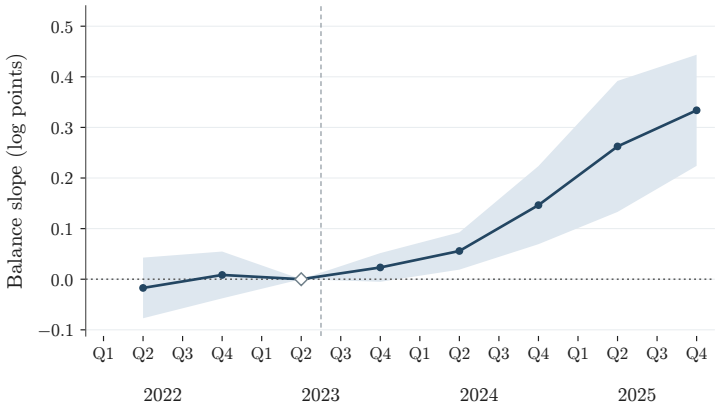
Notes: This event study plots quarter-specific home-bond allocation coefficients η_k. The outcome is a buying institution's LGFV holdings in a province as a percentage of its total bond holdings, winsorized at the 1st and 99th percentiles. It is regressed on Home interacted with quarter indicators. Buyer-by-province, buyer-by-quarter, and destination-province-by-quarter fixed effects enter, and 2023Q2 is omitted. The sample covers 2022Q1–2025Q3. The observation is a buyer, province, and quarter, so the coefficients measure changes in the home gap for an entire province rather than a single city as in Table 11. Home equals one for a local bank's own province, so the comparison includes national banks and local banks' holdings elsewhere. The shaded bands are pointwise 95% confidence intervals based on standard errors clustered by bank city, and the dashed line separates 2023Q2 and 2023Q3.

Figure 10
Bank size and product returns by quarter



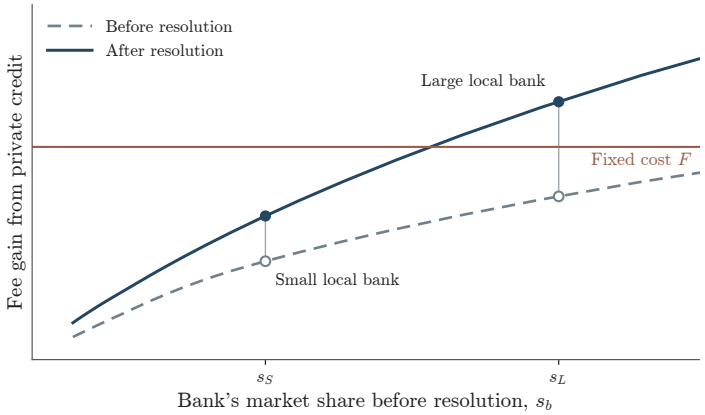
Notes: This event study replaces both refinancing interactions in equation (D.6) with log bank size interacted with quarter indicators. The outcome is monthly log adjusted-NAV return (bp) for local-bank products. Each coefficient measures the change in the size–return slope relative to 2023Q3, per log point of FY2021 parent-bank interest-bearing assets in RMB trillion, without standardization. The regression includes the eleven time-varying controls, institution effects, and bank-home-province-by-month effects, but no initial-home-overweight interactions. Returns are winsorized at the 1st and 99th percentiles of all monthly returns. The sample is 2022Q2–2025Q3 (212,016 NAV-series-months). Shading shows pointwise 95% confidence intervals based on parent-bank clustered errors. The hollow diamond marks the normalized reference quarter, not an estimated zero. The vertical line marks implementation in 2023Q4.

Figure 11
Bank size and total WMP balances



Notes: This event study replaces the size–Post interaction in equation (9) with log bank size interacted with half-year reporting-date indicators. The quarterly axis locates actual midyear and year-end reports at Q2 and Q4. No Q1 or Q3 balances are imputed. The log-balance regression includes bank and half-year fixed effects and no additional controls, for local banks over 2022H1–2025H2 (571 bank-half-years). Each coefficient measures the change in the size–balance slope relative to the June 2023 report (2023Q2), per log point of FY2021 interest-bearing assets in RMB trillion, without standardization. Balances combine a bank's own products and those of its wealth-management subsidiary. Shading shows pointwise 95% confidence intervals based on bank-clustered errors. The hollow diamond marks the reference. The vertical line separates 2023Q2 and Q3. Source: Wind institution-level WMP statistics, corrected with bank disclosures.

Figure 12
The scale threshold for private credit



Notes: This figure illustrates the entry condition in equation (17), using the parameters and equilibrium portfolios in Internet Appendix B.7. The curves vary the reputation of a hypothetical fourth bank facing those portfolios. They plot its optimized gain from private credit net of the expected reputational loss against its bond-only market share before resolution. The dashed curve is the gain before resolution, and the solid curve is the gain afterward. Entry is worthwhile when the gain covers F. The circles evaluate these curves at the small and large local banks' pre-resolution market shares. After resolution, only the gain at the large bank's share exceeds F. Table B1 reports the exact gains for the three equilibrium banks.

Internet Appendix

Reaching for Yield in the Dark

Smaller Banks, Local Government Debt, and Household Savings in China

The appendix contains supporting exhibits (Internet Appendix A), the model solution and proofs (Internet Appendices B and C), and specifications and supplementary results (Internet Appendix D).

Internet Appendix A
Supporting Evidence

A.1. Data and portfolio denominators

Table A1 defines the variables, Table A2 reports sample coverage, and Table A3 summarizes products, holdings, and issuers. Figure A1 compares WMPs with other Chinese asset-management products and shows their 2025 asset mix.

Figure A2 shows the estimated scale of the WMP–LGFV bond link, and Internet Appendix D.2 explains these industry estimates and reconciles them with the portfolio shares in Section 4.1.

A.2. Supporting estimates

Table A4 reports the home allocations by redemption terms discussed in Section 4.1, and Table A5 relates relative performance to subsequent investor flows (Section 3). Table A6 compares LGFV issuance yields with those of other issuers. Across 20,172 unique RMB fixed-rate bonds issued at par in 2022Q1–2023Q3, LGFV yields exceed other nonfinancial and financial issuers' yields by 122 and 149 basis points. Controlling for issue month, maturity, issue size, and call provisions leaves gaps of 91 and 108 basis points, and adding ratings gives 88 and 110.

Table A1
Variable definitions

Variable	Definition
$\mathbb{1}\{\text{Local bank}\}_b$	One for banks other than state-owned and nationwide joint-stock banks.
Within-category home bias$^n_{iq}$	Home holdings / all holdings within class n, minus the home province's share of nationwide class-n holdings. Requires positive class holdings (equation (3)).
Portfolio home overweight$^n_{iq}$	Home class-n holdings / all included debt holdings, minus nationwide investment in class n in the home province / nationwide included debt holdings. Includes products holding none of class n. Without n, it sums across classes.
Initial home overweight$_a$	Product-assets-weighted pre-program mean of portfolio home overweight for issuing institution a, 2022Q1–2023Q3.
Log bank size$_b$	Natural log of FY2021 parent-bank interest-bearing assets (RMB trillion), without standardization.
Initial home LGFV bond share$_b$	Home LGFV bonds / included debt holdings, holding-value weighted over 2022Q1–2023Q2, without subtracting market weights.
Home LGFV credit share$_{iq}$	Home-province LGFV bonds and private credit to home-province LGFV borrowers / disclosed holdings (all reported positions, including equity and funds), in percent.
Disclosed bonds and private credit	All disclosed bond and private credit positions, including government and policy-bank bonds, which included debt holdings exclude.
$\mathbb{1}\{\text{Restricted redemption}\}_i$	One for products without daily redemption.
Monthly return	Log return computed from month-end adjusted NAVs in consecutive months, in basis points.
Reported return volatility	Log standard deviation of the product's monthly reported returns over the twelve months before the outcome quarter.
Net flow$_{iq}$	Quarterly asset growth net of investment gains, as a percentage of beginning assets.
$\mathbb{1}\{\text{Home buyer}\}_{ac}$	Equals one when issuing institution a is a local bank or its subsidiary and invests in issuer city c of the parent bank's headquarters province.
Refinancing$^{GDP}_{pq}$	Cumulative special-refinancing issuance / 2020 provincial GDP (percentage points). Final refinancing$^{GDP}_p$ is its value at the last observed date.
Refinancing$^{debt}_{pq}$	Cumulative special-refinancing issuance / 2020 provincial LGFV debt (fraction), the refinancing intensity of the allocation and return tests.
$\mathbb{1}\{\text{Post}\}$	Equals one from 2023Q4 in quarterly data and from October 2023 in monthly data.
Log WMP balances$_{bs}$	Log outstanding WMP balance of bank b at half-year end s, combining its own and its subsidiary's products.

Notes: Holdings cover 2022Q1–2025Q3. Included debt comprises bonds and identified private credit with known borrower locations, excluding central-government, central-bank, and policy-bank obligations. Issuer credit events include defaults, rating warnings, and guarantor events, not necessarily default on the held bond. Classes n are LGFV bonds, private credit (PC), and other bonds.

Table A2
Sample coverage

Panel A. Overall coverage

Measure	Count
Source holding records	2,288,894
Holding records with a usable product identifier	2,279,252
WMP products	60,396
Product-quarter observations	253,484
Bank headquarters provinces	30
Quarters	15

Panel B. Annual coverage

Year	Source holding records	WMP products
2022	421,072	19,514
2023	579,703	26,021
2024	708,907	30,512
2025 (Q1–Q3)	579,212	29,527

Panel C. Bank categories

Bank group	WMP products	Product-quarters
National banks	27,799	131,535
Large state-owned	9,571	49,593
Nationwide joint-stock	18,228	81,942
Local banks	32,597	121,949
City and provincial commercial	23,817	91,937
Rural banks	8,780	30,012
All banks	60,396	253,484

Notes: Coverage is 2022Q1–2025Q3. A holding record is a product-asset-quarter observation. Source-record counts include 9,642 records with blank or "NA" product registration codes. Usable-record, product, and product-quarter counts exclude these codes. A parent-bank crosswalk groups wealth-management subsidiaries and alternative bank names with their parent banks. Panel B counts products once per year, so annual counts need not sum to full-period counts. Panel C uses the source bank-type codes and assigns subsidiary products to the parent bank's category. The source does not separate city from provincial commercial banks. Rural banks include rural commercial and cooperative banks. Guangfa Bank's wealth-management subsidiary is classified with nationwide joint-stock banks. Estimation samples impose the restrictions stated in each regression table.

Table A3
Descriptive statistics

Variable	Observations	Mean	SD	Median	p90
Panel A. Holding-observation characteristics					
Position weight (%)	2,286,025	7.12	12.03	4.04	11.46
Holding value (RMB million)	2,278,335	59.22	345.74	10.39	113.89
Panel B. Bond characteristics					
Issue yield (%)	1,162,002	4.09	1.19	4.00	5.58
Issue maturity (years)	1,162,491	5.09	3.06	5.00	10.00
Panel C. Product-quarter characteristics in the NAV sample					
Log product assets	153,563	19.41	1.64	19.38	21.51
Product risk rating	153,563	2.15	0.44	2.00	3.00
Restricted redemption (0/1)	153,563	0.88	0.33	1.00	1.00
Panel D. Pre-implementation issuers' FY2021 assets (RMB billion)					
LGFV issuers	1,939	42.79	48.09	28.41	88.39
Other nonfinancial issuers	2,062	134.48	548.95	35.32	243.34
Financial issuers	570	525.79	2,765.50	52.90	560.29

Notes: Panel A describes holding records and Panel B bond holding records with the characteristic available. Panel C counts each product-quarter in the return sample once. Log product assets is $\log(1 + \text{inferred assets in RMB})$, product risk is the numerical rating, and p90 is the 90th percentile. The cross-sectional SD of products' average monthly returns is 10.5 bp. Among local-bank product-quarters before 2023Q4, mean home and non-home LGFV bond shares of included debt are 0.263 (0.722 at the 90th percentile for the home share) and 0.180. Mean issue rates are 5.62% for home LGFV holdings and 4.93% for other-province holdings, and the mean within-product gap is 0.33 pp. Panel D separately describes firms issuing RMB debt in the available Wind issuance file over 2022Q1–2023Q3, not only WMP-held issues. Each issuer receives equal weight and appears once, matched by normalized exact name to reported year-end 2021 total assets. No future balance-sheet values are substituted. The LGFV roster identifies the first group, and source issuer types identify financial institutions and other enterprises. Central and local governments, policy banks, and unclassified issuers are excluded. Assets are observed for 1,939 of 1,964 LGFV issuers (98.7%), 2,062 of 2,167 other nonfinancial issuers (95.2%), and 570 of 723 financial issuers (78.8%).

Table A4
Portfolio home overweight and redemption terms

Asset class	All included debt	LGFV	PC	Other bonds
Dependent variable	Portfolio home overweight (fraction) (1)	Portfolio home overweight (fraction) (2)	Portfolio home overweight (fraction) (3)	Portfolio home overweight (fraction) (4)
$\mathbb{1}\{\text{Local bank}\}$	−0.063	0.009	−0.021	−0.050
	(0.060)	(0.035)	(0.026)	(0.034)
Log bank assets	−0.016	−0.004	0.018	−0.030**
	(0.015)	(0.010)	(0.012)	(0.012)
$\mathbb{1}\{\text{Local bank}\} \times$ restricted redemption	0.200***	0.077***	0.078**	0.045**
	(0.056)	(0.030)	(0.036)	(0.023)
Log product assets	−0.001	0.001	0.000	−0.002
	(0.004)	(0.003)	(0.002)	(0.002)
$\mathbb{1}\{\text{Restricted redemption}\}$	−0.048***	−0.014**	0.010	−0.045***
	(0.009)	(0.007)	(0.008)	(0.008)
Product risk rating	0.015	0.000	0.012	0.003
	(0.014)	(0.008)	(0.012)	(0.008)
Bank nonperforming-loan ratio (%)	−0.074	−0.005	−0.044**	−0.022
	(0.055)	(0.041)	(0.020)	(0.027)
Bank cost–income ratio (%)	−0.001	−0.001	−0.000	−0.000
	(0.001)	(0.001)	(0.001)	(0.001)
Bank net interest margin (%)	0.040	0.026	0.007	0.004
	(0.060)	(0.045)	(0.027)	(0.031)
Home-city GDP (RMB trillion)	0.003	0.023	−0.023**	0.003
	(0.025)	(0.017)	(0.009)	(0.012)
Home-city GDP growth (%)	0.013	0.019***	−0.003	−0.003
	(0.010)	(0.007)	(0.005)	(0.006)
Home-city LGFV debt/GDP	0.189***	0.217***	0.019	−0.049
	(0.069)	(0.048)	(0.036)	(0.058)
Home-city income (RMB 10,000)	−0.048**	−0.044***	0.007	−0.010
	(0.023)	(0.016)	(0.009)	(0.012)
Mean of dependent variable	0.191	0.092	0.031	0.069
Observations	169,997	169,458	168,327	169,997
Quarter fixed effects	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Notes: PC denotes private credit. Estimates use equation (4). The unit of observation is a product-quarter. The dependent variable is the product's home-province holdings of the asset class divided by all its included debt holdings, minus the home province's share of nationwide included debt holdings in the class, as a fraction. Restricted redemption equals one for products without daily redemption, so the local-bank coefficient compares local and national banks among daily-open products and the interaction gives the additional difference among other products. Periodic-open and closed-end products account for 80.2% of local banks' product assets and 47.3% of national banks'. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2022Q1–2025Q3.

Table A5
Relative performance and subsequent investor flows

Dependent variable	Net flow (% initial assets) (1)	Net flow (% initial assets) (2)	Symmetric net flow (pp) (3)	Positive-flow probability (pp) (4)
Prior-quarter return quintile (bottom quintile omitted)				
$\mathbb{1}\{\text{Return quintile} = 2\}$	1.790***	1.793***	1.461***	1.860***
	(0.605)	(0.592)	(0.515)	(0.591)
$\mathbb{1}\{\text{Return quintile} = 3\}$	2.873***	2.860***	2.851***	4.600***
	(0.789)	(0.770)	(0.565)	(0.847)
$\mathbb{1}\{\text{Return quintile} = 4\}$	5.185***	5.396***	4.833***	6.102***
	(1.152)	(1.168)	(0.780)	(1.017)
$\mathbb{1}\{\text{Return quintile} = 5\}$	12.008***	12.604***	8.785***	8.036***
	(2.671)	(2.730)	(1.527)	(1.627)
Log beginning product assets	−7.375***	−7.615***	−4.594***	−3.783***
	(0.982)	(0.976)	(0.567)	(0.333)
Product risk rating	Absorbed	Absorbed	Absorbed	Absorbed
Restricted redemption	Absorbed	Absorbed	Absorbed	Absorbed
Log bank assets	3.970***	4.332***	3.308***	3.263**
	(1.104)	(1.304)	(0.826)	(1.406)
Bank nonperforming-loan ratio (%)	−1.279	−0.522	−0.827	1.228
	(1.958)	(3.322)	(1.543)	(2.371)
Bank cost–income ratio (%)	−0.136**	−0.117	−0.126**	−0.070
	(0.066)	(0.077)	(0.055)	(0.058)
Bank net interest margin (%)	0.898	−0.122	−0.787	2.815
	(1.903)	(1.725)	(1.248)	(2.508)
Home-city GDP (RMB trillion)	−0.751	0.527	−0.669	2.290
	(0.726)	(2.075)	(0.604)	(1.407)
Home-city GDP growth (%)	−0.217	−0.278	−0.069	0.106
	(0.420)	(0.605)	(0.330)	(0.660)
Home-city LGFV debt/GDP	−1.516	−2.940	1.622	4.631
	(2.830)	(3.758)	(1.952)	(5.471)
Home-city income (RMB 10,000)	0.003	−3.326*	−0.306	−2.861**
	(0.645)	(1.914)	(0.478)	(1.326)
$\mathbb{1}\{\text{Local bank}\}$	4.393*	3.779	5.296***	14.732***
	(2.600)	(3.203)	(1.993)	(3.860)
Mean of dependent variable	1.566	1.562	−4.763	42.946
Top-quintile wild-bootstrap p	< 0.001	—	< 0.001	< 0.001
Observations	129,577	129,570	129,577	129,577
Category × quarter fixed effects	Yes	Yes	Yes	Yes
Home province × quarter fixed effects	No	Yes	No	No
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Notes: Each column is one regression of next-quarter flows on prior-quarter return quintiles within risk and redemption categories, with the bottom quintile omitted. Controls are displayed and dated before flows. Category-by-outcome-quarter effects enter, and column 2 adds home-province-by-quarter effects. Parentheses contain parent-bank clustered errors, and the wild-bootstrap p is for the top-quintile coefficient. Net flows in columns 1 and 2 are capped at the 1st and 99th percentiles. Symmetric flows are bounded between −200 and 200 and uncapped, and positive flow is 100 times an indicator of net inflow. Stars indicate 10%, 5%, and 1% significance. Estimation sample: 2022Q3–2025Q3.

Table A6
LGFV bond issuance yields relative to other issuers

Panel A. Fixed-rate bonds issued at par, 2022Q1–2023Q3

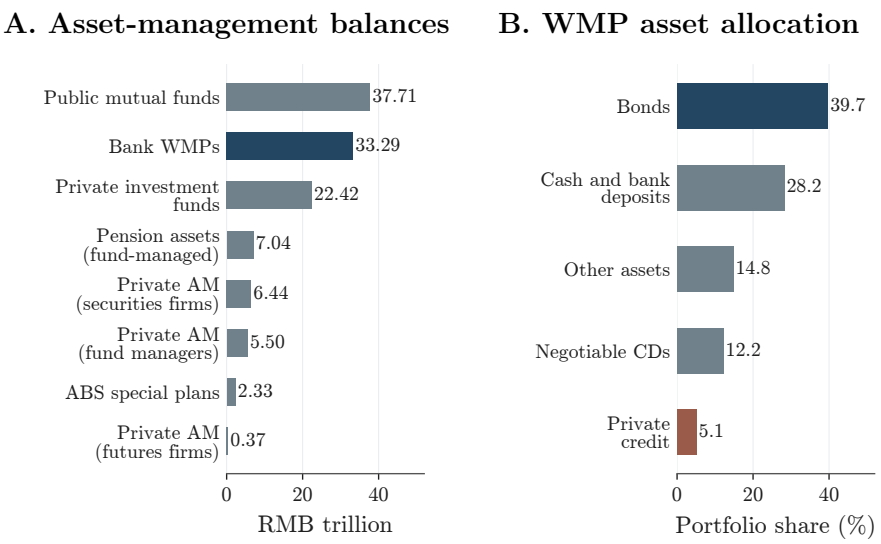
	LGFV	Other nonfinancial	Financial
Mean issuance yield (%)	4.26	3.03	2.77
Median issuance yield (%)	3.92	2.79	2.75
Mean maturity (years)	2.28	1.79	2.12
Bond issues	8,044	10,832	1,296
Issuers	1,581	1,523	205

Panel B. Yield differences and bond characteristics

Dependent variable	Issuance yield (pp) (1)	Issuance yield (pp) (2)	Issuance yield (pp) (3)
$\mathbb{1}\{\text{LGFV issuer}\}$	1.224***	0.910***	0.875***
	(0.069)	(0.089)	(0.093)
$\mathbb{1}\{\text{Financial issuer}\}$	−0.265***	−0.173***	−0.226***
	(0.054)	(0.057)	(0.063)
Maturity (years)	—	0.282***	0.277***
		(0.023)	(0.028)
Maturity squared	—	−0.009***	−0.009***
		(0.001)	(0.001)
Log issue amount	—	−0.280***	−0.264***
		(0.038)	(0.036)
$\mathbb{1}\{\text{Callable}\}$	—	0.378***	0.398***
		(0.099)	(0.099)
LGFV minus financial-issuer yield	1.489***	1.084***	1.101***
	(0.061)	(0.099)	(0.117)
Observations	20,172	20,172	20,172
Month fixed effects	No	Yes	Yes
Reported rating fixed effects	No	No	Yes
Standard errors	Issuer	Issuer	Issuer

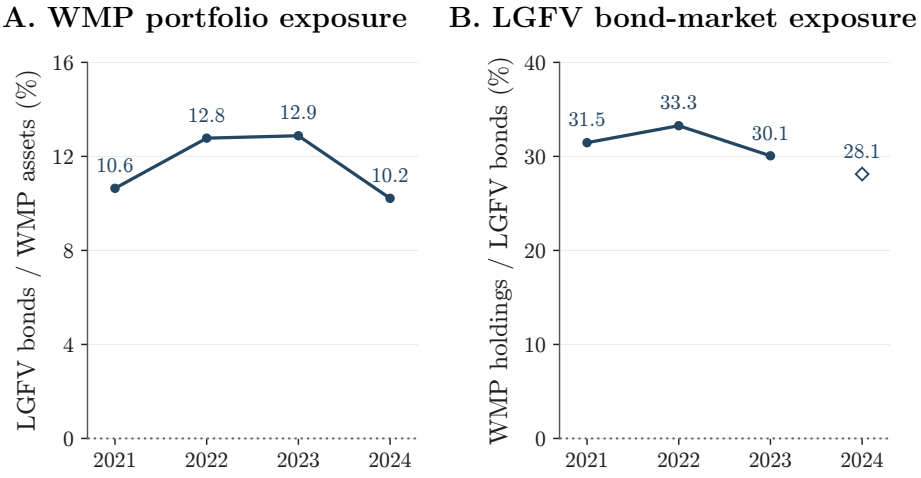
Notes: Panel A describes unique RMB fixed-rate bonds issued at par in 2022Q1–2023Q3. Panel B regresses issuance yields (pp) on issuer-type indicators, with other nonfinancial issuers omitted. All columns use the same sample, and later columns add the displayed issue characteristics, issue-month effects, and reported ratings (including a category for missing ratings). Unlike Panel D of Table A3, the sample does not require observed issuer assets. CDs, convertibles, exchangeables, structured, perpetual, subordinated, floating-rate, and reset-rate bonds are excluded. There is no winsorization. Parentheses contain issuer-clustered errors, and stars indicate 10%, 5%, and 1% significance. For these fixed-rate, at-par issues, the stated rate equals the contractual yield at issuance. Estimation sample: 2022Q1–2023Q3.

Figure A1
Chinese asset-management products and WMP asset allocation, 2025



Notes: Panel A reports assets under management of Chinese asset-management products at the end of 2025, in RMB trillion, and Panel B reports the asset allocation of bank WMPs, in percent. Balances overlap through cross-investment. Private AM denotes private asset-management plans. Sources: Asset Management Association of China and annual WMP market reports.

Figure A2
The scale of the WMP–LGFV bond link, 2021–2024



Notes: Both panels use estimated holdings $\widehat{H}_t = B_t q_t$ from Internet Appendix D.2, which multiply industry WMP bond assets B_t by the LGFV share q_t of bonds in disclosed year-end holdings. Panel A divides by official WMP net assets under management, and Panel B divides by the published LGFV bond balance. The open diamond marks the change in the source of the LGFV bond balance in 2024. Sources: Wind disclosures, annual WMP market reports, Lianhe Ratings (2024), and Zhang (2025).

Internet Appendix B
Model Derivations and Numerical Solution

This appendix states the model of Section 6 in full (Subsection B.1) and derives its results (Subsections B.2–B.6), and Internet Appendix C gives proofs. Subsection B.7 solves the full equilibrium numerically, without the first-order approximations used in the text.

B.1 The model in full

This subsection states the model of Section 6 with all its assumptions and intermediate steps. Equations that also appear in the main text keep their main-text numbers.

Savers and banks. A bank in the model represents its WMP business and any wealth-management subsidiary. The investments below belong to its products and not to its own balance sheet. A unit mass of savers each invests one unit with a bank or an outside saving option. Savers differ in risk aversion γ_h, which is drawn from a nondegenerate distribution G on a bounded positive interval. Saver h obtains utility

$$U_{hb} = \gamma_h a_b + \beta\Big(R_b - \frac{\gamma_h}{2} V_b\Big) + \varepsilon_{hb} \tag{11}$$

from bank b, and utility ε_{h0} from the outside option. Each saver chooses the alternative with the highest utility. Here R_b and V_b are the excess return and return variance that the bank's products report, so $R_b - \frac{\gamma_h}{2} V_b$ is the risk-adjusted reported return to saver h, and $\beta > 0$ measures how strongly choices respond to it. Savers observe these reported returns and their fluctuation but not the portfolios behind them. Flows respond to past returns and reported volatility and are insensitive to holdings risk (Section 3), which motivates this demand channel. The term a_b captures the bank's reputation and sales network. We assume that cautious savers place greater weight on reputation, because a bank's standing matters more to savers who fear losses. We set that weight equal to γ_h. The taste term ε_{hb} captures preferences such as a nearby branch or an existing account.[25] Among savers with risk aversion γ, the share choosing bank b is

$$s_b(\gamma) = \frac{\exp\big(\gamma a_b + \beta(R_b - \gamma V_b/2)\big)}{1 + \sum_j \exp\big(\gamma a_j + \beta(R_j - \gamma V_j/2)\big)}, \tag{12}$$

and the bank's market share is $s_b = \int s_b(\gamma)\, dG(\gamma)$. There are three banks, a national bank N, a large local bank L, and a small local bank S, with reputations $a_N > a_L > a_S$. The sum in the denominator runs over them, and the term 1 represents the outside option. Holding

[25] The taste shocks include the outside-option shock. They are independent of risk aversion and independently and identically distributed type I extreme value across alternatives and savers. This logit assumption yields equation (12). See Berry (1994) and, for bank deposits, Xiao (2020).

portfolios fixed, a stronger reputation attracts more savers. Market share measures scale in the model, and Log bank size$_b$ measures predetermined scale in the empirical comparisons.

Each bank invests its products' funds in one portfolio. It places a share x_b in LGFV bonds, a share n_b in private credit linked to the same borrower sector, and the rest in accessible low-yielding assets with riskless return r_f, with $x_b, n_b \geq 0$ and $x_b + n_b \leq 1$. LGFV bonds stand for the risky credit that supplied most of smaller banks' extra yield (Section 4.1). In the absence of a credit event, bonds and private credit pay excess returns μ_B and μ_{PC} over r_f, and their reported returns have variances σ_B^2 and σ_{PC}^2 and covariance $\sigma_{B,PC} > 0$. The portfolio's reported excess return and variance are

$$R_b = \mu_B x_b + \mu_{PC} n_b, \qquad V_b = \sigma_B^2 x_b^2 + 2\sigma_{B,PC} x_b n_b + \sigma_{PC}^2 n_b^2. \tag{B.1}$$

We assume $\sigma_{PC}^2 > \sigma_B^2 > \sigma_{B,PC}$, so the reported return on private credit moves less than one for one with the bond return. All banks face the same asset returns and risks, so no bank has better information about the borrowers.

LGFV-linked credit also carries a credit risk that reported returns do not show, as Section 4.2 documents. With probability θ, a credit event occurs among the sector's borrowers. Savers then see the losses and lower the bank's reputation by $\kappa \ell_b^2/2$, where $\ell_b = x_b + n_b$ is the bank's exposure to the LGFV sector. The damage rises more than proportionally with exposure, because a product can absorb losses on a small position within its other income but not losses on a concentrated one. WMPs do not guarantee principal, so the losses themselves fall on savers, and the bank bears the credit risk only through its reputation.

A bank earns a fee f per yuan of savings. Initiating private credit requires a one-time fixed cost $F > 0$ for sourcing, evaluation, and contracting. The bank pays this cost when it first arranges private credit and not for each product or contract, and the cost is sunk afterward. Write $\mathcal{S}_b(a)$ for the bank's market share when its reputation is a, holding its portfolio and its rivals' offers fixed. Taking rivals' portfolios as given, the bank chooses its portfolio to maximize

$$f\,\mathcal{S}_b(a_b) - \theta f\left[\mathcal{S}_b(a_b) - \mathcal{S}_b\left(a_b - \frac{\kappa}{2}\ell_b^2\right)\right] - F\,\mathbb{1}\{n_b > 0\}\,e_b, \tag{13}$$

where $e_b = 1$ if the bank has not arranged private credit before and $e_b = 0$ otherwise. The second term is the expected loss of fee income from the damage to its reputation. Entry into private credit must generate enough additional value to pay the fixed cost.[26]

[26] We compare two one-period portfolio-and-entry games, one before and one after an unanticipated reform. The probability θ includes the discounting of fees lost after a credit event. Prior entry carries over through $e_b = 0$, so the model does not assign an additional continuation value to entering before the reform. A Nash equilibrium consists of portfolios and entry decisions that are best responses to the other banks' choices in that state.

Portfolio choice. A bank's portfolio choice depends on the savers most likely to change their choice when its offer improves. Hold rivals' offers fixed and consider a unit increase in the utility offered to type γ, excluding the taste shock. This increase raises the bank's share of that type at rate $s_b(\gamma)\big(1 - s_b(\gamma)\big)$. This response is small when the bank attracts almost none or almost all of that type. Weighting each type by its response gives the average risk aversion of the bank's marginal savers,

$$\bar{\gamma}_b = \frac{\int \gamma \, s_b(\gamma)\big(1 - s_b(\gamma)\big) \, dG(\gamma)}{\int s_b(\gamma)\big(1 - s_b(\gamma)\big) \, dG(\gamma)}. \tag{14}$$

We write $\mathrm{Var}_b(\gamma)$ for the dispersion of risk aversion among these marginal savers, using the same weights. This differs from V_b, the variance of reported portfolio returns.

The same weights determine what a credit event costs. Because savers weight reputation by their risk aversion, a unit loss of reputation costs the bank as many savers as a loss of $\bar{\gamma}_b/\beta$ in reported return. To first order, the expected fee loss in equation (13) equals the fee value of a reduction in reported return of $\bar{\gamma}_b \varphi \ell_b^2/2$, where

$$\varphi = \frac{\theta\kappa}{\beta} \tag{15}$$

measures the credit risk the bank bears through its reputation, in units of reported variance. We hold the composition of marginal savers fixed, which fixes $\bar{\gamma}_b$. The first-order conditions at an optimum with positive holdings of both assets and $x_b + n_b < 1$ are then

$$\mu_B = \bar{\gamma}_b\big(\sigma_B^2 x_b + \sigma_{B,PC} n_b + \varphi \ell_b\big), \qquad \mu_{PC} = \bar{\gamma}_b\big(\sigma_{B,PC} x_b + \sigma_{PC}^2 n_b + \varphi \ell_b\big). \tag{B.2}$$

Each condition equates an asset's excess return to its marginal risk. That risk has two parts, the reported variance that savers see and the credit risk that the bank bears through its reputation. Both are scaled by $\bar{\gamma}_b$, the risk aversion of the bank's marginal savers. A bank with cautious marginal savers has more reputation at stake. Subsection B.7 solves the full problem numerically, letting marginal savers adjust. For interior portfolios, the bond-only regime sets $n_b = 0$ and gives

$$x_b^B = \frac{\mu_B}{\bar{\gamma}_b\,(\sigma_B^2 + \varphi)}. \tag{16}$$

Credit risk that savers do not see restrains a bank's bond holdings in the same way as the variance they do see. The same expression shows why reported volatility matters to a bank. A lower reported variance σ_B^2 raises the bond share, so a bank can hold more bonds without losing savers. Products whose reported returns do not follow market prices have such a lower variance. Section 4.2 shows that smaller banks' products reported less volatility than their holdings implied. Even if reported returns showed no risk at all, the bond share would remain

finite, because $x_b^B \to \mu_B/(\bar{\gamma}_b\varphi)$ as $\sigma_B^2 \to 0$. It would still fall with the caution of the bank's marginal savers.

If the bank arranges private credit, the bond condition gives $x_b = x_b^B - \chi n_b$, where

$$\chi = \frac{\sigma_{B,PC} + \varphi}{\sigma_B^2 + \varphi}, \qquad \omega = \sigma_{PC}^2 + \varphi - \frac{(\sigma_{B,PC} + \varphi)^2}{\sigma_B^2 + \varphi}. \tag{B.3}$$

Because private credit adds exposure to the same borrower sector, the bank reduces its bond share by χ for each unit of private credit, and ω is the risk that private credit adds beyond the bonds it displaces. Substituting into the private credit condition gives

$$\mu_{PC} - \chi\mu_B = \bar{\gamma}_b\, \omega\, n_b. \tag{B.4}$$

The left side is the return that private credit earns beyond the bonds it displaces. Solving,

$$n_b^{PC} = \frac{\mu_{PC} - \chi\mu_B}{\bar{\gamma}_b\, \omega}, \qquad x_b^{PC} = x_b^B - \chi\, n_b^{PC}, \tag{B.5}$$

and private credit raises the bank's risk-adjusted value, per unit of savings and at its marginal savers, by

$$g_b = \frac{\big(\mu_{PC} - \chi\mu_B\big)^2}{2\bar{\gamma}_b\, \omega}. \tag{B.6}$$

We impose $\mu_{PC} > \chi\mu_B$, so private credit earns more than the bonds it displaces and n_b^{PC} and g_b are positive. We also restrict attention to parameters for which both portfolios are interior.[27] With marginal savers held fixed, the bank chooses (x_b^{PC}, n_b^{PC}) if it arranges private credit and $(x_b^B, 0)$ otherwise.

Who serves cautious savers. Reputation determines not only how many savers a bank attracts but also which ones. For any two banks, equation (12) gives

$$\log \frac{s_b(\gamma)}{s_{b'}(\gamma)} = \beta\Big(R_b - R_{b'}\Big) + \gamma\Big[\Big(a_b - a_{b'}\Big) - \frac{\beta}{2}\Big(V_b - V_{b'}\Big)\Big]. \tag{B.7}$$

The bracket combines the reputation gap with the difference in reported risk. We call the case in which the bracket is positive the sorting condition. Under it, more risk-averse savers are relatively more likely to choose the more reputable bank. To obtain transparent comparative statics, we use the small-share approximation. When $s_b(\gamma)$ is small, the marginal-saver weight $s_b(\gamma)(1 - s_b(\gamma))$ is approximately $s_b(\gamma)$, and in the limit a bank's marginal savers have the same distribution of risk aversion as all its savers. Holding portfolios fixed, reputation then

[27] In each state, $0 < x_b^B < 1$, $x_b^{PC} > 0$, $n_b^{PC} > 0$, and $x_b^{PC} + n_b^{PC} < 1$.

raises marginal risk aversion and log market share at the rates

$$\frac{\partial \bar{\gamma}_b}{\partial a_b} = \mathrm{Var}_b(\gamma) > 0, \qquad \frac{\partial \log s_b}{\partial a_b} = \bar{\gamma}_b \tag{B.8}$$

(Proposition B.1). Under the scale condition below, allowing portfolios to adjust preserves these signs. Higher-reputation banks attract more cautious savers, and these savers also value reputation more. These banks therefore both accept less reported variance and have more to lose from a credit event. Lower-reputation banks hold more bonds and therefore more credit risk that their savers do not see. The higher bond share in equation (16) therefore helps compensate for a weaker reputation in attracting savings.

Why private credit requires scale. A bank that has not yet arranged private credit enters only if the additional value it creates covers the fixed cost. Let Φ_b^B and Φ_b^{PC} be the maximized values of the first two terms of equation (13) without and with access to private credit, holding rivals' portfolios fixed. These maximized values use the exact objective and let marginal-saver composition adjust. Breaking ties in favor of entry, the bank enters if and only if

$$\Phi_b^{PC} - \Phi_b^B \geq F. \tag{17}$$

This exact comparison is the entry rule. For intuition, in the small-share case the gain is approximately $f\beta s_b g_b$, so to first order the bank enters when

$$f\beta\, s_b\, g_b \geq F \tag{B.9}$$

(Proposition B.2). Here βg_b is the approximate proportional increase in the bank's net fee income from adding private credit. Multiplying it by the investor base s_b and the fee f gives the additional fee income available to cover F. For a given proportional gain, a larger bank earns more fees while paying the same fixed cost.

Less cautious savers accept more risk in exchange for yield, so private credit adds more value per unit of savings at the small bank. However, the larger bank can offer it to a broader investor base. Under the scale condition, that broader base dominates, and its additional fee income is greater.[28] For an intermediate fixed cost, only the large local bank enters. The small bank also wants to offer higher returns, but doing so through private credit would cost more than the additional fees it earns. Launching a new product does not remove this bank-level cost. Figure 12 illustrates the resulting entry threshold.

[28] For the small-share, first-order gain, Proposition B.2 gives the sufficient condition $\mathrm{Var}_b(\gamma)(1+\beta R_b) < \bar{\gamma}_b^2$ along the reputation comparison, which we call the scale condition. It allows the bond portfolio to adjust and ensures that the investor-base effect outweighs the change in marginal savers' preferences. The condition takes the same form with and without the credit risk borne through reputation. The numerical example checks the exact gains directly.

Debt resolution and the three predictions. Debt resolution strengthened expected support for LGFVs. We represent it as a fall in the probability of a credit event from θ to $(1-\tau)\theta$ with $0 < \tau < 1$, so the credit risk that banks bear through reputation falls from φ to $(1-\tau)\varphi$. Because the excess yield on LGFV bonds compensates investors for this risk, it falls in the same proportion, from μ_B to $(1-\tau)\mu_B$. The program thus takes over part of the credit risk and the yield together. The return on private credit and all reported variances are unchanged. This simplification matches Section 5.1, where private-credit yields fell far less than bond yields. The predictions hold marginal-saver composition fixed and use the sorting, scale, and interior-portfolio conditions. Subsection B.7 solves a numerical equilibrium with changing saver composition. Throughout, Δ denotes the post-resolution change, and μ_B and φ denote their pre-resolution values.

Prediction 1. Smaller banks take more credit risk that savers do not see. Before entry into private credit, the optimal bond share and its response to reputation are

$$x_b^B = \frac{\mu_B}{\bar{\gamma}_b(\sigma_B^2 + \varphi)}, \qquad \frac{dx_b^B}{da_b} = -\frac{\mu_B}{\bar{\gamma}_b^2(\sigma_B^2 + \varphi)}\frac{d\bar{\gamma}_b}{da_b} < 0. \tag{B.10}$$

Under the sorting and scale conditions, $d\bar{\gamma}_b/da_b > 0$. Higher reputation thus raises marginal risk aversion and lowers the optimal bond share. Smaller banks hold more LGFV credit and therefore more of the credit risk their savers do not see. Their marginal savers accept more reported risk, and a credit event would cost them fewer savers. All banks face the same returns and risks, so the gradient arises without any information advantage, as in Table 9. We test the all-bank size gradient in Table 4. Table 3 shows that smaller banks hold riskier debt throughout their portfolios and that home LGFV bonds are the largest source of their extra yield. Table 7 shows the same gradient among products with the same redemption terms and risk rating, together with the lower risk that smaller banks' products report.

Prediction 2. Scale determines which banks substitute into private credit. For a first-time entrant, the small-share, first-order entry benchmark gives

$$\begin{aligned} n_b^* &= \mathbb{1}\Big\{s_b/\bar{\gamma}_b \geq C(\tau)\Big\}\, n_b^{PC}, \\ C(\tau) &= \frac{2F\,\omega_\tau}{f\beta\Big[\mu_{PC} - \chi_\tau(1-\tau)\mu_B\Big]^2}, \end{aligned} \tag{B.11}$$

where χ_τ and ω_τ are equation (B.3) evaluated at the post-resolution credit risk $(1-\tau)\varphi$. Here s_b is the bond-only market share at the comparison date, and n_b^{PC} is the conditional position in equation (B.5) after resolution. As τ rises, the entry cutoff $C(\tau)$ falls for two reasons. The bond yield falls, and support lowers the credit risk that private credit shares with bonds, which reduces both the bonds that private credit displaces and the risk it adds. Under the scale condition, the large local bank can cross this cutoff while the small bank remains below it. Earlier users do not pay again and expand n_b^{PC} as bonds become less

attractive, as equation (B.23) shows. The allocation tests measure expansion across size groups and combine existing users and new participants (Tables 13 and 14). The difference also appears in freshly chosen portfolios (Table 15).

Prediction 3. Smaller local banks lose more product return and savings business. Comparing the two initially bond-only local banks when only the large bank enters,

$$\begin{aligned} \Delta R_L - \Delta R_S &= \lambda \frac{\mu_B^2}{\sigma_B^2 + \varphi}\Big(\frac{1}{\bar{\gamma}_S} - \frac{1}{\bar{\gamma}_L}\Big) + \Big(\mu_{PC} - \chi_\tau(1-\tau)\mu_B\Big) n_L^{PC} > 0, \\ \lambda &= 1 - \frac{(1-\tau)^2(\sigma_B^2 + \varphi)}{\sigma_B^2 + (1-\tau)\varphi} \in \Big(0, \tau(2-\tau)\Big]. \end{aligned} \tag{18}$$

Here λ is the proportional fall in a bond-only bank's reported return. The smaller bank loses more because it held more bonds before resolution, which is the first term. Only the larger bank replaces part of that income with private credit, which is the second. The small bank therefore loses the return advantage it used to compete against the larger bank's reputation. In the small-share benchmark with the national bank's offer held fixed, the deterioration in its relative offer also reduces its relative share of savings (Proposition B.5). Because smaller banks both cut their LGFV holdings and lose savers, a larger part of the LGFV credit held by WMPs moves to larger banks' products, as in Section 5.3. We examine the return implications using the initial home LGFV bond share, private-credit expansion, and bank size (Table 17, Panel A). The business comparisons relate the same measures to balances in newly launched WMPs (Table 17, Panel B) and bank size to total WMP balances (Table 16). Figures 10 and 11 trace the size-related differences over time.

B.2 Savers and market shares

Savers' risk aversion γ has a nondegenerate distribution G on $[\underline{\gamma}, \overline{\gamma}]$, where $0 < \underline{\gamma} < \overline{\gamma} < \infty$. Utility is given by equation (11). Bounded support permits differentiation under the integral. Under the logit assumption on savers' tastes, the share of savers with risk aversion γ choosing bank b is equation (12). Savers' choices depend on reported returns and variances, so the credit risk in a bank's exposure ℓ_b affects its market share only through the reputation damage of a credit event. Holding rivals' portfolios fixed,

$$\begin{aligned} \frac{\partial s_b(\gamma)}{\partial x_b} &= \beta s_b(\gamma)\Big(1 - s_b(\gamma)\Big)\Big[\mu_B - \gamma\Big(\sigma_B^2 x_b + \sigma_{B,PC} n_b\Big)\Big], \\ \frac{\partial s_b(\gamma)}{\partial n_b} &= \beta s_b(\gamma)\Big(1 - s_b(\gamma)\Big)\Big[\mu_{PC} - \gamma\Big(\sigma_{B,PC} x_b + \sigma_{PC}^2 n_b\Big)\Big], \qquad \frac{\partial s_b(\gamma)}{\partial a_b} = \gamma\, s_b(\gamma)\Big(1 - s_b(\gamma)\Big). \end{aligned} \tag{B.12}$$

The small-share approximation treats $s_b(\gamma)$ as small for every bank and saver type, so $s_b(\gamma)(1 - s_b(\gamma)) \approx s_b(\gamma)$. A bank's marginal savers then have approximately the same

distribution as its savers, and the two distributions coincide in the small-share limit used for the analytic comparative statics. Subsection B.7 instead retains the full demand system, including a large national-bank share. For any two banks and whatever their shares, equation (12) gives equation (B.7).

B.3 Portfolio choice

Integrating equation (B.12) over G gives

$$\frac{\partial s_b}{\partial x_b} = \beta W_b\Big[\mu_B - \bar{\gamma}_b\Big(\sigma_B^2 x_b + \sigma_{B,PC} n_b\Big)\Big], \qquad \frac{\partial s_b}{\partial n_b} = \beta W_b\Big[\mu_{PC} - \bar{\gamma}_b\Big(\sigma_{B,PC} x_b + \sigma_{PC}^2 n_b\Big)\Big],$$
$$\frac{\partial s_b}{\partial a_b} = W_b \bar{\gamma}_b,$$

where $W_b = \int s_b(\gamma)(1 - s_b(\gamma))\, dG(\gamma) > 0$ and $\bar{\gamma}_b$ is defined in equation (14). The variance used in the sorting and scale conditions is $\mathrm{Var}_b(\gamma) = W_b^{-1} \int (\gamma - \bar{\gamma}_b)^2 s_b(\gamma)(1 - s_b(\gamma))\, dG(\gamma)$. To first order in the reputation loss, the expected fee loss in equation (13) is $\theta f(\kappa \ell_b^2/2)\, \partial s_b/\partial a_b = f\beta W_b \bar{\gamma}_b \varphi \ell_b^2/2$, with φ in equation (15). We differentiate the objective with the weights $W_b \bar{\gamma}_b$ in the reputational term held fixed and divide by $f\beta W_b$. This gives equation (B.2) for interior portfolios. Conditional on the investment regime, the fixed cost does not vary with the portfolio.

Let $z_b = (x_b, n_b)'$, $\mu = (\mu_B, \mu_{PC})'$, $\iota = (1, 1)'$, and let Σ be the covariance matrix of reported returns. The effective covariance matrix $\widetilde{\Sigma} = \Sigma + \varphi\, \iota\iota'$ adds the credit risk borne through reputation, which is common to bonds and private credit because both are exposed to the LGFV sector. The first-order conditions of the exact objective are necessary but not generally sufficient for a best response because that objective need not be globally concave. The feasible sets are compact and the objective is continuous, so a maximizer exists within each regime, and Subsection B.7 checks the computed choices numerically.

Holding marginal savers fixed gives the quadratic objective $Q_b(z) = \mu' z - \bar{\gamma}_b z' \widetilde{\Sigma} z/2$. The maintained restrictions imply $\det \Sigma = \sigma_B^2 \sigma_{PC}^2 - \sigma_{B,PC}^2 > 0$, so $\widetilde{\Sigma}$ is positive definite and Q_b is strictly concave. Its constrained maximizer is unique, and its interior choices are equations (16) and (B.5). The bond-only value is $\mu_B^2 / \Big(2\bar{\gamma}_b(\sigma_B^2 + \varphi)\Big)$. With χ and ω in equation (B.3), $0 < \chi < 1$ and $\omega > 0$. After optimizing the bond position, the additional value of private credit is $(\mu_{PC} - \chi\mu_B)n - \bar{\gamma}_b \omega n^2/2$. Its interior maximum is g_b in equation (B.6). If $\mu_{PC} \le \chi\mu_B$ and the bond-only optimum is interior, the optimal private credit position and its gain are zero. If a portfolio constraint binds, the constrained solution replaces the interior formulas.

In the full demand system, a portfolio changes which savers are marginal and how much reputation the bank has at stake, so an interior best response satisfies equation (B.2) only approximately. The two regimes generally have different values $\bar{\gamma}_b^B$ and $\bar{\gamma}_b^{PC}$. The cross-regime

formula $x_b^{PC} = x_b^B - \chi n_b^{PC}$ and quadratic gain g_b apply with a common, fixed $\bar{\gamma}_b$. Exact entry instead compares maximized values, allowing the weights to differ. Subsection B.7 computes each portfolio with the marginal savers it attracts.

B.4 Sorting by reputation

Proposition B.1 (Sorting). *In the small-share case, holding portfolios fixed,*

$$\frac{\partial \bar{\gamma}_b}{\partial a_b} = \mathrm{Var}_b(\gamma), \qquad \frac{\partial \log s_b}{\partial a_b} = \bar{\gamma}_b. \tag{B.13}$$

Along an interior bond-only optimum, holding rivals' offers fixed and allowing the bank's bond share to respond,

$$\frac{d\bar{\gamma}_b}{da_b} = \frac{\mathrm{Var}_b(\gamma)}{1 - \beta\sigma_B^2 \, (x_b^B)^2 \, \mathrm{Var}_b(\gamma)/\bar{\gamma}_b}, \qquad \frac{dx_b^B}{da_b} = -\frac{x_b^B}{\bar{\gamma}_b} \frac{d\bar{\gamma}_b}{da_b}, \tag{B.14}$$

with $d\bar{\gamma}_b/da_b > 0$ and $dx_b^B/da_b < 0$ when the denominator is positive. In the small-share case, if $a_b - a_{b'} > \beta\sigma_{PC}^2/2$, then $\bar{\gamma}_b > \bar{\gamma}_{b'}$ for any feasible portfolios, and if both banks have interior bond-only choices, $x_b^B < x_{b'}^B$.

In equation (B.14), the portfolio response reinforces the sorting. A bank that holds fewer bonds reports less return risk, which attracts still more cautious savers. Savers do not observe the credit risk borne through reputation, so it does not enter this feedback. It nevertheless lowers the bond share at every reputation. Outside the small-share case, the ordering still holds for all of a bank's savers but need not hold for its marginal savers, whose weights $s_b(\gamma)(1 - s_b(\gamma))$ fall where a bank serves most of a group. The national bank has a large share in the numerical equilibrium of Subsection B.7, and that equilibrium confirms the ordering of $\bar{\gamma}_b$ for its parameters. The condition in the last statement of Proposition B.1 ensures that the reputation gap outweighs any difference in reported risk, because no feasible portfolio has reported variance above σ_{PC}^2. In the data, FY2021 bank assets proxy for the reputation and sales network that determine a bank's share.

B.5 Scale and entry into private credit

Access to private credit changes the risk-adjusted reported return offered to saver type γ by $\Delta v_b(\gamma) = \Delta R_b - \frac{\gamma}{2}\Delta V_b$. In this subsection, Δ compares the bank's exact optimal portfolios with and without access and not the two resolution states. This change need not be positive for every saver type. Holding rivals' portfolios fixed, the change in fee income is

$$f\left(s_b^{PC} - s_b^B\right) = f \int s_b(\gamma) \, \frac{(1 - s_b(\gamma))\left(e^{\beta \Delta v_b(\gamma)} - 1\right)}{1 - s_b(\gamma) + s_b(\gamma) e^{\beta \Delta v_b(\gamma)}} \, dG(\gamma), \tag{B.15}$$

where $s_b(\gamma)$ is its bond-only share before the portfolio change. The gain from access is $\Pi_b = \Phi_b^{PC} - \Phi_b^B$. It subtracts from this change in fee income the change in the expected reputational loss in equation (13). Since access expands the feasible set, $\Pi_b \geq 0$. A first-time entrant pays the cost if and only if $\Pi_b \geq F$, breaking ties in favor of entry. A bank that has arranged private credit before pays no further cost and holds it whenever access strictly improves its optimum. This is the exact comparison in equation (17).

Proposition B.2 (Scale and entry)**.** *In the small-share case, the first-order approximation to the gain is $\widetilde{\Pi}_b = f\beta s_b g_b$, with s_b the bond-only market share and g_b in equation (B.6). Along an interior bond-only optimum in a fixed competitive environment, allowing the bond share to respond to reputation,*

$$\frac{d\log\widetilde{\Pi}_b}{da_b} = \bar{\gamma}_b - \left(\frac{1}{\bar{\gamma}_b} + \beta\varphi\,(x_b^B)^2\right)\frac{d\bar{\gamma}_b}{da_b}, \tag{B.16}$$

which is positive whenever $\mathrm{Var}_b(\gamma)\,(1+\beta R_b) < \bar{\gamma}_b^2$, where $R_b = \mu_B x_b^B$ is the reported excess return. If this condition holds along the interior optimum for all reputations between a_S and a_L, then $\widetilde{\Pi}_L > \widetilde{\Pi}_S$. In this first-order entry benchmark, the large bank enters whenever the small bank does, and for $\widetilde{\Pi}_S < F \leq \widetilde{\Pi}_L$ only the large bank enters.

The condition bounds the dispersion of risk aversion among marginal savers relative to its mean. Holding portfolios fixed, it is $\mathrm{Var}_b(\gamma) < \bar{\gamma}_b^2$, and the factor $1+\beta R_b$ accounts for the portfolio response in equation (B.14). The credit-risk term in equation (B.16) arises because the bank holds fewer bonds than would maximize its market share, so the fall in its bond share as reputation rises costs it some savers. The condition that makes the derivative positive nonetheless takes the same form as in a model without credit risk borne through reputation. Under it, the proportional increase in the investor base exceeds the proportional decline in the value of private credit to marginal savers. The exact entry comparisons below use Π_b rather than $\widetilde{\Pi}_b$.

B.6 Debt resolution, substitution, and market shares

Debt resolution lowers the probability of a credit event from θ_0 to $\theta_1 = (1-\tau)\theta_0$ and the bond's excess return from $\mu_{B,0}$ to $\mu_{B,1} = (1-\tau)\mu_{B,0}$, with $0 < \tau < 1$. Subscripts 0 and 1 denote values before and after resolution. The credit risk borne through reputation falls from φ_0 to $\varphi_1 = (1-\tau)\varphi_0$. The return on private credit and the covariance matrix of reported returns are unchanged. We write χ_t and ω_t for equation (B.3) evaluated at φ_t, so χ_1 and ω_1 are the χ_τ and ω_τ of Subsection B.1, and $\mu_{B,0}$ and φ_0 are its μ_B and φ. The propositions hold the composition of each bank's marginal savers fixed across the two states, so $\bar{\gamma}_b$ is fixed. Subsection B.7 lets it adjust.

Proposition B.3 (Bond retreat). *A bank without private credit changes its bond share by*

$$\Delta x_b = -\nu\, x_{b,0}^B, \qquad \nu = \frac{\tau\, \sigma_B^2}{\sigma_B^2 + (1-\tau)\varphi_0} \in (0, \tau], \tag{B.17}$$

which is proportional to its initial bond share, and smaller than the proportional fall in the bond's yield whenever $\varphi_0 > 0$.

Because the change is proportional to the initial position, a larger initial position produces a stronger response to resolution. This exposure interaction is examined in Section 5.1. The retreat is smaller than the fall in the yield because support also lowers the credit risk that restrained bond holdings. This benchmark applies to banks that stay bond-only, holding marginal savers fixed. The exposure measure in Section 5.1 is Initial home overweight$_a$. It also includes other home credit and subtracts a nationwide benchmark, so it proxies initial exposure to the affected investment.

Write $g_{b,0}$ and $g_{b,1}$ for the per-unit value g_b of private credit before and after resolution, and $\Pi_{b,0}$ and $\Pi_{b,1}$ for the corresponding exact gains, evaluated at equilibrium portfolios.

Proposition B.4 (Entry after debt resolution). *Suppose the exact pre-resolution gains satisfy* $\Pi_{S,0} \le \Pi_{L,0}$, *as the scale condition suggests. If* $\mu_{PC} > \chi_0 \mu_{B,0}$, *the gain from private credit rises with resolution,*

$$\frac{\partial g_b}{\partial \tau} = n_b^{PC}\left(\chi\, \mu_{B,0} - \mu_B \frac{\partial \chi}{\partial \tau}\right) - \frac{\bar{\gamma}_b}{2}\left(n_b^{PC}\right)^2 \frac{\partial \omega}{\partial \tau} > 0, \tag{B.18}$$

where $\partial\chi/\partial\tau = -\varphi_0(\sigma_B^2 - \sigma_{B,PC})/\left(\sigma_B^2 + (1-\tau)\varphi_0\right)^2 < 0$, $\partial\omega/\partial\tau = -\varphi_0(\sigma_B^2 - \sigma_{B,PC})^2/\left(\sigma_B^2 + (1-\tau)\varphi_0\right)^2 < 0$, χ, ω, *and* μ_B *are evaluated at* τ, *and* n_b^{PC} *is the conditional position in equation* (B.5). *At a given market share, the first-order gain* $\widetilde{\Pi}_b = f\beta s_b g_b$ *therefore rises. If the bond-only market share also changes,* $d\widetilde{\Pi}_b/d\tau = f\beta\left(s_b\, dg_b/d\tau + g_b\, ds_b/d\tau\right)$. *The exact gains support entry by the national bank before resolution, entry by the large local bank only afterward, and no entry by the small local bank whenever*

$$\max\{\Pi_{L,0},\, \Pi_{S,1}\} < F \le \min\{\Pi_{L,1},\, \Pi_{N,0}\}. \tag{B.19}$$

For an entrant, the combined share of bonds and private credit changes by

$$\Delta(x_b + n_b) = -\nu\, x_{b,0}^B + (1-\chi_1)\, n_{b,1}, \tag{B.20}$$

which is positive when the new private credit position is large relative to the bond retreat.

The interval in condition (B.19) is nonempty when both upper bounds exceed its lower bound. This requires enough post-resolution value to make entry worthwhile for the large

local bank, while the small bank's gain remains below the cost. Subsection B.7 verifies these comparisons with endogenous market shares rather than inferring them from the yield change alone.

The portfolio expression behind Prediction 2 combines the first-order entry benchmark with the private credit choice at fixed marginal savers. After resolution,

$$\begin{aligned} n^{*}_{b,1} &= \mathbb{1}\{\widetilde{\Pi}_{b,1} \geq F \text{ or } e_b = 0\}\, n^{PC}_{b,1}, \\ n^{PC}_{b,1} &= \frac{\mu_{PC} - \chi_1(1-\tau)\mu_{B,0}}{\bar{\gamma}_b\, \omega_1}. \end{aligned} \tag{B.21}$$

Here $\widetilde{\Pi}_{b,1} = f\beta s^{B}_{b,1} g_{b,1}$, with $s^{B}_{b,1}$ the post-resolution bond-only market share, and e_b records whether the bank has already paid the fixed cost. The exact problem instead compares maximized values using equation (17) and computes each portfolio with its own marginal savers. Substituting $g_{b,1}$ into the first-order entry benchmark gives, for a first-time entrant,

$$\frac{s_b}{\bar{\gamma}_b} \;\geq\; \frac{2F\,\omega_1}{f\beta\Big(\mu_{PC} - \chi_1(1-\tau)\mu_{B,0}\Big)^2}. \tag{B.22}$$

In this approximation, s_b is the bank's bond-only share at the date of the comparison. Holding the portfolio fixed, equation (B.13) gives $\partial \log(s_b/\bar{\gamma}_b)/\partial a_b = \bar{\gamma}_b - \mathrm{Var}_b(\gamma)/\bar{\gamma}_b$, positive when $\mathrm{Var}_b(\gamma) < \bar{\gamma}_b^2$. Proposition B.2 gives the stronger condition when that portfolio adjusts. Holding market shares and marginal savers fixed as resolution intensifies, the right side of equation (B.22) falls. A lower bond return and a lower shared credit risk make private credit more valuable and lower the entry cutoff. The full numerical solution lets both market shares and marginal savers change.

For a bank that already used private credit, $e_b = 0$ and the entry condition is no longer binding. Subtracting its pre-resolution conditional position from equation (B.21), with marginal-saver composition fixed and both portfolios interior, gives

$$\Delta n^{PC}_{b} = \frac{1}{\bar{\gamma}_b}\left[\frac{\mu_{PC} - \chi_1\mu_{B,1}}{\omega_1} - \frac{\mu_{PC} - \chi_0\mu_{B,0}}{\omega_0}\right] > 0, \tag{B.23}$$

because resolution raises the numerator and lowers ω. Bonds and private credit are substitutes in the portfolio because they share credit risk and their reported returns covary positively. A lower bond yield and a lower shared credit risk raise the optimal private credit position without a second payment of the fixed cost. The model thus separates expansion by existing users from scale-dependent first entry. The observed increase in private credit can contain both.

Proposition B.5 (Returns and market shares)**.** *Suppose $\bar{\gamma}_L > \bar{\gamma}_S$, as under the sorting condition, neither local bank holds private credit before resolution, and only the large bank enters afterward. Holding marginal savers fixed and under the interior-portfolio conditions,*

the entrant's reported return changes relative to the small bank by

$$\Delta R_L - \Delta R_S = \lambda\,\mu_{B,0}\Big(x^B_{S,0} - x^B_{L,0}\Big) + \Big(\mu_{PC} - \chi_1\mu_{B,1}\Big)n_{L,1} > 0, \tag{B.24}$$

with λ in equation (18). In the small-share case, to first order in utility changes and holding the national bank's offer fixed, the entrant's market share grows faster than the small bank's.

$$\Delta \log s_L - \Delta \log s_S \simeq \beta\Big[\eta\left(J_{S,0} - J_{L,0}\right) + \Delta J^{PC}_L\Big] > 0, \tag{B.25}$$

where $J_b = R_b - \bar{\gamma}_b V_b/2$ is the risk-adjusted reported return at the bank's marginal savers, $\eta = 1-(1-\tau)^2 h\Big((1-\tau)\varphi_0\Big)/h(\varphi_0) \in (0,1)$ with $h(\varphi) = (\sigma_B^2+2\varphi)/(\sigma_B^2+\varphi)^2$, and $\Delta J^{PC}_L \geq g_{L,1}$ is the entrant's gain in J from private credit after resolution.

Both terms are positive in each comparison. The small bank held more bonds (Proposition B.1), so it loses more reported return, and only the large bank gains from private credit. With $\varphi_0 = 0$, $\lambda = \eta = \tau(2-\tau)$ and the expressions reduce to those of a model without credit risk borne through reputation.

The national bank already holds private credit before resolution, so it is not part of this entrant–nonentrant comparison.

B.7 Numerical solution of the full equilibrium

The propositions use the small-share case, first-order approximations, and a fixed composition of marginal savers. We now compute a three-bank equilibrium of the full demand system for an illustrative parameterization, with the exact expected reputational loss in equation (13). Savers' risk aversion is lognormal with log mean 1.0 and log standard deviation 0.6, truncated to $[0.4, 15]$. We set $\beta = 60$, $f = 1$, $\mu_B = 0.012$, and $\mu_{PC} = 0.022$. Reported returns have $\sigma_B^2 = 0.006$, $\sigma_{PC}^2 = 0.034$, and $\sigma_{B,PC} = 0.002$. The probability of a credit event is $\theta = 0.3$ and the reputation damage parameter is $\kappa = 1.2$, so $\varphi = 0.006$. The bond's effective risk is $\sigma_B^2 + \varphi = 0.012$, and half of it is credit risk that reported returns do not show. Reputations are $a_N = 0$, $a_L = -0.75$, and $a_S = -1.25$, and resolution lowers the bond return and the probability of a credit event by $\tau = 0.5$. Each bank's portfolio is a numerically computed best response to its rivals' portfolios, with market shares obtained by integrating equation (12) over the distribution of risk aversion. The fixed cost $F = 0.0080$ lies in the interval of condition (B.19).

Table B1
Full equilibrium of the model

	Before resolution			After resolution		
	National	Large local	Small local	National	Large local	Small local
Bond share x_b	0.229	0.593	0.779	0.113	0.225	0.433
Private credit share n_b	0.125	0	0	0.169	0.286	0
Market share s_b	0.494	0.079	0.035	0.479	0.080	0.028
Marginal savers' risk aversion $\bar{\gamma}_b$	3.24	1.99	1.61	3.24	1.96	1.63
Reported excess return R_b (bp)	55.1	71.2	93.5	43.9	76.4	26.0
Gain from private credit Π_b	0.0130	0.0064	0.0038	0.0232	0.0101	0.0054

Notes: The table reports the numerically computed Nash equilibrium of the model in Section 6 with the illustrative parameters in the text. The gain compares optimized values with and without private credit access. It is net of the expected reputational loss and holds rivals' portfolios fixed. The fixed cost is 0.0080. Before resolution, only the national bank's gain exceeds the cost. Afterward, the large local bank's gain does as well, the small bank's does not, and the national bank keeps access without paying again. In the counterfactual equilibrium restricting all three banks to bonds, their pre-resolution bond shares are 0.310, 0.592, and 0.777.

All three predictions hold in this numerical example (Table B1). Bond shares fall with reputation, and the more reputable banks' marginal savers are more risk averse, so the small bank carries the most credit risk that its savers do not see. In the counterfactual with all three banks restricted to bonds, resolution lowers their bond shares by 0.105, 0.242, and 0.341, or 34%, 41%, and 44%. The first-order benchmark in Proposition B.3 predicts a one-third fall at every bank. The exact falls are close to it at the national bank and larger at the local banks, which held more bonds. The exact gain from private credit is larger at more reputable banks and rises after resolution, and only the large local bank newly crosses the fixed cost. The large bank's reported return rises by 73 basis points relative to the small bank's, and the log of its market share relative to the small bank's rises by 0.24. The small bank's share of the LGFV credit held by the three banks' products falls from 10.9% to 6.4%. Before resolution, the scale-condition ratio $\mathrm{Var}_b(\gamma)(1+\beta R_b)/\bar{\gamma}_b^2$ is 0.33 at the large local bank and 0.31 at the small one. Both ratios are below one, so the scale condition holds. The exact comparisons establish the entry pattern for this example.

We check each bank before and after resolution, with and without private credit access. Holding rivals' equilibrium portfolios fixed, a search of the feasible set $\{x_b, n_b \geq 0,\ x_b + n_b \leq 1\}$ in steps of 0.002 finds no higher value. Constrained optimizations from six starting portfolios agree within 10^{-12}. The solutions are feasible, have first-order residuals below 10^{-7}, and have negative-definite Hessians within each regime. Increasing integration accuracy from 400 to 800 Gauss–Legendre nodes leaves every reported table entry unchanged. These checks support the computed best responses. When we set $\varphi = 0$ and add $\varphi \iota\iota'$ to the reported covariance matrix, the same code reproduces the equilibrium of a model in which all risk is visible to savers.

Internet Appendix C
Proofs

C.1. Proof of Proposition B.1

In the small-share case, $s_b(\gamma) = \exp(u_b(\gamma))/D(\gamma)$, with $u_b(\gamma) = \gamma a_b + \beta(R_b - \gamma V_b/2)$ and $D(\gamma)$ the denominator in equation (12), whose dependence on bank b's own offer is negligible. The weights defining $\bar{\gamma}_b$ are then proportional to $\exp(\gamma t_b)\, m_b(\gamma)$, where $t_b = a_b - \beta V_b/2$ and $m_b(\gamma) = e^{\beta R_b}\, dG(\gamma)/D(\gamma)$ does not depend on a_b. This is an exponential tilting of a fixed distribution, so the derivative of its mean with respect to the tilt t_b is its variance, which gives $\partial\bar{\gamma}_b/\partial a_b = \mathrm{Var}_b(\gamma)$ holding the portfolio fixed. Similarly, $\partial \log s_b/\partial a_b = \int \gamma\, s_b(\gamma)\, dG / \int s_b(\gamma)\, dG = \bar{\gamma}_b$.

For a bank without private credit, $V_b = \sigma_B^2 (x_b^B)^2$ is the reported variance and $x_b^B = \mu_B/\big(\bar{\gamma}_b(\sigma_B^2 + \varphi)\big)$. Then $d\bar{\gamma}_b/da_b = \mathrm{Var}_b(\gamma)\, dt_b/da_b$, with $dt_b/da_b = 1 - \beta\sigma_B^2 x_b^B\, dx_b^B/da_b$ and $dx_b^B/da_b = -(x_b^B/\bar{\gamma}_b)\, d\bar{\gamma}_b/da_b$. Substituting the last expression into the first and solving gives equation (B.14).

For the final statement, equation (B.7) shows that $s_b(\gamma)/s_{b'}(\gamma)$ increases with γ when $(a_b - a_{b'}) - \frac{\beta}{2}(V_b - V_{b'}) > 0$. Every feasible portfolio has $0 \le V \le (x\sigma_B + n\sigma_{PC})^2 \le \sigma_{PC}^2$, so $a_b - a_{b'} > \beta\sigma_{PC}^2/2$ guarantees the inequality whatever the portfolios. The distribution of bank b's savers then dominates that of bank b''s savers in the likelihood-ratio order. In the small-share case, a bank's marginal savers have the same distribution as its savers, so $\bar{\gamma}_b > \bar{\gamma}_{b'}$. If neither bank holds private credit, equation (16) gives $x_b^B < x_{b'}^B$. □

C.2. Proof of Proposition B.2

In the small-share case, $\partial s_b/\partial a_b = s_b\bar{\gamma}_b$, so to first order in the reputation loss the bank's objective is $f s_b - f\beta s_b\bar{\gamma}_b\varphi\ell_b^2/2$. For a change from the bond-only optimum, the integrand in equation (B.15) is approximately $s_b(\gamma)\big(e^{\beta\Delta v_b(\gamma)} - 1\big)$, and linearizing in $\beta\Delta v_b$ gives a share change of $\beta s_b\big(\Delta R_b - \frac{\bar{\gamma}_b}{2}\Delta V_b\big)$, with the weights evaluated at the bond-only optimum. The reputational term changes by $-f\beta s_b\bar{\gamma}_b\varphi\,\Delta(\ell_b^2)/2$. The first-order gain is therefore $f\beta s_b\,\Delta Q_b$, with $Q_b(z) = \mu' z - \bar{\gamma}_b z'\widetilde{\Sigma} z/2$. Under the interior conditions, its maximal improvement is g_b in equation (B.6), so the optimized first-order gain is $\widetilde{\Pi}_b = f\beta s_b g_b$.

For the derivative, g_b is proportional to $1/\bar{\gamma}_b$, so $d\log g_b/da_b = -(1/\bar{\gamma}_b)\, d\bar{\gamma}_b/da_b$. The bond-only share responds to reputation directly and through the bond position. In the small-share case, $\partial \log s_b/\partial x_b = \beta\big(\mu_B - \bar{\gamma}_b\sigma_B^2 x_b^B\big) = \beta\bar{\gamma}_b\varphi x_b^B$ at the bond-only optimum, by equation (B.2). With $dx_b^B/da_b = -(x_b^B/\bar{\gamma}_b)\, d\bar{\gamma}_b/da_b$,

$$\frac{d\log s_b}{da_b} = \bar{\gamma}_b + \beta\bar{\gamma}_b\varphi x_b^B\frac{dx_b^B}{da_b} = \bar{\gamma}_b - \beta\varphi\,(x_b^B)^2\frac{d\bar{\gamma}_b}{da_b}.$$

Adding the two derivatives gives equation (B.16). Equation (B.14) gives $d\bar{\gamma}_b/da_b = \mathrm{Var}_b(\gamma)/(1-k_b)$, where $k_b = \beta\sigma_B^2(x_b^B)^2\mathrm{Var}_b(\gamma)/\bar{\gamma}_b$. Because $\bar{\gamma}_b x_b^B(\sigma_B^2+\varphi) = \mu_B$, we have $\bar{\gamma}_b(x_b^B)^2(\sigma_B^2+\varphi) = R_b$ and $k_b \leq \beta R_b \mathrm{Var}_b(\gamma)/\bar{\gamma}_b^2$. The condition $\mathrm{Var}_b(\gamma)(1+\beta R_b) < \bar{\gamma}_b^2$ therefore implies $k_b < 1$. Multiplying equation (B.16) by $\bar{\gamma}_b(1-k_b) > 0$, the derivative is positive if and only if

$$\bar{\gamma}_b^2 - \beta\sigma_B^2(x_b^B)^2\bar{\gamma}_b\mathrm{Var}_b(\gamma) > \mathrm{Var}_b(\gamma) + \beta\varphi(x_b^B)^2\bar{\gamma}_b\mathrm{Var}_b(\gamma),$$

that is, $\bar{\gamma}_b^2 > \mathrm{Var}_b(\gamma)\Big(1+\beta\bar{\gamma}_b(x_b^B)^2(\sigma_B^2+\varphi)\Big) = \mathrm{Var}_b(\gamma)(1+\beta R_b)$. Integrating along an interior optimum under the stated condition gives $\widetilde{\Pi}_L > \widetilde{\Pi}_S$. Comparing these first-order gains with F establishes the entry ordering in the approximation. Exact entry requires the comparison of maximized values in equation (17). □

C.3. Proof of Proposition B.3

By equation (16) with $\bar{\gamma}_b$ fixed,

$$x_{b,1}^B = \frac{(1-\tau)\mu_{B,0}}{\bar{\gamma}_b\Big(\sigma_B^2+(1-\tau)\varphi_0\Big)} = \frac{(1-\tau)(\sigma_B^2+\varphi_0)}{\sigma_B^2+(1-\tau)\varphi_0}\,x_{b,0}^B.$$

Hence $\Delta x_b = -\nu x_{b,0}^B$ with $\nu = 1-(1-\tau)(\sigma_B^2+\varphi_0)/\Big(\sigma_B^2+(1-\tau)\varphi_0\Big) = \tau\sigma_B^2/\Big(\sigma_B^2+(1-\tau)\varphi_0\Big)$. Since $\sigma_B^2 > 0$ and $0 < \tau < 1$, $0 < \nu \leq \tau$, with equality if and only if $\varphi_0 = 0$. □

C.4. Proof of Proposition B.4

Let $y = (1-\tau)\varphi_0$, so $dy/d\tau = -\varphi_0$. Differentiating equation (B.3),

$$\frac{d\chi}{dy} = \frac{\sigma_B^2-\sigma_{B,PC}}{(\sigma_B^2+y)^2} > 0, \qquad \frac{d\omega}{dy} = \frac{(\sigma_B^2-\sigma_{B,PC})^2}{(\sigma_B^2+y)^2} > 0,$$

where the second follows because $d\omega/dy = \Big[(\sigma_B^2+y)^2 - 2(\sigma_{B,PC}+y)(\sigma_B^2+y) + (\sigma_{B,PC}+y)^2\Big]/(\sigma_B^2+y)^2$. These give the expressions for $\partial\chi/\partial\tau$ and $\partial\omega/\partial\tau$ in the proposition. Write $A = \mu_{PC} - \chi\mu_B$ with $\mu_B = (1-\tau)\mu_{B,0}$. Then $\partial A/\partial\tau = \chi\mu_{B,0} - \mu_B\,\partial\chi/\partial\tau > 0$, so A rises with τ and $A \geq \mu_{PC} - \chi_0\mu_{B,0} > 0$. Since $g_b = A^2/(2\bar{\gamma}_b\omega)$ and $n_b^{PC} = A/(\bar{\gamma}_b\omega)$,

$$\frac{\partial g_b}{\partial\tau} = \frac{A}{\bar{\gamma}_b\omega}\frac{\partial A}{\partial\tau} - \frac{A^2}{2\bar{\gamma}_b\omega^2}\frac{\partial\omega}{\partial\tau} = n_b^{PC}\frac{\partial A}{\partial\tau} - \frac{\bar{\gamma}_b}{2}\Big(n_b^{PC}\Big)^2\frac{\partial\omega}{\partial\tau},$$

which is equation (B.18). Both terms are positive. Because the first-order gain $f\beta s_b g_b$ increases with g_b, it is larger after resolution at any given share and composition of marginal savers. The same argument shows that $n_b^{PC} = A/(\bar{\gamma}_b\omega)$ rises with τ, which gives equation (B.23).

For the entry pattern, every F in the interval (B.19) satisfies $\Pi_{N,0} \geq F$, $\Pi_{L,0} < F \leq \Pi_{L,1}$, and $\Pi_{S,1} < F$. The maintained pre-resolution ordering $\Pi_{S,0} \leq \Pi_{L,0}$ also implies $\Pi_{S,0} < F$. Thus only the national bank enters before resolution. Afterward, the large local bank enters and the small bank remains out. All gains are evaluated against rivals' portfolios in the corresponding regime. The interval is nonempty precisely when $\max\{\Pi_{L,0}, \Pi_{S,1}\} < \min\{\Pi_{L,1}, \Pi_{N,0}\}$.

For an entrant, equation (B.5) and Proposition B.3 give $x_{b,1} = (1-\nu)x^B_{b,0} - \chi_1 n_{b,1}$, so $\Delta(x_b + n_b) = -\nu x^B_{b,0} + (1-\chi_1)n_{b,1}$, which is equation (B.20). □

C.5. Proof of Proposition B.5

A bank without private credit reports $R_b = \mu_B x^B_b = \mu_B^2/\left(\bar{\gamma}_b(\sigma_B^2 + \varphi)\right)$. After resolution, its reported return is $(1-\tau)^2\mu_{B,0}^2/\left(\bar{\gamma}_b(\sigma_B^2 + (1-\tau)\varphi_0)\right) = (1-\lambda)R_{b,0}$, with λ in equation (18). Because $(1-\tau)^2\sigma_B^2 < \sigma_B^2$ and $(1-\tau)^2\varphi_0 \leq (1-\tau)\varphi_0$, $\lambda > 0$, and because $\sigma_B^2 + \varphi_0 \geq \sigma_B^2 + (1-\tau)\varphi_0$, $\lambda \leq \tau(2-\tau)$. After resolution, the entrant holds $x^B_{L,1} - \chi_1 n_{L,1}$ in bonds and $n_{L,1}$ in private credit, so its reported return is $\mu_{B,1}x^B_{L,1} + (\mu_{PC} - \chi_1\mu_{B,1})n_{L,1}$, and its change is $-\lambda\mu_{B,0}x^B_{L,0} + (\mu_{PC} - \chi_1\mu_{B,1})n_{L,1}$. Subtracting the small bank's change, $-\lambda\mu_{B,0}x^B_{S,0}$, gives equation (B.24). The second term is positive because the entrant holds private credit only when $\mu_{PC} > \chi_1\mu_{B,1}$, and the first is positive because $x^B_{S,0} > x^B_{L,0}$ by Proposition B.1.

In the small-share case, $\log s_b(\gamma) = u_b(\gamma) - \log D(\gamma)$. Holding the national bank's offer fixed, $D(\gamma)$ does not change to first order, because the local banks' own shares are small. A change in the bank's portfolio then changes $\log s_b$ by β times the change in $J_b = R_b - \bar{\gamma}_b V_b/2$, the risk-adjusted reported return at the average risk aversion of its savers. For a bank without private credit, $J_b = \mu_B x^B_b - \bar{\gamma}_b\sigma_B^2(x^B_b)^2/2 = \mu_B^2 h(\varphi)/(2\bar{\gamma}_b)$. Writing $q = 1-\tau$, the post-resolution value is proportional to $q^2 h(q\varphi_0)$, and

$$\frac{d\log\left(q^2h(q\varphi_0)\right)}{dq} = \frac{2}{q} - \frac{2q\varphi_0^2}{(\sigma_B^2 + 2q\varphi_0)(\sigma_B^2 + q\varphi_0)} > 0,$$

because $(\sigma_B^2 + 2q\varphi_0)(\sigma_B^2 + q\varphi_0) > q^2\varphi_0^2$. Resolution therefore lowers J_b by the fraction $\eta \in (0,1)$, and since $J_{b,0}$ is proportional to $1/\bar{\gamma}_b$, $J_{S,0} > J_{L,0}$. For the entrant after resolution, $Q_b = J_b - \bar{\gamma}_b\varphi_1\ell_b^2/2$, so the gain in J from private credit is $\Delta J_L^{PC} = g_{L,1} + \bar{\gamma}_L\varphi_1\left(\ell^2_{PC} - \ell^2_B\right)/2 \geq g_{L,1}$, because $\ell_{PC} = x^B_{L,1} + (1-\chi_1)n_{L,1} > \ell_B = x^B_{L,1}$. Subtracting the small bank's change, $-\eta J_{S,0}$, from the entrant's, $-\eta J_{L,0} + \Delta J_L^{PC}$, gives equation (B.25). □

Internet Appendix D
Specifications and Supplementary Evidence

Subsection D.1 identifies the borrowers that receive private credit, and Subsection D.2 reconciles market-wide exposure with shares of included debt holdings. Subsection D.3 gives the designs behind the tests of holdings risk, reported risk, reaching for yield, and borrower funding, and Subsection D.4 states the specifications that the main text describes in words. Table D1 reports initial credit portfolios by bank type and size, and Figure D1 shows the gap between private credit and bond rates of the same borrower. Table D2 reports alternative specifications for Table 7, and Tables D3 and D4 report investor flows in and after the 2022 sell-off. Table D5 reports the part of the loss of the home LGFV return premium that bond income explains, together with sample and NAV checks.

D.1. Private-credit borrowers and the LGFV sector

To identify who receives private credit financing, we extract borrower names from asset descriptions, locate them using company registration records, and match them to LGFV bond issuers over 2022Q1–2025Q3. These matches account for about 60% of local banks' home private credit value before the program and 59% afterward. Adding a screen for borrower names with terms typical of LGFVs raises the shares to 79% and 86%. For the largest FY2021 size tercile, the corresponding shares with the keyword screen are 77% and 85%. Most of the shift into private credit therefore remains within the LGFV sector.

D.2. The scale of the WMP–LGFV bond link

A bond-only denominator gives a simpler benchmark for the LGFV shares in Section 4.1. We sum reported LGFV bond values and divide by the value of all bonds held by each bank group over 2022Q1–2023Q3, retaining government and policy-bank bonds and excluding private credit. This gives 38.5% for local banks and 13.2% for national banks. In the product-quarters of Table D1, local banks' LGFV bonds account for 40.4% of all bonds, 47.5% of the bonds in included debt holdings, and 43.1% of included debt holdings. As shares of all disclosed assets over 2022Q1–2025Q3, LGFV bonds account for 16.4% of local banks' reported portfolio weights and 5.4% of national banks'.

For market context, we reconstruct the year-end 2021–2024 top-ten holdings cross-sections from the original Wind exports. We retain one representative record per product registration code and deduplicate positions within that record, preserving identical investments made by distinct products. We restrict amounts to yuan-denominated positions, identify bonds by asset type, and match issuers to the January 4, 2026 LGFV roster. Unidentified bond issuers remain in the denominator but not the LGFV numerator. In 2024, the retained disclosures

contain RMB 0.3859 trillion of LGFV bonds out of RMB 2.3459 trillion of bonds, including negotiable certificates of deposit. This ratio of sums is 16.45%, and it is not an equally weighted average of product ratios.

For each year t, let q_t be this disclosed LGFV-to-bond ratio, B_t the industry's bond assets including negotiable certificates of deposit, and A_t its net assets under management, from the annual WMP market reports. Estimated LGFV bond holdings are $\widehat{H}_t = B_t q_t$, and their share of WMP assets is $\widehat{H}_t/A_t$. The annual WMP market report published in 2025 gives year-end 2024 values of $B_t = 18.60$ and $A_t = 29.95$ trillion yuan. Using unrounded holdings gives $\widehat{H}_t = 3.06$ trillion yuan, or 10.2% of net assets under management. Equivalently, the rounded inputs give $16.45\% \times (18.60/29.95) = 10.22\%$. The extrapolation applies the disclosed bond mix to the industry's bond assets. It excludes private credit and indirect exposure through funds and other investment vehicles.

WMPs' share of outstanding LGFV bonds is $\widehat{H}_t/O_t$, where O_t is the published year-end LGFV bond balance. Lianhe Ratings (2024) reports Wind balances of RMB 9.81, 10.62, and 11.48 trillion for 2021–2023. Zhang (2025) reports RMB 10.88 trillion for 2024, implying that WMPs held 28.1% of outstanding LGFV bonds that year. Figure A2 marks the 2024 point with an open diamond to show the change in source. These ratios describe two different economic exposures. Under the extrapolation, LGFV bonds account for about one-tenth of WMP assets in 2024, but WMPs hold more than a quarter of the LGFV bond market. A tenth of the assets of a large savings industry is thus more than a quarter of these borrowers' bond financing.

D.3. Samples and constructions behind the main tests

Risk held and risk reported. Table 7 uses products that the filings classify as fixed-income, with at least nine monthly returns between April 2022 and June 2023 and with disclosed holdings in 2022Q2–2023Q2. Monthly returns are changes in log distribution-adjusted NAV in basis points, as in equation (5). They cover every month in which the product has NAVs at the ends of consecutive months. The sell-off return is the product's mean monthly return in November and December 2022. Reported returns are winsorized at the 1st and 99th percentiles of all monthly NAV returns and implied returns at the 1st and 99th percentiles of their own distribution, so that the sell-off months remain in both. Redemption types are closed-end, periodic-open, and daily-open and come from the product filings. The risk rating is the product's modal rating. Holdings measures average the product's disclosed holdings over the quarters in which it reports. The home LGFV credit share combines LGFV bonds and private credit to LGFVs of the bank's headquarters province as a percentage of disclosed holdings. The all-in spread is defined in Section 4.1. It is computed position by position, as are the implied returns below. The issue spread of a bond is its coupon minus the Treasury yield of the nearest half-year maturity on its issue date, and the product-level spread weights

the bonds held by value. The spread relative to the same official rating uses medium-term notes, short-term financing bills, and corporate bonds issued since 2021. It subtracts from each bond's issue spread the mean spread of bonds with the same issue quarter, issuer rating, issuer administrative level, maturity group, and placement type. Log bank size is the FY2021 measure of Section 4.1, and products are weighted by the inverse of the number of products of their bank in the sample.

Implied returns. The implied return of a product in a month sums over its disclosed positions. Each term is the position's share of disclosed value times the month's price change of the market series assigned to the position. The shares are averaged over the quarters in which the product reports. These are price changes, which leave out interest income. LGFV bonds move by minus their modified duration times the sum of the month's changes in the Treasury yield of their maturity and in the LGFV spread of the issuer's province. Government and policy-bank bonds move by the Treasury term alone. Modified duration is that of a par bond with the asset's coupon and remaining maturity. Other bonds move with the China Securities Index (CSI) credit-bond index of their rating (AAA, AA+, or AA and below) and remaining-maturity bucket (up to 1, 1–3, 3–5, 5–7, and 7–10 years), scaled by the ratio of the bond's duration to that of the bucket. These other bonds include bank capital bonds and securities without the standard-bond flag that we find in the bond descriptions. The price part of an index is its monthly log return minus its median monthly log return over 2021–2023, which removes coupon accrual. Certificates of deposit move with the index for up to one year, at AAA for national banks and at AA+ otherwise. Bank preferred shares move with the 3–5 year AAA index, and asset-backed securities and public bond funds move with the 1–3 year AAA index. Private credit moves like an LGFV bond of the borrower's province when the data vendor classifies the borrower as an LGFV, when the borrower issues LGFV bonds, or when its name passes the screen of Subsection D.1. Other private credit moves with the AA+ index of its maturity bucket. In both cases the maturity is the balance-weighted remaining maturity of the borrower's contracts that are active at the quarter end. Asset-management plans, trust plans, insurance asset-management products, and other wrappers move with the 1–3 year AA+ index, because most of them are fixed-income mandates. Cash, deposits, repurchase agreements, interbank lending, money-market funds, and futures margin do not move, and neither do equity-labeled positions, which are 0.4% of disclosed value. Reported returns also contain interest income net of fees, so we add to each product's implied return its mean difference between the reported return and the price change over the months other than November and December 2022 in which it reports a return. The two series then agree on average outside the sell-off. The adjustment is constant within a product, so it leaves implied volatility unchanged. Implied volatility is the standard deviation of the implied return over the months in which the product reports a return, and the implied sell-off return is its mean in November and December 2022. The reported minus implied sell-off return thus measures how much more the reported return exceeded the price change of the holdings in

the sell-off than in other months. Table D2 reports alternative specifications. They weight products equally, control for the shares of each asset class, keep products whose disclosed holdings are at least 60% bonds, mark the wrappers with the AAA index instead, or mark only directly held bonds, so that every other asset contributes nothing. With that measure the implied-volatility gradient is more than three times as large, because larger banks hold more of the assets that it leaves unpriced. In Figure 3, Panels A to C use banks with at least five products in the sample, and Figure 5 averages products within a bank and then banks within each group.

Product design and ratings. Table 8 uses the products of Table 7. Its Panel B compares closed-end and periodic-open products with daily-open products of the same bank, with bank effects in column 1 and bank-by-risk-rating effects in column 2. Column 6 of Panel B of Table 7 uses all fixed-income products with a monthly return between April 2022 and June 2023 and a redemption type in the filings. It includes redemption-type effects and assigns each product to its parent bank by the issuer named in its filing. The bond-rating statistics use the medium-term notes, short-term financing bills, and corporate bonds issued between 2021Q1 and 2023Q2 with an issuer rating.

Comparison cells. Panel A of Table 9 uses local banks' LGFV bond holdings. Branch presence comes from a bank-branch registry matched to banks by name and headquarters city, and a bank has presence in an issuer city if it has at least one branch there in that quarter. The coupon outcome subtracts from each bond's coupon the mean coupon of all held LGFV bonds with the same issue year and maturity group, so it compares bonds issued under similar market conditions. Issuer administrative level and rating are from the issuance data. Credit events are defaults, rating warnings, and guarantor events recorded for the issuer between 2022Q4 and 2025Q4. A cell enters only if it contains at least two banks. Panel B uses tier-2 and perpetual bonds of other banks, classifies issuers as city or rural banks by name, and compares bank-quarters with the same headquarters city and quarter. Panel C identifies negotiable certificates of deposit by name and code, classifies issuers as national banks or as city and rural banks by name, excludes the bank's own certificates, and compares bank-quarters with the same headquarters city, quarter, and maturity group.

Own-city holdings and local presence. Table 10 keeps local banks headquartered in cities with at least two sample banks and compares banks from the same city in the same quarter. Own-city branches come from the same registry as branch presence, and a bank with no recorded branch in its headquarters city counts as zero. Own-city LGFV bonds are those issued by LGFVs of the bank's headquarters city, and the county-level share uses the issuer's administrative level from the issuance data.

Borrower funding. In Table 18, X_f^H (X_f^L) is the fraction of borrower f's WMP-held bonds or private credit in 2022Q4–2023Q1 that was held by local banks without a subsidiary whose initial home LGFV bond share was above (below) the median. The borrower panel is balanced over 2023Q2–2025Q2, with zero holdings for groups that hold none of the borrower's

debt. Column 5 uses new bonds issued by the same borrowers between 2023Q2 and 2025Q2, with coupons between 0.5% and 12% and known maturity. The new-issue premium estimates in Subsection D.4 use the same specification for the coupon minus the CSI valuation yield at listing.

Provincial spreads on home holdings. The spread comparison of Section 5.3 weights each province's month-end LGFV spread over Treasury yields by the disclosed value of the home LGFV bonds and private credit to home LGFVs that local banks without a subsidiary held over 2022Q4–2023Q2.

D.4. Specifications and details not displayed in the main text

This subsection states the regressions that the main text describes in words and gives the details of the displayed ones, in the order of the main text.

Flows (Section 3). The flow regression with holdings risk follows equation (1). It forms return quintiles within risk rating, redemption terms, and quarter. It includes the eleven lagged controls, a local-bank indicator, and category-by-quarter effects, and errors cluster by parent bank.

Allocation and home tilt (Section 4.1). In equation (2), the sample is 2022Q1–2023Q3. Category zeros remain, and shares are not benchmark-adjusted. Fixed size replaces contemporaneous bank assets, and quarter effects enter without bank effects or a local-bank indicator. In equation (3), the product controls are size, risk, withdrawal terms, disclosure count, and disclosure coverage. The bank controls are its average product term and risk, its nonperforming loans, its cost–income ratio, and its net interest margin, and the city control is home-city GDP growth. Errors cluster by parent bank. The two home measures use different denominators. For example, consider a WMP with RMB 100 in included debt holdings. It holds RMB 20 in LGFV bonds, and RMB 10 of these are from home. Its within-LGFV home share is 50%, but its home LGFV allocation is 10% of included debt holdings. Each measure subtracts the matching nationwide benchmark. Neither denominator is total WMP assets. In equation (4), the controls are those in equation (2) plus time-varying log bank assets. Redemption is not among them, because it enters separately. The sample is 2022Q1–2025Q3. Quarter effects enter, and errors cluster by parent bank. In equation (5), the pre-program home-minus-non-home LGFV contrast is 5.354 monthly basis points per unit share (bank-clustered standard error 1.773, $p = 0.003$). We calculate it from the same regression and its covariance matrix. At a 20-point allocation difference, the contrast is about 13 annual basis points.

Risk held and risk reported (Section 4.2). In Table 7, the coefficients on log bank size for the home LGFV credit share, the issue spread, the all-in spread, and log implied volatility are -5.39, -22.4, -12.8, and -0.074. A bank half as large differs by $\ln 2$ times each coefficient with the opposite sign. Among the LGFV issuers of Figure 4 rated AA+, average issue

spreads range from 72 basis points in Guangdong to 449 in Tianjin. Holding the issue quarter, maturity, placement type, and issuer rating fixed, LGFV bonds were issued at spreads 36 basis points higher (standard error 13) than other bonds. Column 6 of Panel B uses every fixed-income product with a monthly return in the period, whether or not it has nine returns and disclosed holdings. Each product's rating is its modal rating in the returns data.

Reaching for yield (Section 4.3). In Table 9, bonds are weighted by value within a bank. Risk outcomes are indicators multiplied by 100. For other cities of the home province, the coefficients per log point of bank size are −34.3 basis points for the coupon, −13.2 percentage points for county-level issuers, and −4.1 points for issuers rated below AAA. In column 5, they are −39.4 basis points for other cities of the home province and −29.3 for other provinces. Multiplying by $\ln 2$ gives the comparisons for a bank half as large. For the capital bonds of other banks, the coefficients are −19.2 percentage points and −24.7 basis points per log point, with coupons measured relative to issue-year and instrument-type means. Three-quarters of the city and rural banks' capital bonds held by local banks without a wealth-management subsidiary were issued in other provinces. Certificates of deposit issued by city and rural banks pay higher yields than those of national banks.

Same-borrower financing rates (Section 5.1). We estimate the gap between private credit and bond yields separately for each quarter. For contract or bond issue j of borrower f in month m, we estimate

$$\begin{aligned}\text{Yield}_{jfm} = \alpha_{fm} &+ \sum_q \beta_q \, \mathbb{1}\{\text{Private credit}\}_j \mathbb{1}\{q(m) = q\} \\ &+ \sum_q \gamma_q \, \text{Maturity}_j \mathbb{1}\{q(m) = q\} + e_{jfm}.\end{aligned} \tag{D.1}$$

The outcome Yield_{jfm} is the annualized contractual borrowing rate, expressed in percent. The indicator $\mathbb{1}\{\text{Private credit}\}_j$ identifies private credit, Maturity_j is maturity in years, and $q(m)$ is the quarter containing month m. Borrower–month effects compare the two financing channels of the same legal borrower and exclude affiliates. A separate maturity slope in each quarter allows the relationship between maturity and yield to change over time. Thus β_q measures the quarter's private-credit-minus-bond yield gap in percentage points. The sums include every sample quarter, so they already span the private-credit and maturity main effects. Quarter indicators are absorbed by borrower–month effects. There are 17,303 observations for 589 borrowers over 2021Q1–2025Q3. Errors cluster by borrower province. Mixed-channel borrower-months identify the direct comparison, while other months inform the maturity controls. To present the event study relative to 2023Q2, Figure D1 plots

$$\delta_q = \beta_q - \beta_{q_0}, \qquad q_0 = \text{2023Q2}, \qquad \delta_{q_0} = 0. \tag{D.2}$$

Zero on the plot means that the gap is unchanged from the 1.90 percentage points of 2023Q2 and not that yields are equal. The gap reaches 2.86 points in 2024Q4, which is 0.96 points above the reference. Confidence intervals use the covariance of both gap estimates. The plot therefore shows how the relative financing terms of the same borrowers moved.

Retreat from home LGFV bonds and initial home overweight (Section 5.1). Comparing local-bank products in the same home province and quarter, we estimate

$$\begin{aligned}\text{Home LGFV bond share}_{iq} = \beta\, &\text{Initial home overweight}_a \times \text{Refinancing}^{debt}_{p_b,q} \\ &+ \gamma' \text{Controls}_{iq} + \alpha_a + \alpha_{p_b,q} + \varepsilon_{iq}.\end{aligned} \tag{D.3}$$

The outcome is the product's home LGFV bond share in percentage points. Initial home overweight is the issuing institution's product-assets-weighted 2022Q1–2023Q3 average portfolio home overweight, summed across bonds and private credit (Section 4.1). Refinancing intensity is $\text{Refinancing}^{debt}_{p_b,q}$, the cumulative special-refinancing bond issuance divided by 2020 LGFV debt in the parent bank's home province. Institution and home-province–quarter effects hold persistent portfolio differences and common regional conditions fixed. A negative β means that initially more home-oriented institutions reduce home bonds more as refinancing rises. Both interaction variables are fractions. The eleven time-varying controls enter, and errors and wild-bootstrap draws cluster by home province. The refinancing denominator is LGFV debt, not GDP as in the spread tests. Measuring initial home overweight before the announcement or allowing it separate quarterly paths gives similar estimates. The event study in Figure 9 repeats the home-versus-other-buyer comparison of equation (8) for an entire destination province rather than a single city, so its changes are not directly comparable in size to the per-city estimate. Adjustment can occur through sales or by letting bonds mature without replacing them.

Bank size and the shift into private credit (Section 5.2). For product i of local bank b in quarter q, we estimate

$$\begin{aligned}\text{Private credit share}_{iq} = \beta_M\, &\mathbb{1}\{\text{Middle tercile}\}_b \times \mathbb{1}\{\text{Post}\}_q \\ &+ \beta_L\, \mathbb{1}\{\text{Largest tercile}\}_b \times \mathbb{1}\{\text{Post}\}_q \\ &+ \gamma' \text{Controls}_{iq} + \alpha_b + \alpha_q + u_{iq}.\end{aligned} \tag{D.4}$$

The outcome is home or total private credit as a percentage of the product's included debt holdings. Banks remain in their FY2021 asset terciles, and Post begins in 2023Q4. The smallest tercile is the comparison group. Bank effects remove persistent allocation differences, and quarter effects remove common changes. Thus β_L measures how much more the largest group's private-credit share changes after implementation than the smallest group's share. We weight each product-quarter by the value of its included debt holdings, giving larger portfolios more weight. The sample covers 2022Q1–2025Q3. The eleven controls are those in

equation (5). Errors cluster by parent bank. Fixed group indicators and Post are absorbed by the fixed effects. The comparison that conditions on initial home overweight (Table 14) allows initial home overweight and log FY2021 bank assets to predict post-program adjustment separately. It retains institution effects, the eleven controls, and home-province–quarter effects.

Return losses and private-credit substitution (Section 5.2). For equation (10), we winsorize monthly log returns at the 1st and 99th percentiles of all monthly returns. We then average them across NAV series within product-month and across products within bank-month. Post begins in October 2023. The initial shares are raw fractions and not deviations from market weights, and size is log FY2021 interest-bearing assets. A negative β_L means that a higher initial home LGFV bond share predicts a larger return loss, given subsequent substitution and size. A positive β_P means that banks that expanded private credit preserved more return, given their initial portfolios and size. Including initial private credit distinguishes expansion from simply having held more of the instrument before the program. The size–Post coefficient is 0.900 monthly basis points per log point of FY2021 assets (bank-clustered standard error 0.735). It is 1.808 (standard error 0.744) when size enters with the controls alone in column 1 of Table 17. A further specification drops the private-credit interactions and includes bank effects, subsidiary-status-by-month effects, and the eight lagged controls. In it, a ten-point higher initial home LGFV bond share predicts about 10 basis points more annualized return decline (wild $p = 0.050$). The relationship survives province-by-month effects (wild $p = 0.010$) and separate quarterly size slopes.

Balances in newly launched WMPs (Section 5.2). New-product balances sum the balances of observed products launched in quarter q. They are measured at the end of $q + 1$, when products are roughly three to six months old. The measure reflects both the number of products launched and their balances. We estimate

$$\begin{aligned}\text{Log new-product balances}_{bq} = {} & \beta_L \,\text{Initial home LGFV bond share}_b \times \mathbb{1}\{\text{Post}\}_q \\ & + \beta_P \,\Delta\text{Home PC share}_b \times \mathbb{1}\{\text{Post}\}_q \\ & + \beta_S \,\text{Log bank size}_b \times \mathbb{1}\{\text{Post}\}_q \\ & + \beta_H \,\text{Initial home PC share}_b \times \mathbb{1}\{\text{Post}\}_q \\ & + \gamma' \,\text{Controls}_{b,q-1} + \alpha_b + \alpha_q + u_{bq}.\end{aligned} \tag{D.5}$$

Initial shares, private-credit expansion, and bank size have the same definitions as in equation (10), and the same eight bank and home-city controls lag one quarter. Bank effects α_b absorb the fixed bank-characteristic levels, and launch-quarter effects α_q absorb common changes, including Post. Post starts with launches in 2023Q4. Each bank–launch quarter receives equal weight, and errors cluster by bank. Column 4 of Table 17 reports this specification, and column 3 keeps only bank size and the controls. The sample covers launches in 2022Q2–2025Q2 except 2023Q3, and missing bank–launch quarters are not assigned zero

balances. The size coefficient is 0.138 (standard error 0.109) and is not statistically significant, against 0.292 (standard error 0.091) in column 3. The sample uses the bank–launch quarters with sufficient pre- and post-program holdings to measure expansion. Ending the sample in June 2024 gives a coefficient on the initial home LGFV bond share of −1.613 (standard error 0.505, $p = 0.001$). Tables report unstandardized coefficients, and percentage effects in the text use $100[\exp(\beta\Delta x) - 1]$.

Bank size and product returns (Section 5.2). For product returns, each observation is a NAV-series–month, as in equation (5). The event study in Figure 10 builds on the regression

$$\begin{aligned}\text{Monthly return}_{im} = {} & \beta_S \,\text{Log bank size}_b \times \text{Refinancing}^{debt}_{p_b,m} \\ & + \beta_B \,\text{Initial home overweight}_a \times \text{Refinancing}^{debt}_{p_b,m} \\ & + \gamma' \,\text{Controls}_{i,h(m)} + \alpha_a + \alpha_{p_b,m} + \varepsilon_{im}.\end{aligned} \tag{D.6}$$

The size measure Log bank size_b, $\text{Initial home overweight}_a$, and home-province intensity retain their earlier definitions. The outcome $\text{Monthly return}_{im}$ is in basis points and winsorized at the 1st and 99th percentiles of all monthly returns. The eleven controls are measured at the preceding quarter end $h(m)$. Issuing-institution effects remove persistent performance differences, and home-province-by-month effects compare banks facing the same regional conditions.

Figure 10 replaces both refinancing interactions in equation (D.6) with log bank size interacted with quarter indicators and omits 2023Q3. The monthly return outcome, eleven controls, and fixed effects are unchanged, and initial home overweight is not included. The sample covers 2022Q2–2025Q3. Observations are unweighted, and errors cluster by parent bank. Institution effects absorb fixed size, and province–month effects absorb the quarter indicators. The pre-program coefficients show no trend. The plotted coefficients remain in monthly units. They are 5.69 and 7.96 basis points per log point of assets in 2024Q3 and 2025Q2. Multiplying by $\ln 2 \times 12$ gives the annualized doubling comparisons.

Total WMP balances (Section 5.2). In equation (9), size is log FY2021 interest-bearing assets in RMB trillion. A positive β_S means that balances grow proportionally more after implementation at initially larger banks. Column 2 of Table 16 replaces common half-year effects with province-by-half-year effects, and column 3 replaces continuous size with fixed size-tercile indicators interacted with Post. Fixed size is absorbed by bank effects and Post by time effects. The bank-level panel enters without additional controls. For an asset ratio a, the implied relative growth difference is $100[\exp(\beta_S \log a) - 1]\%$. Missing reports are not zeros. Balances measure the scale of the WMP business. They combine investment returns, product entry and maturity, and investor transactions. The raw size–Post coefficient is 0.152 log balance points per log point of assets, and 0.144 within the same province and half-year. The event study in Figure 11 replaces the size–Post interaction with size interacted with each half-year reporting date and omits June 2023. It retains the baseline's bank and

half-year effects, bank-clustered errors, and absence of additional controls. The quarterly axis shows actual reports at Q2 and Q4, without interpolating Q1 or Q3 balances. The raw log-balance coefficients are 0.146 and 0.334 per log point of assets in December 2024 and December 2025, and the doubling comparisons are $100[\exp(0.146 \ln 2) - 1] \approx 11\%$ and $100[\exp(0.334 \ln 2) - 1] \approx 26\%$. The late-period difference exceeds the average 11% Post comparison because that average also includes the earlier reports when the gap was much smaller.

Borrower funding (Section 5.3). High-reliance and low-reliance banks are the local banks without a subsidiary whose initial home LGFV bond share in equation (10) was above and below the median. Let X_f^H and X_f^L be the fractions of borrower f's WMP holdings that these two groups held in 2022Q4–2023Q1. For borrower f in quarter q, we estimate

$$\begin{aligned}\text{Holdings}_{fq}^G = {} & \beta_H \, X_f^H \times \mathbb{1}\{\text{Post}\}_q + \beta_L \, X_f^L \times \mathbb{1}\{\text{Post}\}_q \\ & + \eta_H \, X_f^H \times \mathbb{1}\{\text{Late}\}_q + \eta_L \, X_f^L \times \mathbb{1}\{\text{Late}\}_q + \alpha_f + \alpha_{p(f),q} + \varepsilon_{fq}.\end{aligned} \tag{D.7}$$

The outcome is the inverse hyperbolic sine of the borrower's bonds held by the WMPs of holder group G, in RMB 10,000. Post begins in 2023Q4 and Late in 2024Q3, after the June 2024 guidance. Borrower effects α_f remove persistent differences in WMP funding, and borrower-province-by-quarter effects $\alpha_{p(f),q}$ compare borrowers in the same province and quarter, which absorbs provincial refinancing. The coefficient β_H compares a borrower funded entirely by high-reliance banks with one that neither group funded, and $\beta_H - \beta_L$ tests whether a higher initial home LGFV bond share mattered beyond being a local bank without a subsidiary. The sample covers borrower-quarters in 2023Q2–2025Q2, and errors cluster by borrower. Panel B of Table 18 repeats the regression for private credit. After June 2024, total WMP holdings of the borrowers of high-reliance banks fell by a further 0.48 log points. For a borrower funded entirely by high-reliance banks, private credit held by local banks without a subsidiary falls by 3.96 log points and national banks' holdings rise by 2.28, compared with 1.61 and 1.14 for borrowers of low-reliance banks.

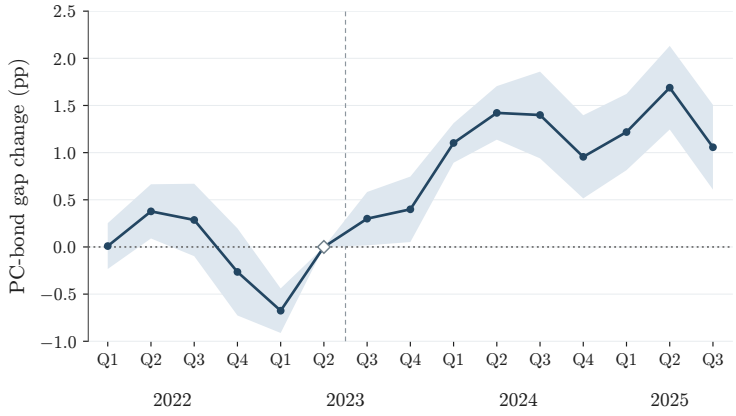
These borrowers also became a larger part of national banks' WMP portfolios. We say that local banks without a subsidiary mainly funded a borrower when they held at least half of its WMP holdings before the program. The share of national banks' LGFV bond holdings invested in such borrowers rose from 10.1% in 2022Q4 to 19.3% in 2025Q2. National banks earned more on these positions than on the rest of their LGFV bond holdings, with value-weighted coupons of 3.74% against 3.37% in 2025Q2. Column 5 of Table 18 relates the coupons of the borrowers' new bonds to the same funding shares. It compares bonds of the same borrower and province-month and controls for maturity. In the same specification, the coupon's premium over the CSI valuation yield at listing also fell for X_f^H, by 15.0 basis points (standard error 3.7) after implementation and by a further 9.0 (2.9) after June 2024.

Table D1
Initial credit portfolios by bank type and bank size

Share (%)	All provinces		Home province	
	LGFV bonds	PC	LGFV bonds	PC
Panel A. Bank type, all available banks				
National banks	12.6	24.1	0.4	2.1
Local banks	43.1	9.3	24.7	6.1
Panel B. FY2021 size terciles, local banks present in both periods				
Smallest tercile	38.6	1.1	25.1	0.7
Middle tercile	53.3	5.3	37.7	3.5
Largest tercile	40.6	11.4	20.7	7.4

Notes: PC denotes private credit. Entries are value-weighted percentages of included debt holdings over 2022Q1–2023Q3, which exclude deposits, funds, other investment vehicles, and central-government, central-bank, and policy-bank obligations. Home means the parent bank's headquarters province. Panel A contains 34,595 national-bank and 36,194 local-bank product-quarters. Panel B uses the local banks and fixed FY2021 size terciles of Panel C of Table 12. Table 4 reports the controlled comparisons. Before implementation, local banks hold 65.0% of the sample's LGFV bonds and national banks hold 82.7% of its PC. Afterward these shares are 62.6% and 68.7%.

Figure D1
The private credit–bond rate gap within the same borrower



Notes: This within-borrower event study plots $\delta_q = \beta_q - \beta_{2023Q2}$ from equation (D.2). It is the change in the annual contractual gap between private credit and bond rates of the same borrower, in percentage points. Gap levels come from equation (D.1), which includes borrower-by-calendar-month effects and maturity interacted with every quarter. The sample is 17,303 equally weighted financing observations of 589 borrowers in 2021Q1–2025Q3. Shading gives pointwise 95% confidence intervals clustered by borrower province, the hollow diamond marks the reference, and the dashed line separates 2023Q2 from 2023Q3. Coverage falls to 58, 35, and 21 borrowers in 2025Q1–Q3.

Table D2
Alternative specifications for bank size, risk held, and risk reported

Specification	Baseline	Products weighted equally	Asset-mix controls	Bonds at least 60% of holdings	Plans marked at AAA	Only bonds marked
	(1)	(2)	(3)	(4)	(5)	(6)
Home LGFV credit share (pp)	−5.394***	−5.730***	−4.048***	−6.012***	—	—
	(0.874)	(1.092)	(0.755)	(1.097)		
Issue spread of bonds held (bp)	−22.374***	−22.578***	−21.968***	−19.168***	—	—
	(3.511)	(4.478)	(3.630)	(4.167)		
Issue spread relative to same rating (bp)	−17.079***	−11.900***	−18.553***	−16.085***	—	—
	(4.084)	(3.445)	(4.402)	(5.262)		
All-in spread of holdings (bp)	−12.792***	−10.570***	−13.556***	−17.082***	−14.642***	—
	(2.866)	(3.893)	(2.437)	(3.576)	(2.941)	
Log volatility implied by holdings	−0.074***	−0.087***	−0.024*	−0.038*	−0.077***	−0.253***
	(0.016)	(0.018)	(0.013)	(0.020)	(0.015)	(0.060)
Log SD of monthly return	0.154***	0.070**	0.158***	0.191***	—	—
	(0.036)	(0.035)	(0.035)	(0.051)		
Log reported minus log implied SD	0.228***	0.158***	0.183***	0.229***	0.231***	0.406***
	(0.040)	(0.033)	(0.039)	(0.055)	(0.040)	(0.069)
Return, Nov.–Dec. 2022 (bp)	−6.228***	−1.697	−7.941***	−8.963***	—	—
	(1.669)	(1.949)	(1.635)	(2.174)		
Reported minus implied, Nov.–Dec. 2022 (bp)	−11.656***	−8.394***	−10.124***	−13.200***	−12.205***	−16.357***
	(2.320)	(2.070)	(2.168)	(3.050)	(2.226)	(2.155)
Mean monthly return (bp)	−1.413***	−1.309***	−1.405***	−1.302***	—	—
	(0.297)	(0.373)	(0.310)	(0.322)		
Observations	4,165	4,165	4,165	1,406	4,165	4,165
Redemption type × risk rating fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank	Parent bank

Notes: Each entry is the coefficient on log bank size from a separate regression, and column 1 repeats Table 7. Column 2 weights products equally. Column 3 controls for the shares of disclosed holdings in cash and deposits, certificates of deposit, private credit, funds, asset-management plans, and other securities. Column 4 keeps products whose disclosed holdings are at least 60% bonds. Column 5 marks plans and other wrappers with the one-to-three-year AAA credit-bond index instead of the AA+ index, and column 6 marks only directly held bonds. The observations row refers to the home LGFV credit share. Except in column 2, each bank receives equal total weight. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance. Estimation sample: April 2022–June 2023.

Table D3
Net flows by reported return in the 2022 sell-off quarter

Reported return, 2022Q4	Products	Median reported return (bp)	Share of beginning assets (%)	Net flow (% of beginning assets)
Loss of more than 0.5%	303	−86.6	13.5	−25.3
Loss of 0.1% to 0.5%	160	−33.8	12.5	−27.5
Loss of less than 0.1%	46	−4.8	3.3	−37.1
Gain of less than 0.5%	196	23.2	34.3	−17.6
Gain of 0.5% to 1%	114	56.7	36.3	8.9
Gain of more than 1%	7	144.5	0.0	39.4
All daily-open products	826	−25.5	100.0	−10.9

Notes: The table covers daily-open WMPs in 2022Q4, which is the quarter of the bond sell-off. It groups them by the return each product reported for the quarter. Net flow is end-of-quarter assets minus beginning assets grown by the reported return, which is the asset growth net of investment gains used in Table 1. It is summed within each group and divided by the group's beginning assets. Daily-open products allow withdrawals on any business day, so their outflows reflect savers' redemption decisions. The median reported return is over the quarter. In the preceding quarter of 2022Q3, daily-open products took in net flows of 6.2% of beginning assets.

Table D4
Flows after the 2022 sell-off, by reported and implied losses

	Net flow in 2023Q1 (% of beginning assets)		
	All products	Daily- and periodic-open	Daily-open
	(1)	(2)	(3)
Reported return, Nov.–Dec. 2022 (per 10 bp a month)	1.16***	1.35**	8.22*
	(0.38)	(0.57)	(4.36)
Return implied by holdings, Nov.–Dec. 2022 (per 10 bp a month)	−0.45	−0.01	−0.79
	(0.28)	(0.37)	(1.20)
Mean reported return (bp a month)	−26.0	−29.4	−19.1
Mean implied return (bp a month)	−63.6	−58.1	−39.0
Observations	5,296	3,732	481
Controls	Yes	Yes	Yes
Risk rating × redemption type fixed effects	Yes	Yes	Yes
Bank fixed effects	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank

Notes: The unit of observation is a product of any type with disclosed holdings and at least nine monthly returns between April 2022 and June 2023 that reports assets in 2022Q4 and 2023Q1. The dependent variable is its net flow in 2023Q1. Net flow is asset growth net of investment gains as a percentage of beginning assets, as in Table 1. The two returns enter jointly. The reported return is the product's mean monthly reported return in November and December 2022, and the implied return is its mean monthly return in the same months when its disclosed holdings are marked to market prices (Internet Appendix D.3). The regressions include the eleven lagged controls of Table 1, fixed effects for risk rating by redemption type, and bank fixed effects, so products are compared with other products of the same bank. Parentheses contain parent-bank clustered errors. Stars indicate 10%, 5%, and 1% significance.

Table D5
Home credit exposure, bond income, and product returns

The dependent variable is the monthly product return (bp).

Panel A. Bond-income controls

Specification	Observed rates No yield control (1)	Observed rates Add yield control (2)	50% coverage No yield control (3)	50% coverage Add yield control (4)
Home LGFV bond share	8.831***	3.759**	9.021***	3.692**
	(1.964)	(1.663)	(2.083)	(1.815)
Home LGFV bond share × 𝟙{Post}	−6.731***	−2.886*	−7.245***	−3.222*
	(1.662)	(1.568)	(1.701)	(1.654)
Home private credit share	8.759***	6.895***	8.087**	6.370**
	(1.450)	(1.402)	(3.297)	(3.044)
Home private credit share × 𝟙{Post}	−3.639*	−2.227	1.280	1.926
	(1.892)	(1.859)	(3.346)	(3.142)
Mean bond issue yield (pp)	—	2.659***	—	2.773***
		(0.510)		(0.521)
Mean bond issue yield × 𝟙{Post}	—	−1.334***	—	−1.414***
		(0.452)		(0.487)
Implied effects from the same regression				
Post-program home-PC premium	5.120***	4.668***	9.367***	8.295***
	(1.490)	(1.353)	(2.978)	(2.637)
Product, bank, and city controls	Yes	Yes	Yes	Yes
Observations	524,670	524,670	449,162	449,162
Month fixed effects	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Panel B. NAV and sample checks

Specification	Mature products (1)	Add volatility (2)	Product-month average (3)	No winsorizing (4)
Home LGFV bond share	9.048***	8.442***	8.990***	8.512***
	(1.777)	(2.269)	(2.040)	(1.938)
Home LGFV bond share × 𝟙{Post}	−6.553***	−5.285***	−7.254***	−6.609***
	(1.811)	(1.978)	(1.662)	(1.718)
Home private credit share	6.951***	7.012***	7.859***	7.598***
	(1.514)	(1.412)	(1.559)	(1.467)
Home private credit share × 𝟙{Post}	−2.816	−2.619	−4.418**	−3.637**
	(1.865)	(1.644)	(1.797)	(1.559)
Implied effects from the same regression				
Post-program home-PC premium	4.136***	4.393***	3.441**	3.961***
	(1.502)	(1.217)	(1.365)	(1.089)
Product, bank, and city controls	Yes	Yes	Yes	Yes
Observations	380,984	383,198	448,645	567,855
Month fixed effects	Yes	Yes	Yes	Yes
Standard errors	Parent bank	Parent bank	Parent bank	Parent bank

Notes: All regressors of Table 6 enter. Month effects absorb Post. In Panel A, columns 1–2 keep products with observed bond issue rates, and columns 3–4 require issue rates for at least 50% of included debt holdings. Columns 2 and 4 add mean bond issue yield and its Post interaction. In Panel B, column 1 uses mature products with age and remaining-maturity controls, column 2 adds volatility and its Post interaction, column 3 averages NAV series equally within product-month, and column 4 uses returns without winsorization. Parentheses contain bank-clustered errors. Stars indicate 10%, 5%, and 1% significance.